Operations and Maintenance (O&M) of the Central Utility Plant (CUP) at the NIST Boulder Campus

SOURCES SOUGHT

Original Set Aside: N/A

Set Aside: N/A

Classification Code: J -- Maintenance, repair & rebuilding of equipment

NAICS Code: 811 -- Repair and Maintenance/811310 -- Commercial and Industrial Machinery and Equipment (except Automotive and Electronic) Repair and Maintenance

Synopsis: ***PLEASE OPEN ATTACHMENTS FOR COMPLETE SOURCES SOUGHT ANNOUNCEMENT INFORMATION***

In accordance with (IAW) FAR 15.201(c)(3) this is a SOURCES SOUGHT FOR MARKET RESEARCH. THIS

SOURCES SOUGHT IS NOT A REQUEST FOR PROPOSALS, QUOTATIONS OR BIDS. This announcement constitutes a Sources Sought (market survey). This notice is a request for information to provide data for planning purposes only and does not constitute an Invitation for Bids, a Request for Proposals, a Solicitation, a Request for Quotes, or an indication the Government will contract for the services contained in this Sources Sought. A solicitation is not currently available. This Sources Sought is a market research tool used by the acquisition team to determine potential and eligible businesses that have the capability and interest in providing the required commercial services described herein, prior to determining the method of acquisition. This also serves as a Request for Comments on the draft performance work statement (PWS) for the planned project.

The U.S. Department of Commerce, National Institute of Standards & Technology (NIST), Acquisition Management Division (AMD), Team E-Boulder seeks information from PRIME CONTRACTORS that are capable and interested in performing the work on a commercial service contract at the NIST Boulder, Colorado Campus Site.

This request is not to be construed as a commitment on the part of the Government to award a contract, nor does the Government intend to pay for any information submitted as a result of this notice. The Government does not reimburse respondents for any cost associated with this source sought or reimburse expenses incurred to interested parties for preparing a response as a result of this posting. The Government is not obligated to and will not pay for any information or comments received from potential sources as a result of this Sources Sought and Request for Comments announcement. Any responses received will not be used as a proposal. Respondents will not be notified of the Sources Sought evaluation results, nor will questions be addressed by the Government.

The results of this Sources Sought will be utilized to assess the amount of interest and capability in the current market, and to determine if any Small Business Set-aside opportunities exist for this potential requirement. All Small Business Set-aside categories will be considered. This market research tool will be used to identify potential and eligible service contractors, of all sizes, prior to determining the method of acquisition and level of announcement for issuance of a solicitation.

The Request for Comments portion of this Sources Sought provides contractors the opportunity to make any comments, including recommendations, regarding the draft PWS for the potential project. Comments will be used for consideration in the development of the PWS. The Government will not provide responses to any questions or Requests for Information regarding this project as a part of this Sources Sought and Request for Comments.

Project Description:

1. NIST is soliciting for capability statements from all interested parties to perform as a Contractor for the Operation and Maintenance of Central Utility Plant, which includes preventative maintenance of all associated equipment, for the NIST Boulder, Colorado Campus. This also includes performing preventative and scheduled maintenance activities at the field sites in Ft. Collins, Colorado, and Kekaha, Hawaii.

2. The North American Industry Classification System (NAICS) code for the proposed acquisition is 811310, Commercial and Institutional Machinery and Equipment (except Automotive and Electronic) Repair and Maintenance, and the Small Business size standard is $12.5 million.

3. Planned acquisition type: Commercial Services, Firm-Fixed-Price Contract. The acquisition is planned to be performed in accordance with FAR PART 12 Acquisition Commercial Items and FAR PART 15 Contracting by Negotiation.

4. See the following attachments:
   - **a.** DRAFT Performance Work Statement, 20 May 2021, “00 CUP PWS 2021-05-20.pdf”
   - **b.** PWS Attachment 1 – Mechanical Room and Preventative Maintenance (MR_PM), “01 Mechanical Room and PM List 4-12-2023.pdf”
   - **c.** PWS Attachment 2 – Spare Parts Inventory, “02 Spare Parts Inventory.pdf”
   - **d.** PWS Attachment 3 – Monthly Report, “03 CUP Monthly Report.pdf”
   - **e.** PWS Attachment 5 – Job Plans Revised, “05 PWS Attach 5 Job Plans 4-12-23.pdf”
   - **f.** PWS Attachment 6 – Daily Report, “06 Daily Report.pdf”
   - **g.** PWS Attachment 8 – NIST General Instructions, “08 NIST General Requirements.pdf”
   - **h.** PWS Attachment 9 – NIST Safety Plan Requirements Checklist, “09 Safety Plan Checklist.pdf”
   - **i.** PWS Attachment 10 – Primer on Safety Plan Require/Addendums COVID-19, “10 Primer on Safety Plan Requirements Addendums for COVID19.pdf”

- **j.** PWS Attachment 11 – Emergency Generators, “11 Emergency Generator Information.pdf”
- **k.** PWS Attachment 12 – Lighting Summary WWVh Field Site Kekaha, HI, “12 WWVH lighting summary.pdf”
- **l.** PWS Attachment 13 – Contract Transition Requirements, “13 Transition 11 Sep.pdf”
- **m.** Comment Log for Draft PWS.

Sources Sought: Interested and capable PRIME CONTRACTORS are requested to electronically submit a Capabilities Statement of no more than twenty (20) pages in length inclusive of the following information:

1. Company name and address. The name, position title, telephone number and email address of a company point of contact (POC).
2. UEI/CAGE code registered in the system for Award Management (SAM) at www.sam.gov.
3. Indicate whether your company is a large business or a small business under the NAICS code specified in this announcement.
4. Company Profile description.
5. Statement of Capability stating your company's skills, experience knowledge and equipment to perform the type of work identified in the PWS. This should include documentation of past specialized experience and technical competence in performing similar projects, up to five (5) recent and relevant past projects for which your company was the prime contractor. For the purposes of the Sources Sought, "similar" is defined as a high-pressure steam plant with the output capacity of at least 100 thousand pounds per hour of steam at a minimum of 100 psi, and a chilled water plant utilizing multiple chillers of at least 1200 tons in size. In addition to the above, simi lar projects may include conducting preventative maintenance for commercial building equipment such as air handlers, roof top packaged HVAC (natural gas heating and DX cooling), small, packaged boilers (1,500,000 to 2,500,000 BTU's), small (25 -50 ton) chillers, cooling towers, pumps, heat exchangers, and exhaust fans.
6. Recent experience is experience performed within the last five (5) years. For each project experience, identify the following:
    - **a.** Did your company perform as a prime or sub-contractor?
    - **b.** Description of the work performed by your company. Identify /describe the work types (i.e. steam plant, chilled water plant, preventative maintenance for commercial building equipment, etc.)
    - **c.** Approximately what dollar amount and percentage, and what types of work d id your company perform directly? What percentage and types of work did your company sub-contract out?
    - **d.** Provide a contract number, the customer organization Government/Agency or private firm), a POC, and a current telephone number and email address for the customer POC.
7. Can your company provide the services as a PRIME contractor?
8. Would your company provide the services per the Draft PWS as a part of a Joint Venture, Partnership, Mentor/Protégé or other legal arrangement? If so, please identify the legal arrangement.

9. Would your company sub-contract work in the performance of the PWS? If so, what percentage of work would your company perform as a Prime Contractor?
10. Any other relevant information that is not listed above which the Government may consider in developing the service requirement and market research.
11. Company sales brochures or marketing packages will NOT be considered.

Request for Comments: All comments provided in response to the Sources Sought shall be submitted electronically on the attached Request for Comments sheet. Comments will be used for consideration in the development of the PWS. The Government will not provide responses to any questions, Requests for Information, regarding this project as a part of this Sources Sought and Request for Comments.

Submittal Instructions: All responses to this Sources Sought shall be submitted via email by the required Response Date to Primary Point of Contact with a courtesy copy to the Alternate Point of Contact in this announcement.

DRAFT PERFORMANCE WORK STATEMENT AND RELATED ATTACHMENTS

The following attachments are provided as part of this sources sought:

1 Series – Introduction of NIST					
	1.1 - Intent				
	1.2 - Shift Information				
	1.3 - Contract Transition Requirements				
2 Series – Key Personnel					
	2.1 - Key Personnel Requirements				
	2.2 - Uniforms				
	2.3 - Employee appearance				
	2.4 - Employee PPPE				
	2.5 - Employee Communication				
3 Series – Running of the Central Plant					
	3.1 - Operation and Maintenance of CUP at the NIST site in Boulder, CO				
	3.2 - Materials and Equipment supplied by the contractor				
	3.3 - Materials and Equipment supplied by the government				
4 Series -Preventative Maintenance					
	4.1 - Preventative Maintenance				
	4.2 - Ft. Collins				
	4.3 - Hawaii				
5 Series – Reporting Requirements					
6 Series –Major Repairs					

Spec Item	Title	Performance Objective	Related Information	Performance Standard	Acceptable Quality Level
1.1	Intent	NIST intends to award a Performance Based Service Contract for the services identified in this Performance Work Statement (PWS). The contract will have Firm-Fixed Price (FFP) line items for the following tasks: 1. 24/7/365 Operation and Maintenance Services at the Central Utility Plant (CUP) in Boulder, CO. 2. Preventative and Scheduled Maintenance at the WWV/WWVB Field Site in Ft. Collins, CO. 3. Preventative and Scheduled Maintenance at the WWV/WWVH Field Site in Kekaha, HI.	In addition to the FFP line items for Operation and Maintenance of the CUP, and the preventative maintenance at the other two sites, NIST further intends to provide for Time and Material (T&M) contract line items for necessary repairs outside the scope of maintenance. Funds in these line items are available ONLY for approved repairs, replacements, and/or rental of equipment to provide 24/7/365 operations of all systems associated with the CUP, all steam systems including piping, traps, expansion joints, condensate systems and all other Misc. components of a steam system, all chillers and chilled water supporting components, all air compressors and compressed air supporting components, any supporting building 42 or tunnel services such as domestic water, electrical and fire protection, all equipment at all three NIST sites that is maintained by the contractor shall be repaired under the repair line items. All work must be performed in compliance with *General Instructions for Conducting Work at the NIST Boulder Campus, and the Fort Collins, CO and Kauai, HI Field Sites - FMD-B, Current Version.*	N/A	N/A
1.2	Shift Information	The CUP shall be staffed with 3 shifts of staff per 24 hours.	For safety concerns and operational continuity with NIST personnel the shifts shall be 8 hours long. Shift 1 shall be: 7:00am - 3:00pm Shift 2 shall be: 3:00pm-11:00pm Shift 3 shall be: 11:00pm-7:00am All staff members will be hard badged with 24 hour access	The plant shall be staffed 24/7/365.	100%
1.2.1	Shift Information	Contractor shall fully staff the central plant with badged employees.	Contractor shall follow the multiple step badging process of the NIST Boulder site.	Contractor shall submit required badging paperwork for employees to NIST 45 days prior to effective date of the contract or any personnel changes during the life of the contract.	100%
1.3	Contract Transition	The contractor shall fully cooperate with contract transition requirements containedin PWS Attachemetn 13 - Contract Transition Requirements	Services under this contract are vital to the Government and must be continued without interruption. In accordance with contract clause 52.237-3, Continuity of Services, both the incoming and outgoing contractors shall provide services to facilitate an orderly and efficient transition to the contractor selected for the award of this contract and the subsequent follow-on Central Utility Plant (CUP) Operations and Maintenance contract. The transition period between contractors shall be the first 30 days of the new contract and the final 30 days of the existing contract. During the 30 day transition the incumbent contractor shall cooperate and provide support with the transition of the successor contractor as outlined below.	The contractor shall fully cooperate and provide services to facilitate an orderly and efficient transition at the start of this contract as the sucessor contractor and at the end of this contract as the incumbent contractor.	100%
2	Key Personnel	The contractor shall provide the appropriate personnel to perform the necessary work as listed below.	Key personnel positions must be employees of the prime contractor. Key personnel cannot perform as a sub-contractor.		
2.1.1	Key Personnel (Project Manager)	The Contractor shall provide a Project Manager who will be responsible for the performance of the work.	The Project Manager and alternate will have full authority to act for the Contractor on all contract matters relating to the daily operation of this contract. Must have experience managing other projects of similar size and scope. May act as security point of contact responsible for all security aspects of the contract.	The name and contact information of this person, and alternates who will act for the Contractor when the manager is absent, will be designated in writing to the contracting officer and COR within 10 days after award. These people will serve as an emergency contact and be available 24/7/365.	100%
2.1.2	Key Personnel (Alternate Project Managers)	The contractor shall provide an alternate project manager.	The Project Manager and alternate will have full authority to act for the Contractor on all contract matters relating to the daily operation of this contract. May act as security point of contact responsible for all security aspects of the contract.	The name and contact information of this person, and alternates who will act for the Contractor when the manager is absent, will be designated in writing to the contracting officer and COR within 10 days after award. These people will serve as an emergency contact and be available 24/7/365.	
2.1.3	Key Personnel (Project Engineer)	The Contractor shall provide a dedicated Project Engineer who is a stationary engineer experienced in the operation of large mechanical systems such as boilers, chillers, air compressors and central plant operations.	It shall be the duty of this person to be on the NIST Boulder site and coordinate the delivery of quality service, offer technical assistance, make suggestions for systems improvement and implement quality assurance by submitting the daily logs. See section 2.1.4 for qualifications	This Project Engineer shall be available during the core hours of (0700 – 1500), Monday through Friday. Daily logs shall be submitted by 9 a.m. to the COR via email utilizing the format given by NIST.	Available 80% of the time

Spec Item	Title	Performance Objective	Related Information	Performance Standard	Acceptable Quality Level
2.1.4	Key Personnel (Stationary Engineers)	Stationary Engineers shall have a valid Stationary Engineer certification/license and possess or obtain a valid Stationary Engineer Certificate from the City and County of Denver. The stationary engineer will be responsible for utilizing and monitoring the NIST Direct Digital Control Building Management System.	The stationary engineer will be responsible for utilizing and monitoring the NIST Direct Digital Control Building Management System. CUP personnel (Stationary engineer) will have to be competent to operate this system and diagnose problems of specified equipment using it. Stationary Engineers must have four-years of verifiable work in stationary engineering work consisting of a minimum of three-years operating high-pressure steam boilers producing over 10 hp and steam pressure over 15 p.s.i. or water heating boilers when the water temperature exceeds 250 degrees F in the system, and a minimum of one-year operating refrigeration Equipment requiring an operator. The digital control systems used by the Government is Johnson Controls/Metasys. The NIST System Administrator will provide an 8 hour orientation to the NIST Metasys Controls system upon award of the contract.	Stationary Engineers must possess or be able to obtain a City and County of Denver Stationary Engineer Certificate within 180 days of the contract award date. Stationary Engineers are required to maintain a valid City and County of Denver Stationary Engineer Certificate through the life of the contract. New hire personnel after the award date will be required to have the proper qualifications before holding the position of Stationary Engineer and will have 180 days in which to obtain City and County of Denver Stationary Engineer Certificate. Proof of qualifications for stationary engineer's certification/license shall be provided as shown below: 1) No later than 30 days after contract award or after applicable employee starts work 2) Upon Receipt of City and County of Denver certification (no later than 180 days after applicable employee starts work). 3) Annually not later than 30 days after receipt of any option modification.	100%
2.1.5	Key Personnel - HVAC Mechanics	HVAC mechanics shall have a minimum of 5 years of experience and hold a Universal CFC License	The Contractor shall ensure employees have current and valid professional certifications before starting work under this contract. The Contractor shall provide, to the NIST COR, a photocopy of the certification (CFC license) of each worker that handles refrigerants on the NIST Site before such work commences.	All workers handling refrigerants on the NIST site must be certified as at least Class III Technicians as required by 40 CFR part 82, subpart F. Proof of qualifications for HVAC Mechanics CFC License shall be provided as shown below: 1) No later than 30 days after contract award or after applicable employee starts work 2) Annually not later than 30 days after receipt of any option modification.	100%
2.2	Uniforms	The contractor employees (including management and supervision) shall wear a distinctive uniform for easy identification.	Uniforms will consist of shirts and work pants as appropriate for the job and for the gender of the employee. No shorts or skirts are allowed due to safety concerns. The shirt portion of the uniform will have, at a minimum, the contractor's company name, and be any combination of colors. All contractor employees shall be in uniform within 30 days after effective date of the contract.	All contractor employees shall be in uniform when on duty at NIST.	100%
2.3	Employee appearance	Employees are required to dress neatly and be clean.	It is the Contractor's responsibility to ensure all employees practice good hygiene, dress appropriately.	All clothing is neat and free of stains, holes, tears, etc.	100%

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2.4	Employee PPE	Contractor shall wear PPE during operations and maintenance of this contract	The contractor will be responsible to provide the appropriate PPE.	The Contractor shall ensure all Contractor and sub-contractor employees wear appropriate PPE as determined by the Contractors site-specific safety plan and job hazard analysis. Contractor shall also adhere to posted hazard signs by wearing the appropriate PPE for the area.	100%
2.5	Employee Communication	Employees shall be able to communicate in writing and speaking effectively in English.	N/A	All contractor employees speak, read, write, and understand English.	100%
3	Operation and Maintenance of CUP at the NIST site in Boulder, CO	Contractor shall provide management and staffing services to operate the Central Utility Plant (CUP), Bldg. 42 and all utilities located in the tunnel system that is attached to building 42.	The CUP Shall provide steam, laboratory-grade compressed air, and chilled water to buildings 1, 2, 3, 24 and 81 and chilled water only to building 92 on the NIST campus. Contractors shall staff to operate building 42 CUP building 24/7/365. It is required that one person per shift shall serve as the shift supervisor and shall possess a valid Stationary Engineers Certificate from the City and County of Denver and at a minimum one qualified HVAC mechanic.	The CUP shall be maintained according to all Colorado boiler and pressure vessel guidelines, in effect as of the date of the solicitation, and remain operational 24/7/365.	100%
3.1	Operational	There are several pieces of specialized equipment in the CUP that require a factory trained technician to perform the required preventive maintenance.	Four (4) Carrier Chiller's: 1498 ton capacity each Four (4) Ingersoll Rand Air compressors 100 h.p. each. Four (4) Ingersoll Rand Desiccant air dryer systems.	Carrier Factory trained technicians for the chillers or Ingersoll Rand factory trained technicians for the air compressors/dryers to perform the required preventive maintenance. Submit to COR certification of factory trained technician 30 days prior to the preventative maintenance taking place.	100%
3.1.1	Operational	Contractor shall adhere to a contractor developed quality control plan	The contractor shall provide to the NIST COR a contractor quality control plan to be adhered to through the duration of the contract.	Contractor shall submit required quality control plan to NIST COR within 30 days of contract award. Any changes to quality control plan shall be communicated to NIST 30 days prior to implementation.	100%
3.1.2	Operational Requirements	The Central Utility Plant operates 24/7/365 with no unscheduled outages within the contractors control.	Many research laboratories operate and have equipment running 24/7/365. Any disruption of services provided by the CUP may cause severe loss of valuable information. The coordination of activities between the COR, the contractor and the site building occupants is critical in an effort to minimize disruption of the research and damage to ongoing experiments.	The COR will require 14 days notice prior to, in order to provide notice to customers who will be affected.	100%
3.1.3	Operation and Maintenance of CUP	The COR shall always be notified in the event of an unplanned outage.	COR phone number will be given to contract at kickoff meeting. Notification shall consist of the problem, current actions being taken and anticipated time frame for equipment to be restored.	In the event of an unplanned outage the contractor shall notify the COR by phone within the hour of the outage.	100%

Spec Item	Title	Performance Objective	Related Information	Performance Standard	Acceptable Quality Level
3.1.4	Operation and Maintenance of CUP	The Contractor is responsible for operating the plant to meet the daily plant load requirements for Steam, Chilled Water and laboratory-grade air, to include seasonal changes.	The contractor needs to determine the equipment that shall be needed to provide the adequate steam, chilled water and lab grade quality compressed air according to seasons and site demand. The plant shall be maintained with the following equipment operating or on ready standby: The Summer and winter Boiler requirements are the same one boiler operating and one boiler warmed up on standby. The load requirements for the chillers is two chillers operating and one chiller on standby summer and winter for the current site load conditions. The air compressor load varies according to scientific activity but the requirements are that one compressor and drier always operating with a second air compressor and drier on standby cycling on and off per load demands there shall also be a third compressor and drier always available in ready standby. There should never be more than one boiler, one chiller, one air compressor and one air drier in need of repair or down for preventive maintenance at one time. Contractor shall be responsible for operating and maintaining all existing equipment that exists on the asset list at time of award. The current operation requirements listed above are subject to change with load requirement changes.	The CUP shall be operated and meeting the daily requirements 24/7/365 (no unplanned outages of steam, chilled water or compressed air)	100%
3.1.5	Operation and Maintenance of CUP	It is mandatory that a Stationary Engineer with a current certificate be on duty 24/7/365.	NIST Boulder shall require for safety purposes that the CUP shall have at least 1 Stationary Engineer on duty at the CUP at all times.	The Stationary Engineer must remain in the CUP (Building 42) at all times or be in the vicinity of the adjacent cooling towers and generators at Building 42.	95%
3.1.6	Operation and Maintenance of CUP	It is mandatory that an HVAC Mechanic be on duty 24/7/365.	For the HVAC Mechanic - On duty shall be deemed as being on the NIST Boulder site. It is acceptable for the HVAC mechanic to be in the CUP, doing rounds of mechanical rooms, preventative maintenance, handling emergencies or responding to address alarms. During normal business hours the HVAC mechanic will respond to alarms within the CUP. Outside of normal business hours the HVAC mechanic will respond to alarms throughout the Boulder site.	The HVAC Mechanic must remain on the NIST Boulder site to respond to alarms within 15 minutes.	95%
3.1.7	Operational Requirements	Contractor shall provide a means to communicate between the stationary engineer and the HVAC mechanic.	Contractor is responsible for providing the equipment. This could be cell phones or radios, this is a requirement for safety concerns. Contractor is responsible for ensuring the equipment is operational or providing a replacement as needed. NOTE: There are places on the site that do not have cell phone service.	Stationary engineer must be able to communicate with HVAC mechanic on duty at all times.	95%
3.1.8	Operational Requirements	The Contractor shall perform routine operating inspections for the equipment in the CUP.	Routine operating inspections shall consist of visual inspections, log readings, routine adjustments, calibration and an assessment of operational conditions to ensure proper operation at optimum efficiencies. Documentation shall be maintained in an Excel spreadsheet and made available to the COR as requested. Any operational discrepancies or mechanical malfunction will be immediately reported to COR.	Daily readings, one per shift, shall be provided electronically to the Government by the 5th business day of each month as part of the Monthly Report.	90%
3.1.9	Operational Requirements	The Contractor shall perform routine operating inspections for the NIST Boulder Site.	Once per 8 hour shift (does not include normal hours of operation, 0700 to 1530 Monday Through Friday) the HVAC mechanic shall perform walks through all mechanical rooms within the buildings on the NIST Boulder site. See PWS Attachment 1, Mechanical Room and Preventative Maintenance (MR_PM) List.	The contractor shall document rounds in Daily Report (PWS Attachment 3). Daily Reports shall be submitted by 9 a.m. to the COR via email.	90%

Spec Item	Title	Performance Objective	Related Information	Performance Standard	Acceptable Quality Level
3.1.10	Operation and Maintenance of CUP	The contractor shall respond to any Metasys alarms, calls from police dispatch or notification from COR 24/7/365.	Contractor will be responsible for calls from 1500 to 0700 during week days and 24/7 during the weekends, 365 days per year. This shall also include government holidays, early releases or site shutdowns due to inclement weather when federal personnel are off site. The government is responsible for calls from 0700-1500 each normal business day. The contractor will be responsible to provide 4 hours of support per call. Support shall be defined as general support for any issue on the NIST site such as water leaks, broken equipment, check and verification of temporary equipment etc. If parts are readily available or can be provided by NIST immediately the contractor shall perform the repair within the 4 hours. The government is responsible for immediate response from 0700-1500 each normal business day.	The Contractor's response time for the NIST Boulder Site during other-than-normal hours of operation (1500 to 0700) and holidays/early dismissals/site closures will be no more than 15 minutes from the time of receipt of notification by the COR or designee.	95%
3.1.11	Operation and Maintenance of CUP	Normal Hours - Metasys alarms monitoring, and notification 24/7/365	Contractor shall be responsible to monitor, review, acknowledge and communicate any alarms received from 0700 to 1500 each day. Communication of alarm shall be to Metasys Administrator, their backups or the COR. No physical response by the contractor for the alarms is required from 0700 to 1500.	Alarms shall be communicated to Metasys Administrator or COR within 15 minutes.	95%
3.1.12	Operation and Maintenance of CUP	After Hours - Metasys alarms monitoring and response 24/7/365	Contractor shall be responsible to monitor, review, acknowledge, and physically respond to any alarms received from 1500 to 0700 each day.	Alarms shall be responded to within 15 minutes.	95%
3.1.13	Operation and Maintenance of CUP	Metasys Alarm Reporting	Contractor shall document all alarms and associated responses on the Daily Report (PWS Attachment 3).	Alarms shall be logged on the Daily Report and provided electronically to the Government daily.	95%
3.1.14	Operational Requirements	The Contractor shall provide all labor, supplies, repair parts, tools, and equipment required to provide preventative maintenance services on all equipment within the CUP.	See PWS Attachment 1, Mechanical Room and Preventative Maintenance (MR_PM) List and PWS Attachment 4, Job Plans pertaining to the CUP equipment. This is for the equipment in the CUP and does not include building infrastructure such as the walls, electrical and lighting. On the equipment list there are CUP Carrier brand Chillers and Ingersoll Rand air compressors and air dryers. These units are categorized as specialized and critical equipment. Therefore NIST is requiring all maintenance performed on these units be performed only by factory certified Carrier and Ingersoll Rand technicians. All PMs are issued in the NIST Maximo system within the first 3 business days of the month and are required to be closed out by the last business day of the same month.	Contractor shall complete preventative maintenance in accordance with PWS Attachment 5, Job Plans. Contractor shall utilize the NIST Maximo system to record completion of the PMs. PMs must be completed within the same month of being issued by the government. Any documentation from the factory certified technician shall be provided to COR within 5 business days of completion of the PM. Documentation of the service technician certification (if required) to be utilized for the critical equipment PMs shall be provided to the COR prior to work beginnings.	95%
3.1.15	Operational Requirements (Annual Repairs)	Annual steam boiler inspection	Steam boilers must be opened up annually for inspection by a certified third party; the Contractor shall open up all steam boilers annually and assist the inspector's as needed during the inspection process. Contractor shall coordinate date of inspection with COR.	Inspection shall occur between July 1 and August 31 of each year for the 3 steam boilers. A written report and any other supporting documentation from the inspector documenting this requirement must be given to the COR within 7 days of inspection.	100%
3.1.16	Operational Requirements (Annual Repairs)	Steam boilers shall be tuned annually, immediately after the steam boiler inspections.	Contractor is responsible to perform the tuning.	Boiler shall be tuned per the manufacturer's operation and maintenance manual. A written report of the boiler tuning and any other supporting documentation must be given to the COR within 7 days of tuning.	95%
3.1.17	Operational Requirements	Boiler blowdowns	Contractor shall perform boiler (Boilers 17, 18, 19) blowdowns per the results of the water treatment testing on a daily basis.	Boiler shall meet chemical manufacturer's control range for total dissolved solids. A daily written log shall be submitted on boiler maintenance with the monthly deliverables to the COR.	95%

Spec Item	Title	Performance Objective	Related Information	Performance Standard	Acceptable Quality Level
3.1.18	Operational Requirements	Contractor shall provide chemical testing on the boilers, steam, chilled water, condensate and condenser water systems.	Balanced chemistry in the water in the cooling towers and boilers at the CUP must be maintained within predetermined control ranges to maximize equipment service life, control scale and oxygen in the boilers, and control corrosion and deposits in condensate system and algaecide and bacteria in the cooling towers. Contractor must comply with NIST wastewater discharge and release reporting plan.	Contractor shall maintain chemical levels that are set by chemical supplier for industry standard protection in all of our systems. The Contractor shall maintain a daily test log that will be available to the COR upon request. Contractor shall provide a chemical treatment plan to the COR within 30 days of award outlining types of chemicals being used in all systems and the control parameters that shall be maintained. Chemical Testing results shall be provided to the Government by the 5th business day of each month as part of the Monthly Report.	95%
3.1.19	Operational Requirements	Chemical Injection pump operation	The contractor will be responsible for utilizing the government supplied re-build kits to repair chemical injection pumps and to rebuild pumps, replace pumps and replace tubing as necessary.	The contractor shall use the spare parts to provide the repairs as needed.	95%
3.1.20	Operational Requirements	Spare Parts List Management	See PWS Attachment 7, Spare Parts Inventory which provides a list and quantity of the spare parts the Government will provide to support this contract. The Government will replenish only these parts as noted on the monthly inventory.	Minimum quantity levels are maintained to support normal operations. As part of the monthly deliverable to the COR the contractor shall provide a copy of the spare parts list indicating if any items are needing to be ordered.	95%
3.1.21	Operational Requirements	Building 42 Meter Readings	Readings shall be taken daily for the meters on the boilers, air compressors, chillers, cooling towers, makeup water, main water meter and the condenser water blowdown.	Contractor shall perform meter readings during the first 2 hours of each shift for all meters within Building 42. A report showing each daily meter reading shall be provided electronically to the Government by the 5th business day of each month as part of the Monthly	80%
3.1.22	Operational Requirements	Steam Trap Maintenance and Testing	Contractor shall be required to perform all steam trap maintenance in Building 42 (CUP) and main steam tunnel attached to Building 42. There are 11 steam traps within Building 42 and 16 located in the tunnel. Contractor shall be required to test all steam traps annually (the dates will be identified in the monthly PM list)	The contractor shall perform maintenance per steam trap specifications. Contractor shall provide report to COR on trap testing and maintenance. COR shall perform walk of completed work for verification of steam trap operation.	95%
3.1.23	Operational Requirements	Steam Trap Repairs	Contractor shall be required to re-build traps upon any failures found in testing. All parts and materials shall be provided by the government to perform repairs within 30 days of notification by contractor.	Contractor shall perform repair within 2 weeks after the parts are received to meet manufacturer's specifications of steam traps. Repairs shall be documented and shall be provided electronically to the Government by the 5th business day of each month as part of the Monthly Report.	95%
3.1.24	Operational Requirements	All areas shall be kept neat and clean in Building 42.	Custodial services are provided by the Government in all restrooms, offices and common areas in the CUP (B42). The Contractor shall keep all areas neat and clean throughout Building 42.	The space shall be free of debris and kept free of hazards.	90%
3.1.25	Operational Requirements	Building 42 Floor Washdown	Contractor shall wash down the boiler room and chiller room floors.	The floors will be washed down every 2 weeks.	90%

Spec Item	Title	Performance Objective	Related Information	Performance Standard	Acceptable Quality Level
3.1.26	Operational Requirements	When refrigerants are transferred into or out of NIST equipment leak rates will be recorded and monitored by the Contractor.	All workers handling refrigerants on the NIST site must be certified as at least Class III Technicians as required by 40 CFR part 82, subpart F	Regardless of the size of the refrigerant leak or release, the Contractor shall notify the COR, in writing (email is acceptable) within 8 hours of a refrigerant leak being detected or a release occurring. A detailed/written refrigerant transfer inventory shall be provided to the COR within ten (10) working days of the transfer of refrigerant. A follow up detailed report of the leak or release shall be provided to the COR within ten (10) working days. The Contractor shall provide copies of these leak rate records to COR with the monthly reports.	100%
3.1.27	Operational Requirements	The Contractor shall operate Central Utility Plant and perform daily operation, minor repairs and maintenance on boilers, chillers and related equipment in the CUP and tunnel.	The Minor Repair services will provide for cleaning, calibration, repair and replacement of small components to keep the system operational. If the repair service cannot be reasonably accomplished without compromising the integrity of the regular operation, the COR shall be notified to determine if these services can be performed by using the CUP personnel or by a subcontractor to the contractor to perform this repair by using the repair line item included in this contract with the CO or the COR's approval.	All equipment shall meet manufacturer's standard performance requirements. The contractor shall document minor repairs in Daily Report and provide additional information, where appropriate, electronically to the Government by the 5th business day of each month as part of the Monthly Report.	85%
3.1.28	Operational Requirements (Minor Repairs)	The Contractor shall perform Minor Repairs consisting of tasks which are performed on an "as needed" basis. May require minor disassembly and replacement lasting less than four (4) man-hours per item at no additional cost to the government.	Included are: (1) Measurements (2) Adjustments (3) Replacement of routine expendable parts (4) Replacement of defective controls (other than DDC controls), switches and indicator lamps (5) Lubrication of moving parts, on a scheduled basis.	All equipment shall meet manufacturer's standard performance requirements. The contractor shall document minor repairs in Daily Report and provide additional information, where appropriate, electronically to the Government by the 5th business day of each month as part of the Monthly Report.	85%
3.1.29	Operational Requirements (Major Repairs)	Typical Major Repair items are generally identified as those services requiring skill levels not normally associated with Boiler Operators and those requiring over four (4) man-hours of labor. Any major repairs or modifications to equipment and systems are considered beyond the scope of the preventative maintenance specified in this contract and is covered under a separate contract line item for repairs.	The COR will determine what if any corrective measures are needed. In each case, any additional work to be performed is considered a separate task, and will have a defined scope of work that is separate and apart from the preventative maintenance part of this contract. The government reserves the right to openly compete and award such tasks to contractors who are best suitable for that work. If the Government determines repairs are to be completed under this contract the contractor shall prepare an itemized quote that must be authorized by the CO. After authorization is given by the CO the contractor shall complete the repairs and notify the COR of completion. COR shall conduct an inspection of repairs prior to any payments to the Contractor. Examples - This list is not all-inclusive: (a) Boiler shell, welding and repair (b) Boiler refractory (c) Boiler tube replacement (d) Boiler burner replacement (e) Entire boiler replacement or overhaul (f) Valve replacement (over 2") (g) Pipe repair (h) Direct Digital controls, hardware/software (i) Pump coupling alignment (j) Motor replacement (k) Electrical wire replacement and/or panel repair due to "shorted" circuitry (l) Electrical repairs to magnetic starters and/or to disconnect panels (m) Replacement of pump: 1)seals, 2)bearings, 3) shafts, 4) impellers (n) Vibration analysis	If the need for such work is identified during the course of preventive maintenance, then the COR shall be notified within 8 hours.	
3.1.30	Operational Requirements (Major Repairs)	If a major repair is authorized, contractor shall ensure equipment is operational.		Equipment shall meet manufacturer's performance standards.	100%

Spec Item	Title	Performance Objective	Related Information	Performance Standard	Acceptable Quality Level
3.2.1	Materials and Equipment supplied by the Contractor (Chemicals)	Contractor shall provide all chemicals, chemical testing equipment/reagents for boilers, chilled water system and condenser water system at building 42 (CUP).	All equipment is listed in PWS Attachment 1 – Mechanical Room and Preventative Maintenance (MR_PM) List. This is intended to support normal operations of the central plant utilities provided to the site. This does not include situations where repairs are needed due to unforeseen circumstances.	Contractor shall perform maintenance per manufacturer's recommendations. A monthly report of the chemical test results shall be provided electronically to the Government by the 5th business day of each month as part of the Monthly Report.	95%
3.2.3	Materials and Equipment supplied by the Contractor	The Contractor shall supply all hand tools or specialized equipment, materials and expendables required for the performance of the preventative maintenance portion of this contract unless otherwise specified in this document.	Expendable items may include lube, shop towels, rags etc. The supplies or materials shall be per the manufacturers recommendations to ensure maximum efficient operation.	Contractor shall perform the uninterrupted preventative maintenance without government furnished tools or supplies.	95%
3.2.4	Materials and Equipment supplied by the Contractor (Chemicals)	Contractor must provide chemical totes for condenser water system in the basement of CUP building 42. Contractor shall also provide a chemical condenser water controller for Building 42 condenser water system.	This controller shall be installed and maintained by the contractor. Contractor shall also provide chemical mixing containers if not using direct chemical injection that is usually supplied by chemical manufacturer. Contractor shall also provide a way to pump chemicals to basement into totes that are provided by contractor. There is existing piping that the chemicals are currently pumped through into the basement totes. This piping will be available but the totes will not be available as they belong to the current chemical supplier Nalco. Please note, there is no elevator in the Central Plant to move chemicals with. As a point of reference: Nalco chemical products are currently in use.	Chemicals shall be provided and supplied as required per chemical manufacturer's standard recommendations. A monthly report of the chemical test results shall be provided electronically to the Government by the 5th business day of each month as part of the Monthly Report.	95%
3.2.5	Materials and Equipment supplied by the Contractor (Deliveries)	Product and materials procured by the Contractor for performance of this contract shall not be shipped directly to NIST.		Any delivery in support of this contract shall be delivered to NIST and will be subject to the NIST shipping/receiving process on site. All deliveries shall have the contractors point of contact, phone number and location of building 42 to allow for arrangement of delivery or pickup. If a large delivery truck is required, site access would need to be coordinated for a daily paper badge. See PWS Attachment 8, General Instructions paragraph 3.3G for notification requirements.	95%
3.3.1	Furnished By the Government	The Government will make available equipment for use by the contractor. See Solicitation/Contract Attachment J.1, Government Furnished Equipment List.	Any damage to this equipment beyond normal wear and tear shall be the responsibility of the Contractor. Contractor shall be responsible for performing and documenting daily inspections of government furnished equipment by a qualified person before subject equipment is used. In addition, a monthly inspection shall also be performed. The government will provide the manufacturer's inspection checklist for contractor use. Qualified person is defined as "by possession of a recognized degree, certificate, or professional standing, or who by extensive knowledge, training and experience, has successfully demonstrated his ability to solve or resolve problems relating to the subject matter, the work, or the project." Additionally, the qualified person shall be responsible for plugging in all equipment to ensure batteries are properly charged. Moreover, Contractor shall also be responsible for checking battery fluid level and adding distilled water as needed. The government shall provide distilled water for the batteries. If discrepancies are found during the daily inspection, Contractor shall immediately report all issues to the COR so repairs can be made. The government shall make every effort, subject to availability of funds, to keep this equipment operational and shall perform an annual certification inspection. It shall be the responsibility of the Contractor to provide and document equipment specific training to employees who will be utilizing government furnished equipment. If the Contractor is unable or unwilling to train their employees on the government furnished equipment, the Contractor shall be responsible for providing their own equipment that will comply with all functions of this contract.	The Contractor shall ensure that this equipment is used only for official duties required in the performance of this contract. No unauthorized users of the equipment shall be allowed.	100%

Spec Item	Title	Performance Objective	Related Information	Performance Standard	Acceptable Quality Level
3.3.2	Furnished By the Government	The Contractor shall use Government supplied steam traps and re-build kits	N/A	Contractor shall utilize new steam traps and re-build kits to repair or replace steam traps within 24 hours of parts being received. After re-build steam traps shall perform at normal operating temperatures.	100%
3.3.3	Furnished By the Government	The Contractor shall use Government supplied chemical injection pumps for the condenser water system, chilled water system and boiler system.	The Government will provide replacement pumps, rebuild kits and all supply tubing and injection quills. The exception of the building 42 condenser water controller and chemical mixing containers this item shall be furnished, installed and maintained by the contractor. The contractor will be responsible for utilizing the government supplied re-build kits to repair chemical injection pumps and to rebuild pumps, replace pumps and replace tubing as necessary.	Contractor shall repair or replace new chemical injection pumps within 24 hours of parts being received.	95%
3.3.4	Furnished By the Government	The Government shall supply all chemical containment pallets for chemical storage and flammable lockers for flammable materials.	The Contractor shall store all boiler chemicals and other related chemicals in appropriately sized containment, and flammable materials in a certified and labeled flammable locker. Contractor shall meet: OSHA 1910.106 https://www.osha.gov/pls/oshaweb/owadisp.show_document?p_id=9752&p_table=standards and shall meet: NFPA 30 Flammable and Combustible Liquids Code: https://www.nfpa.org/codes-and-standards/all-codes-and-standards/list-of-codes-and-standards/detail?code=30	Contractor shall utilize appropriate chemical containment pallets for the chemicals being stored.	100%
4	Preventative Maintenance	NIST has a requirement for site PM's on all equipment listed in PWS Attachment 1, Mechanical Room and Preventative Maintenance (MR_PM) List. The Contractor shall maintain all equipment according to PWS Attachment 5, Job Plans.	The equipment that shall require maintenance is listed in PWS Attachment 1 – Mechanical Room and Preventative Maintenance (MR_PM) List For any equipment that a job plan is not included in PWS Attachment 4, Job Plans; the contractor shall perform maintenance in accordance with the manufacturers recommendations. Materials provided by the Contractor include (but are not limited to) manufacturer recommended belts, air filters, gaskets, chemicals, glycol, corrosion inhibitor, and lubrication. The Contractor shall notify and coordinate with the COR 14 days in advance for any preventative maintenance. The COR will inform building occupants prior to any major shutdown.	Contractor shall perform scheduled maintenance to ensure equipment maintenance meets performance standards as detailed in this PWS and attachments. Contractor shall provide to the COR a checklist for each piece of equipment that has preventive maintenance performed on it. The checklist shall cover all work performed on the unit to include any readings taken, airflow checks, deficiencies found. This checklist shall be provided electronically to the Government by the 5th business day of each month as part of the Monthly Report.	85%
4.1	Preventative Maintenance	The Contractor shall provide for the Government detailed Standard Operating Procedure (SOP) in order to accomplish preventative maintenance as it is assigned in the NIST Maximo system.	The Contractor shall review the existing information the government has in its computerized Maintenance Management System (MAXIMO). After a detailed review the contractor shall notify COR of any discrepancies. All PM's shall correspond to PWS Attachment 1. The Contractor shall verify the Information provided in PWS Attachment 1-~~4~~to include current location/identification of each piece of equipment described. Contractor shall notify the COR of any discrepancies. The Contractor shall notify the COR of any changes so current information can be maintained in the Maximo system.	The Contractor shall provide a detailed Standard Operating Procedure (SOP) within 45 days of contract being awarded	100%
4.1.1	Preventative Maintenance	Preventative Maintenance performed in laboratories must be performed on schedule.	No unscheduled preventative maintenance will be allowed. The coordination of activities between the COR, the contractor and the site building occupants is critical in an effort to minimize disruption of the research and damage to ongoing experiments. Many research laboratories operate and have equipment running 24/7/365. Any disruption may cause severe loss of valuable information.	The COR will require 14 days notice prior to, in order to provide notice to customers who will be affected.	95%

Spec Item	Title	Performance Objective	Related Information	Performance Standard	Acceptable Quality Level
4.1.2	Preventative Maintenance	Preventative Maintenance performed in accordance with approved procedures	All PM activities shall be recorded utilizing word or excel checklists for each piece of equipment that is a PM is performed on for all 3 NIST sites. Checklist shall detail the individual job plans, all readings taken during the PM (temperatures, pressures etc.) and all parts replaced, filter conditions and oil levels for the specific piece of equipment being worked on. Contractor shall have access to our MAXIMO system to track and pull all PM's and to complete PM's in MAXIMO. Equipment inventories shall be cross checked by the COR at least annually to ensure equipment has not been added to or deleted from the inventory. In the event that equipment changes have occurred, the PM shall be performed on the like-for-like new equipment at no additional cost. If additional non-like for the equipment is added or if equipment is taken out of service the CO will request a price reduction or negotiate a price increase with the Contractor. COR will approve any additions or replacement of non-like or like equipment. In general, there is one month to complete the PM's. They will be issued on the first business day of the month and shall be physically completed as well as completed within the Maximo system by the last business day of the month.	Preventive Maintenance tracking and record keeping shall be done using contractor checklists and the NIST Maximo System. Contractor shall utilize the NIST Maximo system to record completion of the PMs by the last day of the month the PM is issued for the Kekaha, Ft. Collins, and Boulder sites. Contractor shall submit preventative maintenance checklists to COR with monthly COR report.	90%
4.1.3	Preventive Maintenance (Refrigeration Compressors)	Preventative maintenance on Refrigeration Compressors (chillers/cooling towers/condenser water and chilled water piping) on the NIST Boulder Site	Refrigeration Compressors (chillers/cooling towers/condenser water and chilled water piping) – The contractor shall maintain the equipment per PWS Attachment 5, Job Plan JP-C-CCH Balanced chemistry in the water of the cooling towers and chilled water piping must be maintained within predetermined control ranges to maximize equipment service life, and control growth of algae and bacteria in the cooling towers. Control ranges are determined per the chemical manufacturer's specifications. Materials that the Contractor shall provide include, but are not limited to, belts, air filters, chemicals, corrosion inhibitor for chilled water and condenser water systems, cooling tower chemicals and lubrication.	Preventive maintenance shall be performed per the manufacturer's guidelines. The Contractor shall supply the COR with a monthly report documenting the results for all tests performed (daily, weekly etc.) to ensure compliance with the predetermined control ranges. This report shall be provided to the electronically to the Government by the 5th business day of each month as part of the Monthly Report.	90
4.1.4	Preventative Maintenance Generators	Contractor shall perform Preventative Maintenance on the generators listed in PWS Attachment 1, Mechanical Room and Preventative Maintenance (MR_PM) List. PWS Attachment 11 - Emergency Generators provides descriptions of the generators to include model and serial number.	All generator PMs shall be completed in accordance with PWS Attachment 5, Job Plans EMG-A and EMG-M by a technician with a minimum of 5 years experience working on both natural gas and diesel generators.	The preventative maintenance shall be performed in accordance with the job plan to ensure that the generators are always in standby operation readiness. The Contractor shall provide the report documenting the results for all preventive maintenance electronically to the Government by the 5th business day of each month as part of the Monthly Report.	90%
4.1.5	Preventive Maintenance Chemical Testing	Contract shall perform annual Legionnaires Testing (Boulder CUP Only)	Contractor shall test all cooling towers on the NIST site for legionella bacteria. Testing will be issued via Maximo preventative maintenance every July	Contractor shall submit records of annual legionnaires testing of the NIST site within 30 days.	95%

Spec Item	Title	Performance Objective	Related Information	Performance Standard	Acceptable Quality Level
4.1.6	Preventive Maintenance (PAC Units)	Preventive Maintenance on the NIST Packaged Units (PAC) shall be performed per PWS Attachment 5, Job Plans.	Service and supplies for heating and cooling packaged units, split system package units, Liebert split system units, Liebert water cooled units, Data Aire computer room units and air handling units. The technical specifics of the work to be performed are detailed in the attached job plans. The Liebert cooling units are categorized as specialized and critical equipment. Therefore NIST is requiring all maintenance performed on these units be performed only by factory certified Liebert technicians. The Data Aire cooling units are categorized as specialized and critical equipment. Therefore NIST is requiring all maintenance performed on these units be performed only by factory certified Data Aire technicians.	Preventative maintenance shall be completed per PWS Attachment 5 or the manufacturer's recommendations if no Job Plan exists. The COR will require 14 days notice prior to work, in order to provide notice to customers who will be affected. Contractor shall utilize the NIST Maximo system to record completion of the PMs by the last day of the month the PM is issued for the Ft. Collins and Boulder sites. Any documentation from the factory certified technician shall be provided to COR within 5 days of the completion of the PM. Documentation of the service technician to be utilized for the critical equipment PMs shall be provided to the COR prior to the start of PM activities.	90%
4.1.7	Preventive Maintenance (PAC Units)	Preventative Maintenance Certification for Subcontractors shall be given to COR	Any work requiring a factory certified technician shall be documented and provided to the COR.	Contractor shall submit sub contractor certification documentation prior to start of PM activities.	100%
4.1.8	Preventive Maintenance (Boilers and associated pumps and equipment)	Preventative Maintenance on hot water boilers and associated pumps and equipment	Contractor shall utilize PWS Attachment 4, Job Plan JP-C-BLR and PWS Attachment 1 – Mechanical Room and Preventative Maintenance (MR_PM) List, for specific boilers. Contractor shall also re-assemble boilers and perform start up to enter equipment back in service after inspection.	Contractor shall perform all maintenance and safety checks per Job Plan JP-C-BLR to include water testing, supplying chemicals, preparing boilers for annual boiler inspections. Contractor shall notify COR 14 days in advance of scheduled annual inspection date and provide certification report to COR within 7 days of completion of annual inspection.	95%
4.1.9	Preventive Maintenance (Boilers and associated pumps and equipment)	Boiler inspection completed by certified technician	The Contractor shall assist the inspector's as needed during the inspection process.	Contractor shall hire a certified boiler inspector on an annual basis to inspect the four hot water boilers on the Boulder site for an annual inspection between July 1 and August 31 of each year. A written report and any other supporting documentation from the inspector documenting this requirement must be given to the COR within 7 days of the inspection.	95%
4.2	Preventive Maintenance (Ft. Collins)	Perform preventative maintenance services on all equipment included on PWS Attachment 1, Mechanical Room and Preventative Maintenance (MR_PM) List and PWS Attachment 5, Job Plans at the Ft. Collins site.	The Contractor shall provide all supplies, repair parts, tools, and equipment required to perform the work. There are no equipment operation activities at the Fort Collins site, only preventive maintenance and repairs. The Contractor shall plan, schedule, and coordinate all required preventative maintenance activities at the Ft Collins Site with the COR 14 days in advance.	Equipment shall be maintained in accordance with the job plans provided. Contractor shall utilize the NIST Maximo system to record completion of the PMs by the last day of the month the PM is issued for the Ft. Collins site. The COR will require 14 days notice prior to work, in order to provide notice to customers who will be affected.	90%
4.3	Preventive Maintenance (Kekaha)	Provide preventative maintenance services on all included PWS Attachment 1, Mechanical Room and Preventative Maintenance (MR_PM) List and PWS Attachment 5, Job Plans at the Kekaha Hawaii site.	The Contractor shall maintain adequate lightbulb inventory in accordance with the applicable Job Plan and PWS Attachment 12 - Lighting Summary WWVh Field Site Kekaha, HI, and provide all supplies, repair parts, tools, and equipment required to perform the work. There are no equipment operation activities at the Hawaii site, only preventive maintenance and repairs. The Contractor shall plan, schedule, and coordinate all required preventative maintenance activities at the Kekaha Site with the	Equipment shall be maintained in accordance with the job plans provided. Contractor shall utilize the NIST Maximo system to record completion of the PMs by the last day of the month the PM is issued for the Kekaha site.	90%

Spec Item	Title	Performance Objective	Related Information	Performance Standard	Acceptable Quality Level
5	Reporting Requirements	Individual Monthly Reporting Requirements, PWS Attachment 2, Monthly Report	1. Boiler repair log to document any repairs (minor or major) made to the boilers on the NIST site. (See "Boiler Repairs" Tab) 2 Air compressor repair log to document any repairs (minor or major) made to the air compressors on the NIST site. (See " Air Comp Repairs" Tab) 3. Building water usage report (See "Building Water Usage" Tab) 4. DA Tank daily report (See "DA Tank Daily" Tab) 5. Building 42 Gas usage report for boilers and Building 42. (See "Gas Usage Report" Tab) 6. Boiler Daily Readings report (See "Boiler Daily Readings" Tab) 7. Man Hour report (monthly staff used to support this contract as well as any sub contractors used) (See "Man Hour" Tab") 8. Maintenance reports on the PMs completed at the Ft. Collins site and the Hawaii site. (See "Ft Collins & Hawaii" Tab) 9. Documented alarm response log for any out of hours calls responded to from the D.O.C. police or any alarms generated requiring a response. (See "After Hours Log" Tab) 10. Steam trap repairs/replacement report - provide report on any steam trap repairs or replacements completed. (See "Steam Trap" Tab) 11. Monthly system chemical reports showing a summary of all daily logs and status (any chemicals added, water testing of each system and any recommendations (See "Chemical Testing" Tab) - Chemical testing of B-17, 18 and 19 every 8 hours - B-42 closed loop chilled water system monthly - B-42 condenser water system tested every 8 hours 12. Chilled Water repair log to document any repairs (minor or major) made to the chilled water system (chillers, cooling towers, condenser water system, pumps etc.) on the NIST site. (See "Chilled Water" Tab)	The contractor shall submit the monthly report to the Government electronically by the 5th business day of each month as part of the Monthly Report. See PWS Attachment 3.	90%
5.1	Reporting Requirements	Additional reports provided on an as needed basis	There may be additional reports the government asks for during the performance of the contract in order to support Department of Commerce, City of Boulder or the State of Colorado requirements. These shall be part of the base contract and shall be at no extra charge to the government.	Reports shall be given to the COR within 10 days of request	90%
5.2	Reporting Requirements	Weekly Meetings with COR/TPOC and/or CO/CS	Weekly meetings may be conducted to discuss CUP operations and any issues needed. The conference will consist of a minimum of the COR or TPOC, Project Manager, and Stationary Engineer.	The contractor will record the meeting notes and email the CO, COR and TPOC within 48 hours of the end of the meeting.	90%
6	Major Repairs	The contractor shall make repairs necessary to keep all CUP equipment, the mechanical or electrical infrastructure, and for repairs to equipment that preventative maintenance is performed on, operational.	In addition to the FFP CLINs mentioned above, NIST further intends to provide one (1) Time and Material (T&M) CLIN (X004) to support all 3 sites repair needs. Funds in this line item are available ONLY for approved repairs, replacements, and/or rental of equipment to provide 24/7/365 operations of all systems associated with the CUP, all steam systems including piping, traps, expansion joints, condensate systems and all other misc. components of a steam system, all equipment at all three NIST sites that is maintained by the contractor shall be repaired under the Repair CLINs also. The Contracting Officer (CO) is the only individual authorized to approve individual work orders under these line items and approval must be obtained prior to performance of any work in association with these line items. Each requested repair will be issued a work order number at the time pricing is requested. Funds in this repair fund are not guaranteed to the contractor until a quote is accepted by the CO and the work is authorized. Contractor shall report to COR any issues within the scope of this major repair line item. COR shall determine course of action for the issue discovered and shall inform contractor to submit work order. Work order shall be submitted to COR within 7 days unless date is prior coordinated with the COR due to sub-contractor availability.	All equipment shall meet manufacturer's standard performance requirements.	90%

Spec Item	Title	Performance Objective	Related Information	Performance Standard	Acceptable Quality Level
6.1	Major Repair Line Item	The contractor shall provide itemized pricing to NIST for any repairs above the minor repairs listed.	The contractor shall submit a quote, along with a detailed description of the work to be completed. The quote will then be evaluated for price reasonableness. The CO will approve the work based on the quote and concurrence from the COR. The contractor exceeds the amount approved by the CO at their own risk. The quote will include labor (at the hourly rate established in the contract) and material and equipment (at cost, plus direct handling fees that may be charged) to perform authorized repair. The government will not pay any markup, overhead or profit, on subcontractor invoices. FAR 52.212-4, Alternate I, applies to this line item. **The funds on this repair fund are not guaranteed to the contractor.**	Work is performed only after approval from the CO.	100%
6.2	Major Repair Line Item	The contractor shall provide NIST with fair and reasonable pricing.	The contractor shall submit quotes for work performed under this line item when the need for repairs is identified. When a work order requires subcontractor involvement, the subcontractor quote shall be included with the quote to NIST. The contractor shall make an effort to receive at least two subcontractor quotes for subcontractor work, including material and equipment orders, over $10,000. If more than one quote is not possible, the contractor shall explain the effort made to get more than one quote.	More than one subcontractor quote is requested when the work is more than $10,000.	90%

Deliverables

Reference	Title	Performance Objective	Deliverable	Due Date
General Instructions 1.6 E	Employee Identification	Hours of Operation	Employee identification shall include the name, country of citizenship, and employer for each Contractor and Subcontractor's employee working on the DOC site.	These shall be submitted to the COR for review 3 days prior to the work date.
General Instructions 1.8 B	Sanitary Facilities	Toilet rooms in existing buildings may be used by the Contractor when performing work inside DOC owned buildings. When work is being conducted outside the Contractor shall provide the COR with keys to all temporary structures.	Provide the COR with keys to all temporary structures.	When work is being conducted outside
General Instructions 1.9.A.1	Equipment and Material Storage	Prior to the start of work, Contractor shall meet with all Subcontractors to arrange and prepare a plot plan defining staging, storage, field office and traffic areas within lay-down/staging and trailer/parking areas designated.	Plot Plan which defines staging, storage, field office and traffic areas within lay-down/staging and trailer/parking areas designated.	Not less than 3 days prior to the installation of any temporary facilities.
General Instructions 1.10.6	Electricity and Water	Temporary electrical power derived from existing site distribution system will be paid for by the Government. Any other power generation shall be paid for by Contractor.	Contractor to submit after hours lighting plan to COR for approval when required	14 days prior to work
General Instructions 2.1.A.2	Occupational Health and Safety	The Contractor is required to implement and manage a job specific Safety, Health and Environmental plan (Site Specific Safety Plan). NIST will review and evaluate the Contractor’s Site Specific Safety Plan to assure that it is site specific for the work being performed. The NIST Safety Plan Requirements Checklist, PWS Attachment 9 and PWS Attachement 10, Primer on Safety Plan Requirements/Addendums for COVID-19, are provided to assist the contractor in developing their Site Specific Safety Plan.	A Site Specific Safety Plan is to be developed and submitted to the government for review and acceptance	14 days after contract award. No site work may be started until the Site Specific Safety Plan is accepted.
General Instructions 2.5.A	Safety Inspections	If the Contractor is in non-compliance with safety rules and regulations the COR/COSR will notify the Contractor’s Project Manager or Safety Point of Contact verbally and in writing for immediate corrective action.	Immediate corrective action is required and a written report of corrective action(s) taken is required to be returned to the COR/COSR.	Immediate corrective action is required and a written report of corrective action(s) is required within 10 days of occurrence.
General Instructions 2.6.A	Incidents on the Job	The Contractor is responsible for keeping the Government informed of all incidents during the execution of the project.	If an on-the-job incident occurs during the execution of a project at the NIST Boulder site the Contractor shall immediately notify the COR by phone and will submit an incident report to the CO and COR	Notification-Immediately Incident Report - no later than 2 days after the incident occurred.
General Instructions 2.6.B	Incidents on the Job	The Contractor is responsible for keeping the Government informed of all incidents during the execution of the project.	A written report summarizing incidents for the previous month will be sent to NIST via email	This report shall be provided electronically to the Government by the 5th business day of each month as part of the Monthly Report.
General Instructions 2.8.A	Proper Waste Disposal	The Contractor will remove all materials that are not an integral part of the project, including paint, chemicals and debris. Materials will not be disposed of by flushing them into a drainage system.	When hazardous materials from the performance of this contract are disposed of off-site, submit copies of permits from applicable Federal, State, or Municipal authorities, manifests and/or certificates-of-destruction to demonstrate that the material has been disposed as per all applicable regulations.	When hazardous materials from the performance of this contract are disposed of off-site
General Instructions 2.11.B	Fire Prevention/Protection	The Contractor shall obtain a "Hot Work Permit" from the COSR for use of open flame devices, such as blow torches, portable furnaces, tar kettles or gas and electric welding and cutting equipment.	Upon completion of Hot Work, the contractor will return the “yellow” copy to the COSR or COR.	Hot Work permits should be requested at a minimum 2 days prior to the start of work. Upon completion of Hot Work, the contractor will return the “yellow” copy to the COSR or COR.
General Instructions 3.2.A	Contractor’s Security Point of Contact	The Contractor/subcontractor will be responsible for all security aspects of work performed under this contract. Security is an individual responsibility.	The Contractor will provide to the CO and COR the name, address and telephone number of a Security Point of Contact (POC) for receipt and/or notification of security related matters	Within 14 days of contract award and Within 7 days of change of Security POC
General Instructions 3.3.C	Personal Security & Employee Identification	The Contractor will fill out and require each of the Contractor’s/subcontractor’s employees performing contract work to fill out the required paperwork and forms for submission to the Government.	The contractor shall provide to the COR the names of individuals so those contractor employees needing access can be scheduled for the security training and review of their paperwork.	All paperwork must be filled out prior to the scheduled time for submittal of the forms.
General Instructions 3.3.F	Personal Security & Employee Identification	ALL individuals’ names needing access for out of normal business hours	MUST be provided to the COR in writing	2working days prior to the scheduled work.
General Instructions 3.3.G	Personal Security & Employee Identification	The COR MUST have in writing twenty four hours’ notice of deliveries by vendors for this contract.	The company making the delivery, the driver’s name, and the location the delivery is to be made MUST be provided.	24 hours prior to any delivery

Reference	Title	Performance Objective	Deliverable	Due Date
General Instructions 4.1.B	General Description	Many research laboratories operate and have equipment running 24/7. Any disruption may cause severe loss of valuable information.	Prior to any disconnect of power and or shut down of a system/system component, the generation of undue vibrations or noise levels the COR must be notified	a minimum of 14 days prior
General Instructions 4.1.D	General Description	Entrance into laboratories is normally not required. Arrangements must be coordinated with the owner/occupant to ensure work to be performed will not disrupt or damage technical experiments/projects in the specified area will be coordinated.	If entrance into the area is required the contractor shall contact the COR or his/her designee	minimum of 7 days prior
General Instructions 5.1.C.1	Permit Required Operations During Construction or Maintenance Activities	In an effort to ensure safety as a priority, to ensure individuals performing certain operations have the appropriate training and equipment, and to prevent unnecessary utility outages DOC requires that a Roof Access Permit be obtained.	Roofs Access Permit: A roof access permit will only be required if the contractor needs to go outside of the previously established warning lines.	A minimum of 7 days notice to the COR will be required to obtain permit.
General Instructions 5.1.C.2	Permit Required Operations During Construction or Maintenance Activities	In an effort to ensure safety as a priority, to ensure individuals performing certain operations have the appropriate training and equipment, and to prevent unnecessary utility outages DOC requires that a Permit to Penetrate Walls, Floors, Ceilings and Roofs be obtained.	Permit to Penetrate walls, floors, ceilings, and roofs (COSR provides upon request).	Request must be submitted to COSR a minimum of 7 days prior to the work.
General Instructions 5.1.C.3	Permit Required Operations During Construction or Maintenance Activities	In an effort to ensure safety as a priority, to ensure individuals performing certain operations have the appropriate training and equipment, and to prevent unnecessary utility outages DOC requires that a Hot Work Permit be obtained	Hot Work Permit (COSR provides upon request)	Request must be submitted to COSR a minimum of 7 days prior to the work.
General Instructions 5.1.C.4	Permit Required Operations During Construction or Maintenance Activities	In an effort to ensure safety as a priority, to ensure individuals performing certain operations have the appropriate training and equipment, and to prevent unnecessary utility outages DOC requires that a Fire Protection System Impairment Permit be obtained.	Fire Protection System Impairment Permit (Obtain from fireimp-mail@nist.gov)	Request must be submitted to COSR a minimum of 7 days prior to the work.
General Instructions 5.1.C.5	Permit Required Operations During Construction or Maintenance Activities	In an effort to ensure safety as a priority, to ensure individuals performing certain operations have the appropriate training and equipment, and to prevent unnecessary utility outages DOC has the following permits which are required when performing tasks/functions to support this contract/project.	Confined Spaces Permit (COSR provides upon request)	Request must be submitted to COSR a minimum of 7 days prior to the work.
General Instructions 5.1.C.6	Permit Required Operations During Construction or Maintenance Activities	In an effort to ensure safety as a priority, to ensure individuals performing certain operations have the appropriate training and equipment, and to prevent unnecessary utility outages DOC requires that a Confined Space Permit be obtained.	Excavation Permit (COSR provides upon request)	Request must be submitted to COSR a minimum of 7 days prior to the work.
General Instructions 5.1.G	Permit Required Operations During Construction or Maintenance Activities	All organizations performing work on behalf of the DOC that requires Fall Protection shall provide a fall protection plan and a fall rescue plan	All employees affected by the fall protection plan must have attended training IAW 29 CFR 1926.503(b) and provide copies of training certificates upon request. Contractor shall ensure that a licensed PE has designed all fall protection systems.	For roof work, contractor must obtain a Roof Permit from COSR.

Technical				
Technical 2.1.1	Key Personnel (Project Manager)	The Contractor shall provide a Project Manager who will be responsible for the performance of the work.	The Project Manager and alternate will have full authority to act for the Contractor on all contract matters relating to the daily operation of this contract. Must have experience managing other projects of similar size and scope. May act as security point of contact responsible for all security aspects of the contract.	The name and contact information of this person, and alternate(s) who will act for the Contractor when the manager is absent, will be designated in writing to the CO and COR no later than 10 days after contract award. These individuals serve as an emergency contact and be available 24/7/365.
Technical 2.1.2	Key Personnel (Alternate Project Manager(s))	The contractor shall provide at a minimum 1 alternate project manager.	The Project Manager and alternate will have full authority to act for the Contractor on all contract matters relating to the daily operation of this contract. May act as security point of contact responsible for all security aspects of the contract.	The name and contact information of this person, and alternates who will act for the Contractor when the manager is absent, will be designated in writing to the contracting officer and COR within 10 days after award. These people will serve as an emergency contact and be available 24/7/365.
Technical 2.1.3	Key Personnel (Project Engineer)	The Contractor shall provide a dedicated Project Engineer who is a stationary engineer experienced in the operation of large mechanical systems such as boilers, chillers, air compressors and central plant operations..	It shall be the duty of this person to be on the NIST Boulder site and coordinate the delivery of quality service, offer technical assistance, make suggestions for systems improvement and implement quality assurance by submitting the daily logs. See section 2.1.4 for qualifications	Daily logs shall be submitted by 9 a.m. to the COR via email utilizing the format at PWS Attachment 6
Technical 2.1.4	Key Personnel (Stationary Engineers)	Stationary Engineers shall have or be able to obtain a valid Stationary Engineers Certificate from the City and County of Denver within 180 days of contract award. The stationary engineer will be responsible for utilizing and monitoring the NIST Direct Digital Control Building Management System.	The stationary engineer will be responsible for utilizing and monitoring the NIST Direct Digital Control Building Management System. CUP personnel (Stationary engineer) will have to be competent to operate this system and diagnose problems of specified equipment using it. Stationary Engineers must have four-years of verifiable work in stationary engineering work consisting of a minimum of three-years operating high-pressure steam boilers producing over 10 hp and steam pressure over 15 p.s.i. or water heating boilers when the water temperature exceeds 250 degrees F in the system, and a minimum of one-year operating refrigeration equipment requiring an operator. The digital control systems used by the Government is Johnson Controls/Metasys. The NIST System Administrator will provide an 8 hour orientation to the NIST Metasys Controls system upon award of the contract.	Contractor shall provide proof of Stationary Engineer Qualifications/Certification: 1) Within 30 days of contract award or after applicable employee starts work 2) Upon Receipt of City and County of Denver certification (no later than 180 days after applicable employee starts work). 3) Annually-not later than 30 days after receipt of any option modification.
Technical 2.1.5	Key Personnel - HVAC Mechanics	HVAC mechanics shall have a minimum of 5 years of experience and hold a Universal CFC License. All workers handling refrigerants on the NIST site must be certified as at least Class III Technicians as required by 40 CFR part 82, subpart F.	The Contractor shall ensure employees have current and valid professional certifications before starting work under this contract. The Contractor shall provide, to the NIST COR, a copy of the certification (CFC license) of each worker that handles refrigerants on the NIST Site before such work commences.	Proof of qualifications for HVAC Mechanics CFC License shall be provided as shown below: 1) No later than 30 days after contract award or after applicable employee starts work 2) Annually, no later than 30 days after receipt of any option modification.

Technical				
Technical 3.1	Operational	There are several pieces of specialized equipment in the CUP that require a factory trained technician to perform the required preventive maintenance.	Four (4) Carrier Chiller's: 1498 ton capacity each Four (4) Ingersoll Rand Air compressors 100 h.p. each. Four (4) Ingersoll Rand Desiccant air dryer systems.	Carrier Factory trained technicians for the chillers or Ingersoll Rand factory trained technicians for the air compressors/dryers to perform the monthly preventive maintenance. Submit to COR certification of factory trained technician 30 days prior to the preventative maintenance taking place.
Technical 3.1.1	Operational	Contractor shall adhere to a contractor developed quality control plan	The contractor shall provide to the NIST COR a contractor quality control plan to be adhered to through the duration of the contract.	Contractor shall submit required quality control plan to NIST COR within 30 days of contract award. Any changes to quality control plan shall be communicated to NIST 30 days prior to implementation.
Technical 3.1.2	Operational Requirements	The Central Utility Plant operates 24/7/365 with no unscheduled outages within the contractors control.	Many research laboratories operate and have equipment running 24/7/365. Any disruption of services provided by the CUP may cause severe loss of valuable information. The coordination of activities between the COR, the contractor and the site building occupants is critical in an effort to minimize disruption of the research and damage to ongoing experiments.	The COR will require 14 days notice prior to, in order to provide notice to customers who will be affected.
Technical 3.1.3	Operation and Maintenance of CUP	The COR shall always be notified in the event of an unplanned outage.	COR phone number will be given to contract at kickoff meeting. Notification shall consist of the problem, current actions being taken and anticipated time frame for equipment to be restored.	In the event of an unplanned outage the contractor shall notify the COR by phone within the hour of the outage.
Technical 3.1.8	Operational Requirements	The Contractor shall perform routine operating inspections for the equipment in the CUP.	Routine operating inspections shall consist of visual inspections, log readings, routine adjustments, calibration and an assessment of operational conditions to ensure proper operation at optimum efficiencies. Documentation shall be maintained in an Excel spreadsheet and made available to the COR as requested. Any operational discrepancies or mechanical malfunction will be immediately reported to COR.	The contractor shall provide daily readings, one per shift, in the monthly reports (electronic) submitted to COR.
Technical 3.1.9	Operational Requirements	The Contractor shall perform routine operating inspections for the NIST Boulder Site.	Once per 8 hour shift (does not include normal hours of operation, 0700 to 1530 Monday Through Friday) the HVAC mechanic shall perform walks through all mechanical rooms within the buildings on the NIST Boulder site. See PWS Attachment 1, Mechanical Room and Preventative Maintenance (MR_PM) List.	The contractor shall document rounds in daily log and provide to the COR in the monthly reports.
Technical 3.1.11	Operation and Maintenance of CUP	Normal Hours - Metasys alarms monitoring, and notification 24/7/365	Contractor shall be responsible to monitor, review, acknowledge and communicate any alarms received from 0700 to 1500 each day. Communication of alarm shall be to Metasys Administrator, their backups or the COR. No physical response by the contractor for the alarms is required from 0700 to 1500.	Alarms shall be communicated to Metasys Administrator or COR within 15 minutes.

Technical				
Technical 3.1.13	Operation and Maintenance of CUP	Metasys Alarm Reports	Contractor shall provide reports via electronic format. All alarms and associated responses shall be logged and given to the COR on a monthly basis.	Report shall be given electronically to the Government by the 5th business day of each month.
Technical 3.1.14	Operational Requirements	The Contractor shall provide all labor, supplies, repair parts, tools, and equipment required to provide preventative maintenance services on all equipment within the CUP.	See attached equipment list and job plans pertaining to the CUP equipment. This is for the equipment in the CUP and does not include building infrastructure such as the walls, electrical and lighting. On the equipment list there are CUP Carrier brand Chillers and Ingersoll Rand air compressors and air dryers. These units are categorized as specialized and critical equipment. Therefore NIST is requiring all maintenance performed on these units be performed only by factory certified Carrier and Ingersoll Rand technicians. All PMs are issued in the NIST Maximo system within the first 3 business days of the month and are required to be closed out by the last business day of the same month.	Contractor shall complete preventative maintenance in accordance with the job plans. Contractor shall utilize the NIST Maximo system to record completion of the PMs within the same month of being provided the PMs by the government. Any documentation from the factory certified technician shall be provided to COR after completion of the PM within 5 business days. Documentation of the service technician to be utilized for the critical equipment PMs shall be provided to the COR prior to work beginnings.
Technical 3.1.15	Operational Requirements (Annual Repairs)	Annual steam boiler inspection	Steam boilers must be opened up annually for inspection by a certified third party; the Contractor shall open up all steam boilers annually and assist the inspector's as needed during the inspection process. Contractor shall coordinate date of inspection with COR.	Inspection shall occur between July 1 and August 31 of each year for the 3 steam boilers. A written report and any other supporting documentation from the inspector documenting this requirement must be given to the COR within 7 days of inspection.
Technical 3.1.16	Operational Requirements (Annual Repairs)	Steam boilers shall be tuned annually, immediately after the steam boiler inspections.	Contractor is responsible to perform the tuning.	Boiler shall be tuned per the manufacturer's operation and maintenance manual. A written report of the boiler tuning and any other supporting documentation must be given to the COR within 7 days of tuning.
Technical 3.1.17	Operational Requirements	Boiler blowdowns	Contractor shall perform boiler (Boilers 17, 18, 19) blowdowns per the results of the water treatment testing on a daily basis.	Boiler shall meet chemical manufacturer's control range for total dissolved solids. A daily written log shall be submitted on boiler maintenance with the monthly deliverables to the COR.

Technical				
Technical 3.1.18	Operational Requirements	Contractor shall provide chemical testing on the boilers, steam, chilled water, condensate and condenser water systems.	Balanced chemistry in the water in the cooling towers and boilers at the CUP must be maintained within predetermined control ranges to maximize equipment service life, control scale and oxygen in the boilers, and control corrosion and deposits in condensate system and algaecide and bacteria in the cooling towers.	Contractor shall maintain chemical levels that are set by chemical supplier for industry standard protection in all of our systems. A monthly report of the chemical test results shall be supplied to the COR. The Contractor shall maintain a daily test log that will be available to the COR upon request. Contractor shall provide an electronic copy of this test log on a monthly basis to the COR. Contractor shall provide a chemical treatment plan to the COR within 30 days of award outlining types of chemicals being used in all systems and the control parameters that shall be maintained.
Technical 3.1.20	Operational Requirements	Spare Parts List Management	A list and quantity of the spare parts is provided in the Monthly Reports, Spare Parts Inventory. The Government shall provide replacements to maintain inventory. Cis responsible for tracking inventory and notifying the Government when replacements are required.	Minimum quantity levels are maintained to support normal operations. Spare parts Inventory shall be provided electronically to the Government by the 5th business day of each month as part of the Monthly Report.
Technical 3.1.21	Operational Requirements	Building 42 Meter Readings	Readings shall be taken daily for the meters on the boilers, air compressors, chillers, cooling towers, makeup water, main water meter and the condenser water blowdown.	Contractor shall perform meter readings during the first 2 hours of each shift for all meters within Building 42. The contractor shall provide the COR a spreadsheet showing each meter reading on a daily basis for the month. Spreadsheet shall be submitted to COR every 30 days in the monthly report.
Technical 3.1.22	Operational Requirements	Steam Trap Maintenance and Testing	Contractor shall be required to perform all steam trap maintenance in Building 42 (CUP) and main steam tunnel attached to Building 42. There are 11 steam traps within Building 42 and 16 located in the tunnel. Contractor shall be required to test all steam traps annually (the dates will be identified in the monthly PM list)	The contractor shall perform maintenance per steam trap specifications. Contractor shall provide report to COR on trap testing and maintenance. COR shall perform walk of completed work for verification of steam trap operation.

Technical				
Technical 3.1.26	Operational Requirements	When refrigerants are transferred into or out of NIST equipment leak rates will be recorded and monitored by the Contractor.	All workers handling refrigerants on the NIST site must be certified as at least Class III Technicians as required by 40 CFR part 82, subpart F	Regardless of the size of the refrigerant leak or release, the Contractor shall notify the COR, in writing (email is acceptable) within 8 hours of a refrigerant leak being detected or a release occurring. A follow up detailed report of the leak or release shall be provided to the COR within ten (10) working days. A detailed/written refrigerant transfer inventory shall be provided to the COR within ten (10) working days of the transfer of refrigerant. The Contractor shall provide copies of these leak rate records to COR with the monthly reports.
Technical 3.1.27	Operational Requirements	The Contractor shall operate Central Utility Plant and perform daily operation, minor repairs and maintenance on boilers, chillers and related equipment in the CUP and tunnel.	The Minor Repair services will provide for cleaning, calibration, repair and replacement of small components to keep the system operational. If the repair service cannot be reasonably accomplished without compromising the integrity of the regular operation, the COR shall be notified to determine if these services can be performed by using the CUP personnel or by a subcontractor to the contractor to perform this repair by using the repair line item included in this contract with the CO or the COR's approval.	All equipment shall meet manufacturer's standard performance requirements. The contractor shall document minor repairs in daily log and provide to the COR in the monthly reports.
Technical 3.1.28	Operational Requirements (Minor Repairs)	The Contractor shall perform Minor Repairs consisting of tasks which are performed on an "as needed" basis. May require minor disassembly and replacement lasting less than four (4) man-hours per item at no additional cost to the government.	Included are: (1) Measurements (2) Adjustments (3) Replacement of routine expendable parts (4) Replacement of defective controls (other than DDC controls), switches and indicator lamps (5) Lubrication of moving parts, on a scheduled basis.	All equipment shall meet manufacturer's standard performance requirements. The contractor shall document minor repairs in daily log and provide to the COR in the monthly reports.
Technical 3.2.1	Materials and Equipment supplied by the Contractor (Chemicals)	Contractor shall provide all chemicals, chemical testing equipment/reagents for boilers, chilled water system and condenser water system at building 42 (CUP).	All equipment is listed in PWS Attachment 1 – Mechanical Room and Preventative Maintenance (MR_PM) List. This is intended to support normal operations of the central plant utilities provided to the site. This does not include situations where repairs are needed due to unforeseen circumstances.	Contractor shall perform the preventative maintenance at the time stated in the SOP. A monthly report of the chemical test results shall be supplied to the COR.
Technical 3.2.2	Materials and Equipment supplied by the Contractor (Chemicals)	Contractor shall provide all chemicals, chemical testing equipment/reagents for the chilled water system and condenser water system at building 1 room 2058.	All equipment is listed in PWS Attachment 1 – Mechanical Room and Preventative Maintenance (MR_PM) List. Contractor shall also provide all chemical testing equipment and any reagents needed to properly test chemical levels.	Contractor shall perform the preventative maintenance at the time stated in the SOP. A monthly report of the chemical test results shall be supplied to the COR.

Technical

Technical 3.2.4	Materials and Equipment supplied by the Contractor (Chemicals)	Contractor must provide chemical totes for condenser water system in the basement of CUP building 42. Contractor shall also provide a chemical condenser water controller for Building 42 condenser water system.	This controller shall be installed and maintained by the contractor. Contractor shall also provide chemical mixing containers if not using direct chemical injection that is usually supplied by chemical manufacturer. Contractor shall also provide a way to pump chemicals to basement into totes that are provided by contractor. There is existing piping that the chemicals are currently pumped through into the basement totes. This piping will be available but the totes will not be available as they belong to the current chemical supplier Nalco. Please note, there is no elevator in the Central Plant to move chemicals with. As a point of reference: Nalco chemical products are a recommended supplier as they have historically been used.	Chemicals shall be provided and supplied as required per chemical manufacturer's standard recommendations. A monthly report of the chemical test results shall be supplied to the COR.
Technical 4	Preventative Maintenance	NIST has a requirement for site PM's at the NIST Site on all equipment listed on PWS Attachment 1 The Contractor shall maintain all equipment according to PWS Attachment 4, Job Plans.	The equipment that shall require maintenance is listed in PWS Attachment 1 – Mechanical Room and Preventative Maintenance (MR_PM) List. NIST Site, Ft. Collins and Hawaii tabs of the spreadsheet. For any equipment without an attached job plan the contractor shall refer to the manufacturers recommendations for maintenance. Materials provided by the Contractor include (but are not limited to) manufacturer recommended belts, air filters, gaskets, chemicals, glycol, corrosion inhibitor, and lubrication. The Contractor shall notify and coordinate with the COR 30 days in advance for any preventative maintenance. The COR will inform building occupants prior to any major shutdown.	Contractor shall provide to the COR a checklist for each piece of equipment that has preventive maintenance performed on it. The checklist shall cover all work performed on the unit to include any readings taken, airflow checks, deficiencies found. This checklist shall be given to the COR within 14 days after the last day of the month that the PM's were issued.

Technical

Technical 4.1	Preventative Maintenance	The Contractor shall provide for the Government detailed Standard Operating Procedure (SOP) in order to accomplish preventative maintenance as it is assigned in the NIST Maximo system.	The Contractor shall review the existing information the government has in its computerized Maintenance Management System (MAXIMO) and provide back to the government, after a detailed review, shall notify COR of any issues. In addition, the contactor shall provide upon the completion of the first PM tour of all equipment listed a list of spare parts required to be used for each PM frequency. The Government shall provide a standard spreadsheet for the contractor to utilize for this requirement. All PM's shall correspond to the manufacturers requirements and use as a guide the Means Maintenance guide. The Contractor shall verify the equipment list and current location/identification of each piece of equipment described in the attachments. The Contractor shall notify the COR of any changes so current information can be maintained in the Maximo system.	The Contractor shall provide for the Government detailed Standard Operating Procedure (SOP) within 45 days of contract being awarded
Technical 4.1.1	Preventative Maintenance	Preventative Maintenance performed in laboratories must be performed on schedule.	No unscheduled preventative maintenance will be allowed. The coordination of activities between the COR, the contractor and the site building occupants is critical in an effort to minimize disruption of the research and damage to ongoing experiments. Many research laboratories operate and have equipment running 24/7/365. Any disruption may cause severe loss of valuable information.	All preventative maintenance must be coordinated with the COR 30 days in advance using PWS Attachment 1 Mechanical Room and Preventative Maintenance (MR_PM) List. The COR will require 14 days notice prior to, in order to provide notice to customers who will be affected.

Technical				
Technical 4.1.2	Preventative Maintenance	Preventative Maintenance performed in accordance with approved procedures	All PM activities shall be recorded utilizing word or excel checklists for each piece of equipment that is a PM is performed on for all 3 NIST sites. Checklist shall detail the individual job plans, all readings taken during the PM (temperatures, pressures etc.) and all parts replaced, filter conditions and oil levels for the specific piece of equipment being worked on. Contractor shall have access to our MAXIMO system to track and pull all PM's and to complete PM's in MAXIMO. Equipment inventories shall be cross checked by the COR at least annually to ensure equipment has not been added to or deleted from the inventory. In the event that equipment changes have occurred, the PM shall be performed on the like-for-like new equipment at no additional cost. If additional non-like for the equipment is added or if equipment is taken out of service the CO will request a price reduction or negotiate a price increase with the Contractor. COR will approve any additions or replacement of non-like or like equipment. In general, there is one month to complete the PM's. They will be issued on the first business day of the month and shall be physically completed as well as completed within the Maximo system by the last business day of the month.	Contractor shall utilize the NIST Maximo system to record completion of the PMs by the last day of the month the PM is issued for the Kekaha, Ft. Collins, and Boulder sites. Contractor shall submit preventative maintenance checklists to COR with monthly COR report.
Technical 4.1.3	Preventive Maintenance (Refrigeration Compressors)	Preventative maintenance on Refrigeration Compressors (chillers/cooling towers/condenser water and chilled water piping) on the NIST Site	Refrigeration Compressors (chillers/cooling towers/condenser water and chilled water piping) – The contractor shall maintain the equipment per the attached job plan JP-C-CCH Balanced chemistry in the water of the cooling towers and chilled water piping must be maintained within predetermined control ranges to maximize equipment service life, and control growth of algae and bacteria in the cooling towers. Control ranges are determined per the chemical manufacturer's specifications. Materials that the Contractor shall provide include, but are not limited to, belts, air filters, chemicals, corrosion inhibitor for chilled water and condenser water systems, cooling tower chemicals and lubrication.	Preventive maintenance shall be performed per the manufacturer's guidelines. The Contractor shall supply the COR with a monthly report documenting the results for all tests performed (daily, weekly etc.) to ensure compliance with the predetermined control ranges. This monthly report shall be provided in a Microsoft Excel format and emailed to the COR.
Technical 4.1.4	Preventative Maintenance Generators	Contractor shall perform Preventative Maintenance on the generators listed in the Cup Contract PMs equipment spreadsheet.	See attached job plans EMG-A and EMG-M. All generator PMs shall be completed by a technician with a minimum of 5 years experience working on both natural gas and diesel generators.	The Contractor shall supply the COR with a monthly report documenting the results for all preventive maintenance.
Technical 4.1.5	Preventive Maintenance Chemical Testing	Contract shall perform annual Legionnaires Testing	Contractor shall test all cooling towers on the NIST site for legionella bacteria. Testing will be issued via Maximo preventative maintenance every July	Contractor shall submit records of annual legionnaires testing of the NIST site within 30 days.

Technical				
Technical 4.1.6	Preventive Maintenance (PAC Units)	Preventive Maintenance on the NIST Packaged Units (PAC) shall be performed per job plans.	Service and supplies for heating and cooling packaged units, split system package units, Liebert split system units, Liebert water cooled units, Data Aire computer room units and air handling units. The technical specifics of the work to be performed are detailed in the attached job plans. Preventative maintenance shall be completed per the manufacturer's guidelines. The Liebert cooling units are categorized as specialized and critical equipment. Therefore NIST is requiring all maintenance performed on these units be performed only by factory certified Liebert technicians. The Data Aire cooling units are categorized as specialized and critical equipment. Therefore NIST is requiring all maintenance performed on these units be performed only by factory certified Data Aire technicians.	Contractor shall utilize the NIST Maximo system to record completion of the PMs by the last day of the month the PM is issued for the Ft. Collins and Boulder sites. The COR will require 14 days notice prior to work, in order to provide notice to customers who will be affected. Any documentation from the factory certified technician shall be provided to COR within 14 days of the completion of the PM. Documentation of the service technician to be utilized for the critical equipment PMs shall be provided to
Technical 4.1.7	Preventive Maintenance (PAC Units)	Preventative Maintenance Certification for Subcontractors shall be given to COR	Any work requiring a factory certified technician shall be documented and provided to the COR.	Documentation of the service technician to be utilized for the critical equipment PMs shall be provided to the COR prior to the start of PM activities.
Technical 4.1.8	Preventive Maintenance (Boilers and associated pumps and equipment)	Preventative Maintenance on hot water boilers and associated pumps and equipment	See attached job plan JP-C-BLR and equipment list Cup Contract PMs for specific boilers. Contractor shall also re-assemble boilers and perform start up to enter equipment back in service after inspection.	Contractor shall notify COR of scheduled annual inspection date and provide certification report to COR within 7 days of completion of annual inspection.
Technical 4.1.9	Preventive Maintenance (Boilers and associated pumps and equipment)	Boiler inspection completed by certified technician	Steam boilers must be opened up annually for inspection by a certified third party; the Contractor shall open up all steam boilers annually and assist the inspector’s as needed during the inspection process. Contractor shall coordinate date of inspection with COR. Annual inspection shall be conducted between July 1 and August 31 of each year.	A written report and any other supporting documentation from the inspector documenting this requirement must be given to the COR within 7 days of the inspection.

Technical				
Technical 5	Reporting Requirements	Individual Monthly Reporting Requirements (PWS Attachment 2)	1. Boiler repair log to document any repairs (minor or major) made to the boilers on the NIST site. (See "Boiler Repairs" Tab) 2 Air compressor repair log to document any repairs (minor or major) made to the air compressors on the NIST site. (See " Air Comp Repairs" Tab) 3. Building water usage report (See "Building Water Usage" Tab) 4. DA Tank daily report (See "DA Tank Daily" Tab) 5. Building 42 Gas usage report for boilers and Building 42. (See "Gas Usage Report" Tab) 6. Boiler Daily Readings report (See "Boiler Daily Readings Temp and Pressure" Tabs) 7. Man Hour report (monthly staff used to support this contract as well as any sub contractors used) (See "Man Hour" Tab") 8. Maintenance reports on the PMs completed at the Ft. Collins site and the Hawaii site. (See "Ft Collins and Hawaii" Tabs) 9. Documented alarm response log for any out of hours calls responded to from the D.O.C. police or any alarms generated requiring a response. (See "After Hours Log" Tab) 10. Steam trap repairs/replacement report - provide report on any steam trap repairs or replacements completed. (See "Steam Trap" Tab) 11. Monthly system chemical reports showing a summary of all daily logs and status (any chemicals added, water testing of each system and any recommendations (See "Chemical Testing" Tab) - Chemical testing of B-17, 18 and 19 every 8 hours - B-42 closed loop chilled water system monthly - B-42 condenser water system tested every 8 hours 12. Chilled Water repair log to document any repairs (minor or major) made to the chilled water system (chillers, cooling towers, condenser water system, pumps etc.) on the NIST site. (See "Chilled Water" Tab) 13. Spare parts Inventory (See "Spare Parts Inventory" Tab) 13. Maintenance reports on the Generator PMs completed (See "Generator PM" Tab)	The Monthly report shall be provided electronically to the Government by the 5th business day of each month as part of the Monthly Report. Excel file for the reports and document tracking shall be provided during the kickoff meeting.
Technical 5.1	Reporting Requirements	Additional reports provided on an as needed basis	There may be additional reports the government asks for during the performance of the contract in order to support Department of Commerce, City of Boulder or the State of Colorado requirements. These shall be part of the base contract and shall be at no extra charge to the government.	Reports shall be given to the COR within 10 days of request
Technical 5.2	Reporting Requirements	Weekly Meetings with COR / TPOC	Weekly meetings will be conducted to discuss CUP operations and any issues needed. The conference will consist of a minimum of the COR or TPOC, Project Manager, and Stationary Engineer.	The contractor will record the meeting notes and email the COR and TPOC within 48 hours of the end of the meeting.

Asset	Description	Location	Frequency	Frequency Units	Monthly Job Plan	Quarterly Job Plan	Tri-Annual Job Plan	Semi-Annual Job Plan	Annual Job Plan	2 Year Job Plan	4 Year Job Plan	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
42-FCU-001	Fan Coil Unit	42-210	6	Months				JP-C-FCU				X						X					
42-HX-001	Heat EXchanger	42-001	1	Year					JP-C-HX								X						
HX-C-029	Heat EXchanger	42-001	1	Year					JP-C-HX								X						
42-PMP-001	Chilled Water Pump for air compressor	42-1	6	Months				JP-C-PMP								X						X	
42-PMP-002	Chilled Water Pump for air compressor	42-1	6	Months				JP-C-PMP								X						X	
AC-C-070	COMPRESSOR, AIR, INGERSALL RAND	42-001	3	Months		JP-C-AC			JP-C-AC-A		JP-C-AC-4Y		X			X			X			X	
AC-C-071	COMPRESSOR, AIR, INGERSALL RAND	42-001	3	Months		JP-C-AC			JP-C-AC-A		JP-C-AC-4Y		X			X			X			X	
AC-C-076	COMPRESSOR, AIR, INGERSALL RAND	42-001	3	Months		JP-C-AC			JP-C-AC-A		JP-C-AC-4Y		X			X			X			X	
AC-C-077	COMPRESSOR, AIR, INGERSALL RAND	42-001	3	Months		JP-C-AC			JP-C-AC-A		JP-C-AC-4Y		X			X			X			X	
AHU-C-101	PRIMARY COMBUSTION AIR HANDLING UNIT	42-ROOF	6	Months				JP-C-AHU							X						X		
AHU-C-102	AIR HANDLING UNIT 102 SOUTH	42-100	6	Months				JP-C-AHU							X						X		
AHU-C-103	AIR HANDLING UNIT FOR OFFICES	42-211	6	Months				JP-C-AHU							X						X		
AHU-C-104	AIR HANDLING UNIT 104 SOUTH	42-001	6	Months				JP-C-AHU							X						X		
AHU-C-105	AIR HANDLING UNIT 105 SOUTH	42-001	6	Months				JP-C-AHU							X						X		
AHU-C-106	AIR HANDLING UNIT 106 SOUTH	42-100	6	Months				JP-C-AHU							X						X		
AHU-C-110	AIR HANDLING UNIT 110	42-103	6	Months				JP-C-AHU							X						X		
AHU-C-111	AIR HANDLING UNIT SOUTH	42-101	6	Months				JP-C-AHU							X						X		
AHU-C-112	AIR HANDLING UNIT CUP 112	42-103	6	Months				JP-C-AHU							X						X		
BLR-C-017	BOILER STEAM, SUPERIOR BOILER WORKS	42-101	1	Months	JP-C-BLR							X	X	X	X	X	X	X	X	X	X	X	X
BLR-C-018	BOILER STEAM, SUPERIOR BOILER WORKS	42-101	1	Months	JP-C-BLR							X	X	X	X	X	X	X	X	X	X	X	X
BLR-C-019	BOILER STEAM, SUPERIOR BOILER WORKS	42-101	1	Months	JP-C-BLR							X	X	X	X	X	X	X	X	X	X	X	X
BPPK-C-001	BOOSTER PUMP PACKAGE NON-POTABLE WATER	42-106	6	Months				JP-C-PMP								X						X	
CCH-C-021	CENTRIFUGAL COMPRESSOR CHILLER, CARRIER_EVR	42-100	3	Months		JP-C-CCH				JP-CCH-C-2Y			X			X			X			X	
CCH-C-022	CENTRIFUGAL COMPRESSOR CHILLER, CARRIER_EVR	42-100	3	Months		JP-C-CCH				JP-CCH-C-2Y			X			X			X			X	
CCH-C-023	CENTRIFUGAL COMPRESSOR CHILLER, CARRIER_EVR	42-100	3	Months		JP-C-CCH				JP-CCH-C-2Y			X			X			X			X	
CCH-C-024	CENTRIFUGAL COMPRESSOR CHILLER, CARRIER_EVR	42-100	3	Months		JP-C-CCH				JP-CCH-C-2Y			X			X			X			X	
CHWF-C-001	CHILLED WATER, FILTER, LAKOS	42-001	1	Year					JP-C-Filter								X						
CHWF-C-002	CHILLED WATER BY-PASS FILTER - BLACK CYLINDER	42-001	1	Year					JP-C-Filter								X						
42-CT-001	COOLING TOWER	42-X-LW	1	Months	JP-C-CT-M	JP-C-CT-Q			JP-C-CT-A			M,Q	M	M	M,Q,A	M	M	M,Q	M	M	M,Q	M	M
42-CT-002	COOLING TOWER	42-X-LW	1	Months	JP-C-CT-M	JP-C-CT-Q			JP-C-CT-A			M,Q	M	M	M,Q,A	M	M	M,Q	M	M	M,Q	M	M
42-CT-003	COOLING TOWER	42-X-LW	1	Months	JP-C-CT-M	JP-C-CT-Q			JP-C-CT-A			M,Q	M	M	M,Q	M	M	M,Q	M	M	M,Q,A	M	M
42-CT-004	COOLING TOWER	42-X-LW	1	Months	JP-C-CT-M	JP-C-CT-Q			JP-C-CT-A			M,Q	M	M	M,Q	M	M	M,Q	M	M	M,Q,A	M	M
42-PMP-001-3	CONDENSER WATER PUMP, AURORA PUMP	42-X-LW	6	Months				JP-C-PMP-S								S						S	
42-VFD-CT-001-1	VFD for 42-CT-001	42-001	1	Year					JP-C-VFD										A				
42-VFD-CT-001-2	VFD for 42-CT-002	42-001	1	Year					JP-C-VFD										A				
42-VFD-CT-001-3	VFD for 42-CT-003	42-001	1	Year					JP-C-VFD										A				
42-VFD-CT-001-4	VFD for 42-CT-004	42-001	1	Year					JP-C-VFD										A				
42-VFD-PMP-001-3	VFD for 42-PMP-001-3	42-001	1	Year					JP-C-VFD										A				
CT-C-041A remove	COOLING TOWER, EVAPCO	42-X-LW	1	Months	JP-C-CT							X	X	X	X	X	X	X	X	X	X	X	X
CT-C-041B remove	COOLING TOWER, EVAPCO	42-X-LW	1	Months	JP-C-CT							X	X	X	X	X	X	X	X	X	X	X	X
CT-C-042A remove	COOLING TOWER, EVAPCO	42-X-LW	1	Months	JP-C-CT							X	X	X	X	X	X	X	X	X	X	X	X
CT-C-042B remove	COOLING TOWER, EVAPCO	42-X-LW	1	Months	JP-C-CT							X	X	X	X	X	X	X	X	X	X	X	X
CT-C-043A remove	COOLING TOWER, EVAPCO	42-X-LW	1	Months	JP-C-CT							X	X	X	X	X	X	X	X	X	X	X	X
CT-C-043B remove	COOLING TOWER, EVAPCO	42-X-LW	1	Months	JP-C-CT							X	X	X	X	X	X	X	X	X	X	X	X
CT-C-044A remove	COOLING TOWER, EVAPCO	42-X-LW	1	Months	JP-C-CT							X	X	X	X	X	X	X	X	X	X	X	X
CT-C-044B remove	COOLING TOWER, EVAPCO	42-X-LW	1	Months	JP-C-CT							X	X	X	X	X	X	X	X	X	X	X	X
CTF-C-001	CONDENSER WATER, FILTER, LAKOS	42-001	1	Year					JP-C-Filter								X						
CUH-C-101	UNIT HEATER ELECTRIC	42-200	1	Year					JP-C-UH												X		
CUH-C-102	UNIT HEATER ELECTRIC	42-ST-1-1	1	Year					JP-C-UH												X		
DRY-C-076	DRYER, INGERSALL RAND	42-001	3	Months		JP-C-DRY							X			X			X			X	
DRY-C-077	DRYER, INGERSALL RAND	42-001	3	Months		JP-C-DRY							X			X			X			X	
DRY-C-201	DRYER, DESICCANT, INGERSALL RAND	42-001	3	Months		JP-C-DRY							X			X			X			X	
DRY-C-202	AIR DRYER, INGERSOL-RAND	42-001	3	Months		JP-C-DRY							X			X			X			X	
EF-C-121	EXHAUST FAN	42-100	6	Months				JP-C-EF							X						X		
EF-C-122	EXHAUST FAN	42-ROOF	6	Months				JP-C-EF							X						X		
EF-C-123	EXHAUST FAN	42-ROOF	6	Months				JP-C-EF							X						X		
EF-C-124	EXHAUST FAN	42-ROOF	6	Months				JP-C-EF							X						X		
EMG-C-019	EMERGENCY GENERATOR	42-001	1	Months	JP-C-EMG							X	X	X	X	X	X	X	X	XA	X	X	X
EMG-C-021	EMERGENCY GENERATOR	42-X-S	1	Months	JP-C-EMG							X	X	X	X	X	X	X	X	XA	X	X	X
EMG-C-022	EMERGENCY GENERATOR	42-X-S	1	Months	JP-C-EMG							X	X	X	X	X	X	X	X	XA	X	X	X
EMG-C-023	EMERGENCY GENERATOR	42-X-S	1	Months	JP-C-EMG							X	X	X	X	X	X	X	X	XA	X	X	X

FAN-C-814	SUPPLY FAN 814 CUP TUNNEL	42-T-0000	6	Months				JP-C-EF							X						X		
FAN-C-815	SUPPLY FAN 815 CUP TUNNEL	42-T-0000	6	Months				JP-C-EF							X						X		
FCU-C-101	FAN COIL UNIT	42-103	1	Year					JP-C-FCU								X						
IH-C-103	INTAKE HOOD	42-ROOF	6	Months				JP-C-IH							X						X		
IH-C-111	INTAKE HOOD	42-ROOF	6	Months				JP-C-IH							X						X		
IH-C-102	INTAKE HOOD	42-ROOF	6	Months				JP-C-IH							X						X		
IH-C-112	INTAKE HOOD	42-ROOF	6	Months				JP-C-IH							X						X		
PMP-C-011	PUMP CHILLED WATER	42-001	6	Months				JP-C-PMP								X						X	
PMP-C-012	PUMP CHILLED WATER	42-001	6	Months				JP-C-PMP								X						X	
PMP-C-013	PUMP CHILLED WATER	42-001	6	Months				JP-C-PMP								X						X	
PMP-C-014	PUMP CHILLED WATER	42-001	6	Months				JP-C-PMP								X						X	
PMP-C-031	CONDENSER WATER PUMP	42-001	6	Months				JP-C-PMP								X						X	
PMP-C-032	CONDENSER WATER PUMP	42-001	6	Months				JP-C-PMP								X						X	
PMP-C-033	CONDENSER WATER PUMP	42-001	6	Months				JP-C-PMP								X						X	
PMP-C-117	HOT WATER PUMP	42-211	6	Months				JP-C-PMP								X						X	
PMP-C-118	HOT WATER PUMP	42-211	6	Months				JP-C-PMP								X						X	
PMP-C-119	HOT WATER PUMP	42-106	6	Months				JP-C-PMP								X						X	
PMP-C-121	PUMP CHILLED WATER	42-106	6	Months				JP-C-PMP								X						X	
PMP-C-201	SUMP PUMP	42-001	6	Months				JP-C-PMP								X						X	
PMP-C-202	SUMP PUMP	42-001	6	Months				JP-C-PMP								X						X	
PMP-C-208	DA WATER RECYCLE PUMP	42-101	6	Months				JP-C-PMP								X						X	
PMP-C-267	BOOSTER PUMP	42-106	6	Months				JP-C-PMP								X						X	
PMP-C-268	BOOSTER PUMP	42-106	6	Months				JP-C-PMP								X						X	
PMP-C-273	CHILLED WATER PUMP	42-001	6	Months				JP-C-PMP								X						X	
PMP-C-274	PUMP CUP	42-001	6	Months				JP-C-PMP								X						X	
PMP-C-275	PUMP CUP	42-001	6	Months				JP-C-PMP								X						X	
PMP-C-N1A	SUMP PUMP	42-TN-0000	6	Months				JP-C-PMP								X						X	
PMP-C-N1B	SUMP PUMP	42-TN-0000	6	Months				JP-C-PMP								X						X	
PMP-C-S1A	SUMP PUMP	42-T-0600	6	Months				JP-C-PMP								X						X	
PMP-C-S1B	SUMP PUMP	42-T-0600	6	Months				JP-C-PMP								X						X	
PMP-C-S2A	SUMP PUMP	42-T-0800	6	Months				JP-C-PMP								X						X	
PMP-C-S2B	SUMP PUMP	42-T-0800	6	Months				JP-C-PMP								X						X	
PMP-C-S3A	SUMP PUMP	42-T-1000	6	Months				JP-C-PMP								X						X	
PMP-C-S3B	SUMP PUMP	42-T-1000	6	Months				JP-C-PMP								X						X	
PMP-C-S4A	SUMP PUMP	42-T-1300	6	Months				JP-C-PMP								X						X	
PMP-C-S4B	SUMP PUMP	42-T-1300	6	Months				JP-C-PMP								X						X	
PMP-C-S5A	SUMP PUMP	42-T-1600	6	Months				JP-C-PMP								X						X	
PMP-C-S5B	SUMP PUMP	42-T-1600	6	Months				JP-C-PMP								X						X	
PMP-TRP-003	PUMP TRAP	42-T-0000	6	Months				JP-C-PMP								X						X	
Steam Traps	All steam traps in the cup and tunnel	42-1	1	Year				JP-C-STEAM									X						
Test for Legionnaires Disease		42-1	1	Year				JP-C-Legionnaires													X		
TS-C-019	TRANSFER SWITCH	42-211	1	Year					JP-C-TS									X					
TS-C-021	TRANSFER SWITCH	42-001	1	Year					JP-C-TS									X					
VFD-C-122	VFD for CCH-C-021	42-100	1	Year					JP-VFD-C										X				
VFD-C-123	VFD for CCH-C-022	42-100	1	Year					JP-VFD-C										X				
VFD-C-124	VFD for CCH-C-023	42-100	1	Year					JP-VFD-C										X				
VFD-C-125	VFD for CCH-C-024	42-100	1	Year					JP-VFD-C										X				
WH-C-011	WATER HEATER	42-106	1	Year					JP-C-WH											X			
WS-C-001	WATER SOFTENER	42-106	1	Months	JP-C-WS							X	X	X	X	X	X	X	X	X	X	X	X
WS-C-002	WATER SOFTENER	42-106	1	Months	JP-C-WS							X	X	X	X	X	X	X	X	X	X	X	X
01-A-AHU-134	Air Handling Unit	01-ROOF-A	3	Months		JP-C-AHU						X			X			X			X		
01-A-AHU-135	Air Handling Unit	01-ROOF-A	3	Months		JP-C-AHU						X			X			X			X		
01-F-CCH-001	CHILLER, COMPRESSOR, YORK	01F	3	Months		JP-C-CCH							X			X			X			X	
01-S1-PMP-279	CONCENTRATE PUMP, Grundfos, # 200 Series	01-1049	6	Months				JP-C-PMP								X						X	
01-S1-PMP-280	Heating Water Pump	01-1049	6	Months				JP-C-PMP								X						X	
01-S1-PMP-283	Building Chilled Water Primary (loop-1)	01-1049	6	Months				JP-C-PMP								X						X	
01-S1-PMP-284	Building Chilled Water Primary (loop-2)	01-1049	6	Months				JP-C-PMP								X						X	
01-S1-PMP-285	Building Chilled Water Primary (loop-3)	01-1049	6	Months				JP-C-PMP								X						X	
01-S1-PMP-286	Building Chilled Water Primary (loop-stand-by)	01-1049	6	Months				JP-C-PMP								X						X	
01-S2-AHU-138	Air Handling Unit	01-1007	3	Months		JP-C-AHU						C	X			X			X			X	
01-S2-HX-2058-1	Chilled water heat exchanger	01-2058	1	Year					JP-C-HX								X						
01-S2-PAC-001	PAC Services Room 2040	01-2040	6	Months				JP-C-PAC							X						X		
01-S2-PMP-2058-1	Inline Pump	01-2058	6	Months				JP-C-PMP								X						X	
01-S2-PMP-2058-2	Inline Pump	01-2058	6	Months				JP-C-PMP								X						X	
01-W1-AHU-136	Air Handling Unit 136	01-ROOF-W1-NC-N	3	Months		JP-C-AHU							X			X			X			X	

01-W3-AHU-140	Air Handling Unit (AHU 3-1)	01-3311	3	Months		JP-C-AHU							X			X			X			X	
01-W3-fan-831	Exhaust Fan (EF 3-1)	01-ROOF-W3	1	Months	JP-C-EF							X	X	X	X	X	X	X	X	X	X	X	X
01-W3-fan-832	Exhaust Fan (EF 3-2)	01-ROOF-W3	1	Months	JP-C-EF							X	X	X	X	X	X	X	X	X	X	X	X
01-W3-PMP-302	Process Chilled Water Pump #1 PCWP-1	01-3353A	1	Year					JP-C-PMP											X			
01-W3-PMP-303	Process Chilled Water Pump #1 PCWP-2	01-3353A	1	Year					JP-C-PMP					X									
01-W3-WH-026	Electric Hot Water Heater	01-3353A	1	Year					JP-C-WH			x											
01-W6-AHU-139	Air Handling Unit	01-4663	3	Months		JP-C-AHU							X			X			X			X	
01-W6-fan-824	Exhaust Fan - Exhaust	01-ROOF-W6	1	Months	JP-C-EF							X	X	X	X	X	X	X	X	X	X	X	X
01-W6-fan-825	Exhaust Fan - Exhaust	01-ROOF-W6	1	Months	JP-C-EF							X	X	X	X	X	X	X	X	X	X	X	X
01-w6-pmp-293	Heating Hot Water Pump	01-3665	1	Year					JP-C-PMP										X				
01-w6-pmp-294	Heating Hot Water Pump	01-3665	1	Year					JP-C-PMP										X				
01-W6-WH-024	Electric Hot Water Heater - EWH-1	01-3663	1	Year					JP-C-WH											X			
02-PAC-001	PAC Unit (rename of AHU-082)	02-ROOF-UPPER-SOUTH	6	Months				JP-C-PAC							X						X		
05-PAC-001	PAC UNIT	05-ROOF	6	Months				JP-C-PAC							X						X		
05-PAC-002	PAC UNIT	05-ROOF	6	Months				JP-C-PAC							X						X		
05-PAC-003	PAC UNIT	05-ROOF	6	Months				JP-C-PAC							X						X		
21-BLR-0001	BOILER, HOT WATER (ESCO project)	21-1008	6	Months				JP-C-BLR							X						X		
22-BLR-1122-1	Boiler	22-1122	6	Months				JP-C-BLR							X						X		
22-PAC-001	Rooftop PAC Unit	22-ROOF	6	Months				JP-C-PAC							X						X		
22-PAC-002	Rooftop PAC Unit	22-ROOF	6	Months				JP-C-PAC							X						X		
24-PAC-001	Package Air (Carrier)	24-ROOF	6	Months				JP-C-PAC								X						X	
25-BLR-001	BOILER, HOT WATER (ESCO project)	25-1107	6	Months				JP-C-BLR							X						X		
25-CH-001	Carrier Chiller Model 30MPW0155FA01005	25-1107	6	Months				JP-CCH-1								X						X	
25-CT-1107-1	Whaley Cooling Tower	25-1107	1	Months	JP-C-CT											X	X	X	X	X	X		
26-PAC-001	Rooftop PAC Unit	26-ROOF	6	Months				JP-C-PAC							X						X		
26-PAC-002	Rooftop PAC Unit	26-ROOF	6	Months				JP-C-PAC							X						X		
26-PAC-003	Rooftop PAC Unit	26-ROOF	6	Months				JP-C-PAC							X						X		
26-PAC-004	Rooftop PAC Unit	26-ROOF	6	Months				JP-C-PAC							X						X		
26-PAC-005	Rooftop PAC Unit	26-ROOF	6	Months				JP-C-PAC							X						X		
26-PAC-006	Rooftop PAC Unit	26-ROOF	6	Months				JP-C-PAC							X						X		
AHU-055	AIR HANDLING UNIT, SPLIT A/C SYSTEM	01-2058	3	Months		JP-C-AHU							X			X			X			X	
AHU-056	AIR HANDLING UNIT, SPLIT A/C SYSTEM	01-2058	3	Months		JP-C-AHU							X			X			X			X	
AHU-057	AIR HANDLING UNIT, SPLIT A/C SYSTEM	01-2058	3	Months		JP-C-AHU							X			X			X			X	
AHU-058	AIR HANDLING UNIT, SPLIT A/C SYSTEM	01-2058	3	Months		JP-C-AHU						A			X			X			X		
AHU-123	AIR HANDLING UNIT, SELF CONTAINED, AAON	ANNEX E2	3	Months		JP-C-AHU							X			X			X			X	
AHU-124	AIR HANDLING UNIT, SELF CONTAINED, AAON	ANNEX E2	3	Months		JP-C-AHU							X			X			X			X	
AHU-125	AIR HANDLING UNIT SELF CONTAINED, LEIBERT	01-2051	3	Months		JP-C-CRAC							X			X			X			X	
AHU-126	AIR HANDLING UNIT SELF CONTAINED, LEIBERT	01-2053	3	Months		JP-C-CRAC							X			X			X			X	
AHU-132	AIR HANDLING UNIT 132 LIEBERT	01-3430	3	Months		JP-C-CRAC							X			X			X			X	
AHU-133	AIR HANDLING UNIT 133 LIEBERT	01-3430	3	Months		JP-C-CRAC							X			X			X			X	
AHU-I-088	AIR HANDLING UNIT, YORK,# TLKM123270	01F-X	1	Months	JP-C-AHU							X	X	X	X	X	X	X	X	X	X	X	X
AHU-I-089	AIR HANDLING UNIT, YORK,# TLKM123270	01F-X	1	Months	JP-C-AHU							X	X	X	X	X	X	X	X	X	X	X	X
BLR-012	BOILER	02-ROOF	1	Months	JP-C-BLR							X	X	X	X	X	X	X	X	X	X	X	X
CT-002	COOLING TOWER, 50 TON, MARLEY	01-ROOF-W2-C	1	Months	JP-C-CT							X	X	X	X	X	X	X	X	X	X	X	X
PAC-026	PACKAGED AIR CONDITIONER, ROOFTOP, TRANE	01D-ROOF	6	Months				JP-C-PAC							X						X		
PAC-027	PACKAGED AIR CONDITIONER, ROOFTOP, TRANE	01D-ROOF	6	Months				JP-C-PAC							X						X		
PAC-029	PACKAGED AIR CONDITIONER, ROOFTOP, Carrier	26-ROOF	6	Months				JP-C-PAC							X						X		
PAC-031	PACKAGED AIR CONDITIONER, ROOFTOP, Carrier	26-ROOF	6	Months				JP-C-PAC							X						X		
PAC-052	PACKAGED AIR CONDITIONER	02-ROOF	6	Months				JP-C-PAC							X						X		
PAC-053	PACKAGED AIR CONDITIONER	02-ROOF	6	Months				JP-C-PAC							X						X		
PAC-054	PACKAGED AIR CONDITIONER	02-ROOF	6	Months				JP-C-PAC							X						X		
PAC-073	PACKAGED AIR CONDITIONER-ROOFTOP	01C-ROOF	6	Months				JP-C-PAC							X						X		
PAC-078	SPLIT PACKAGE A/C SYSTEM,IN ELEV.CONTROL RM	01-ROOF-MS-P	6	Months				JP-C-PAC							X						X		
PAC-101	PACKAGED AIR CONDITIONER, ROOFTOP, TRANE	21-ROOF	6	Months				JP-C-PAC							X						X		
PAC-107	PACKAGED AIR CONDITIONER, ROOFTOP, AAON	01-ROOF-W4-C	6	Months				JP-C-PAC								X						X	
PAC-108	SPLIT PACKAGE A/C SYSTEM,FAN (02-1804	02-1806 change to 02-1804	6	Months				JP-C-PAC							X						X		
PAC-109	SPLIT PACKAGE A/C SYSTEM,FAN	02-1806	6	Months				JP-C-PAC							X						X		
PAC-110	SPLIT PACKAGE A/C SYSTEM,FAN IN 24-ELEV	24-X-SE	6	Months				JP-C-PAC								X						X	
PAC-115	PACKAGED AIR CONDITIONER, Carrier	01C-ROOF	6	Months				JP-C-PAC							X						X		
PAC-116	PACKAGED AIR CONDITIONER, Carrier	01C-ROOF	6	Months				JP-C-PAC							X						X		
PAC-117	PACKAGED AIR CONDITIONER, Carrier	02A-ROOF	6	Months				JP-C-PAC							X						X		
PAC-118	PACKAGED AIR CONDITIONER, Carrier	02A-ROOF	6	Months				JP-C-PAC							X						X		
PAC-135	PACKAGED AIR CONDITIONER	12-ROOF	6	Months				JP-C-PAC								X						X	
PAC-136	PACKAGED AIR CONDITIONER	91-X-S	6	Months				JP-C-PAC								X						X	
PAC-137	PACKAGED AIR CONDITIONER	91-X-S	6	Months				JP-C-PAC								X						X	
PAC-141	SPLIT A/C SYSTEM	51-101A	6	Months				JP-C-PAC								X						X	

PAC-143	PACKAGED AIR CONDITIONER	51-B	6	Months				JP-C-PAC								X						X	
PAC-144	PACKAGED AIR CONDITIONER-ROOFTOP	111-X-S	6	Months				JP-C-PAC								X						X	
PAC-145	PACKAGED AIR CONDITIONER-ROOFTOP	111-X-S	6	Months				JP-C-PAC								X						X	
PAC-146	PACKAGED AIR CONDITIONER	25-TRLR	6	Months				JP-C-PAC								X						X	
PAC-150	PACKAGED AIR CONDITIONER-ROOFTOP	03A-ROOF	6	Months				JP-C-PAC								X						X	
PAC-151	PACKAGED AIR CONDITIONER-ROOFTOP	01-ROOF-W2	6	Months				JP-C-PAC								X						X	
PAC-152	PACKAGED AIR CONDITIONER-ROOFTOP	01-ROOF-W2	6	Months				JP-C-PAC								X						X	
PAC-153	PACKAGED AIR CONDITIONER-ROOFTOP	01-ROOF-W2	6	Months				JP-C-PAC								X						X	
PAC-154	PACKAGED AIR CONDITIONER-ROOFTOP	01-ROOF-W2	6	Months				JP-C-PAC								X						X	
PAC-155	PACKAGED AIR CONDITIONER-ROOFTOP	01-ROOF-W2-C	6	Months				JP-C-PAC								X						X	
PMP-I-001	PUMP, CHILLED WATER	01F	1	Year					JP-C-PMP										X				
PMP-I-002	PUMP, CHILLED WATER	01F	1	Year					JP-C-PMP										X				
PMP-I-003	PUMP, CHILLED WATER	01F	1	Year					JP-C-PMP										X				
PMP-I-004	PUMP, CHILLED WATER	01F	1	Year					JP-C-PMP										X				
PMP-I-008	PUMP, CHILLED WATER	01F-X	1	Year					JP-C-PMP										X				
PMP-I-009	PUMP, CHILLED WATER	01F-X	1	Year					JP-C-PMP										X				
PMP-I-010	PUMP, CHILLED WATER	01F-X	1	Year					JP-C-PMP										X				
RC-056	SPLIT PACKAGE A/C SYSTEM 02-1804	02-roof	6	Months				JP-C-RC								X						X	
RC-057	SPLIT PACKAGE A/C SYSTEM 02-1806	02-roof	6	Months				JP-C-RC								X						X	
RC-059	SPLIT PACKAGE A/C SYSTEM 02-2810 and 02-1802	02-ROOF-UPPER-SOUTH	6	Months				JP-C-RC								X						X	
BL-H-WWVH	EXTERIOR LIGHTING	WWV-H-PMRF	6	Months				JP-C-Lights							X						X		
CU-H-013	CONDENSOR UNIT, DAKIN	WWV-H-EX-N	3	Months		JP-C-CU						X			X			X			X		
CU-H-014	CONDENSOR UNIT, DAKIN	WWV-H-EX-N	3	Months		JP-C-CU						X			X			X			X		
CU-H-015	CONDENSOR UNIT, DAKIN	WWV-H-EX-N	3	Months		JP-C-CU						X			X			X			X		
CU-H-016	CONDENSOR UNIT, DAKIN	WWV-H-EX-N	3	Months		JP-C-CU						X			X			X			X		
CU-H-017	CONDENSOR UNIT, DAKIN	WWV-H-EX-N	3	Months		JP-C-CU						X			X			X			X		
EF-H-125	EXHAUST FAN	WWV-H-ROOF-NC	6	Months				JP-C-EF							X						X		
EF-H-126	EXHAUST FAN	WWV-H-ROOF-NC	6	Months				JP-C-EF							X						X		
EVU-H-013	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-014	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-015	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-016	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-017	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-018	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-019	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-020	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-021	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-022	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-023	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-024	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-025	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-026	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-027	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-028	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-029	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-030	EVAPORATOR UNIT	WWV-H-1011	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-031	EVAPORATOR UNIT	WWV-H-1012	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-032	EVAPORATOR UNIT	WWV-H-1012	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-033	EVAPORATOR UNIT	WWV-H-1012	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-034	EVAPORATOR UNIT	WWV-H-1012	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-035	EVAPORATOR UNIT	WWV-H-1012	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-036	EVAPORATOR UNIT	WWV-H-1012	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-037	EVAPORATOR UNIT	WWV-H-1010A	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-038	EVAPORATOR UNIT	WWV-H-1010A	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-039	EVAPORATOR UNIT	WWV-H-1010B	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-040	EVAPORATOR UNIT	WWV-H-1010B	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-041	EVAPORATOR UNIT	WWV-H-1010	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-042	EVAPORATOR UNIT	WWV-H-1010	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-043	EVAPORATOR UNIT	WWV-H-1010	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-044	EVAPORATOR UNIT	WWV-H-1010	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-045	EVAPORATOR UNIT	WWV-H-1007	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-046	EVAPORATOR UNIT	WWV-H-1006	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-047	EVAPORATOR UNIT	WWV-H-1004	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-048	EVAPORATOR UNIT	WWV-H-1003	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-049	EVAPORATOR UNIT	WWV-H-1002	3	Months		JP-C-EVU						X			X			X			X		
EVU-H-050	EVAPORATOR UNIT	WWV-H-1009	3	Months		JP-C-EVU						X			X			X			X		
INT-H-WWVH	INTERIOR BUILDING LIGHTING WWV HAWAII	WWV-H-PMRF	6	Months				JP-C-Lights							X						X		

SOLAR-H-001	SOLAR ARRAY HAWAII	WWV-H-EX-N	6	Months				JP-SOLAR							X						X		
AC-W-028	COMPRESSOR, AIR, INGERSALL RAND	WWVB-016	3	Months		JP-C-AC							X			X			X			X	
AC-W-029	COMPRESSOR,COMPRSR AIR, SpeedAire	WWVB-1012	3	Months		JP-C-AC							X			X			X			X	
AHU-W-030	AHU SWAMP COOLER, GOES TO TRANSMITTER	WWVB-EX-S	6	Months				JP-C-AHU								X						X	
AHU-W-040	AHU SWAMP COOLER, GOES TO TRANSMITTER	WWVB-EX-W	6	Months				JP-C-AHU								X						X	
AHU-W-050	AHU SWAMP COOLER, GOES TO TRANSMITTER	WWVB-EX-N	6	Months				JP-C-AHU								X						X	
AHU-W-060	AHU SWAMP COOLER, GOES TO TRANSMITTER	WWVB-EX-E	6	Months				JP-C-AHU								X						X	
AHU-W-131	AHU SWAMP COOLER, UNITED METAL, # CEL-ODM-56, # CEL-51830	WWV-ROOF-U	6	Months				JP-C-AHU								X						X	
EF-W-101	EXHAUST FAN	WWV-ROOF-L	6	Months		JP-C-EF											X						X
EF-W-102	EXHAUST FAN	WWV-ROOF-U	6	Months		JP-C-EF											X						X
EF-W-103	EXHAUST FAN	WWV-ROOF-U	6	Months		JP-C-EF											X						X
EF-W-106	EXHAUST FAN	WWV-ROOF-U	6	Months		JP-C-EF											X						X
EF-W-107	EXHAUST FAN	WWV-ROOF-U	6	Months		JP-C-EF											X						X
EF-W-108	EXHAUST FAN	WWV-ROOF-L	6	Months		JP-C-EF											X						X
EF-W-109	EXHAUST FAN	WWV-ROOF-L	6	Months		JP-C-EF											X						X
EF-W-110	EXHAUST FAN	WWVB-ROOF	6	Months		JP-C-EF											X						X
EF-W-112	EXHAUST FAN	WWVB-019	6	Months		JP-C-EF											X						X
EF-W-113	EXHAUST FAN	WWVB-020	6	Months		JP-C-EF											X						X
EF-W-114	EXHAUST FAN	WWVB-013	6	Months		JP-C-EF											X						X
EF-W-115	EXHAUST FAN	WWVB-ROOF	6	Months		JP-C-EF											X						X
EF-W-116	EXHAUST FAN	WWVB-ROOF	6	Months		JP-C-EF											X						X
EF-W-117	EXHAUST FAN	WWVB-ROOF	6	Months		JP-C-EF											X						X
EF-W-118	EXHAUST FAN	WWVB-015	6	Months		JP-C-EF											X						X
EF-W-127	EXHAUST FAN	WWV-ROOF-U	6	Months		JP-C-EF											X						X
EF-W-128	EXHAUST FAN	WWV-ROOF-U	6	Months		JP-C-EF											X						X
EF-W-129	EXHAUST FAN	WWVB-ROOF	6	Months		JP-C-EF											X						X
EF-W-130	EXHAUST FAN	WWVB-ROOF	6	Months		JP-C-EF											X						X
EF-W-131	EXHAUST FAN	WWVB-ROOF	6	Months		JP-C-EF											X						X
EVU-W-051	EVAPORATOR UNIT	WWV-1007	3	Months		JP-C-EVU								X			X			X			X
EVU-W-052	EVAPORATOR UNIT	WWV-1007	3	Months		JP-C-EVU								X			X			X			X
EVU-W-053	EVAPORATOR UNIT	WWV-1007	3	Months		JP-C-EVU								X			X			X			X
EVU-W-054	EVAPORATOR UNIT	WWV-1007	3	Months		JP-C-EVU								X			X			X			X
EVU-W-055	EVAPORATOR UNIT	WWVB-010	3	Months		JP-C-EVU								X			X			X			X
EVU-W-056	EVAPORATOR UNIT	WWVB-010	3	Months		JP-C-EVU								X			X			X			X
EVU-W-057	EVAPORATOR UNIT	WWVB-013	3	Months		JP-C-EVU								X			X			X			X
EVU-W-058	EVAPORATOR UNIT	WWVB-013	3	Months		JP-C-EVU								X			X			X			X
PAC-W-091	PACKAGED AIR CONDITIONER-ROOFTOP	WWVB-ROOF	3	Months		JP-C-PAC								X			X			X			X
PAC-W-092	PACKAGED AIR CONDITIONER-ROOFTOP	WWV-ROOF-L	3	Months		JP-C-PAC								X			X			X			X
PAC-W-093	PACKAGED AIR CONDITIONER-ROOFTOP	WWV-ROOF-U	3	Months		JP-C-PAC								X			X			X			X
PAC-W-124	SPLIT A/C SYSTEM	WWV	3	Months		JP-C-PAC								X			X			X			X
PAC-W-125	SPLIT A/C SYSTEM	WWV	3	Months		JP-C-PAC								X			X			X			X
PAC-W-126	SPLIT A/C SYSTEM	WWV	3	Months		JP-C-PAC								X			X			X			X
PAC-W-127	SPLIT A/C SYSTEM	WWV	3	Months		JP-C-PAC								X			X			X			X
PAC-W-128	SPLIT A/C SYSTEM	WWVB	3	Months		JP-C-PAC								X			X			X			X
PAC-W-129	SPLIT A/C SYSTEM	WWVB	3	Months		JP-C-PAC								X			X			X			X
PAC-W-130	SPLIT A/C SYSTEM	WWVB	3	Months		JP-C-PAC								X			X			X			X
PAC-W-131	SPLIT A/C SYSTEM	WWVB	3	Months		JP-C-PAC								X			X			X			X
PAC-W-132	PACKAGED AIR CONDITIONER-ROOFTOP	WWVB-ROOF	3	Months		JP-C-PAC								X			X			X			X
PAC-W-133	PACKAGED AIR CONDITIONER-ROOFTOP	WWV-ROOF-L	3	Months		JP-C-PAC								X			X			X			X
PAC-W-134	PACKAGED AIR CONDITIONER-ROOFTOP	WWVB-COOP	3	Months		JP-C-PAC								X			X			X			X
SOLAR-H-001	SOLAR ARRAY HAWAII	WWV-H-EX-N	6	Months				JP-SOLAR							X						X		
UH-W-017	UNIT HEATER, ELECTRIC	WWVB-010	1	Year					JP-U-UH												X		
UH-W-018	UNIT HEATER, ELECTRIC	WWVB-014	1	Year					JP-U-UH												X		
UH-W-019	UNIT HEATER, ELECTRIC	WWVB-014	1	Year					JP-U-UH												X		
UH-W-020	UNIT HEATER, ELECTRIC	WWV-1012	1	Year					JP-U-UH												X		
UH-W-021	UNIT HEATER, ELECTRIC	WWV-1007	1	Year					JP-U-UH												X		
UH-W-022	UNIT HEATER, ELECTRIC	WWV-1013	1	Year					JP-U-UH												X		
UH-W-025	UNIT HEATER, ELECTRIC	WWVB-010	1	Year					JP-U-UH												X		
UH-W-026	UNIT HEATER, ELECTRIC	WWVB-016	1	Year					JP-U-UH												X		
UH-W-027	UNIT HEATER, ELECTRIC	WWV-1013	1	Year					JP-U-UH												X		
UH-W-028	UNIT HEATER, ELECTRIC	WWV-1008	1	Year					JP-U-UH												X		
UH-W-039	UNIT HEATER, ELECTRIC	WWV-WH	1	Year					JP-U-UH												X		
VFD-W-127	NO MODIFIER DRIVE, VFD	WWV-1008	1	Year					JP-VFD-C							x							
VFD-W-128	NO MODIFIER DRIVE, VFD	WWV-1008	1	Year					JP-VFD-C							x							
VFD-W-129	NO MODIFIER DRIVE, VFD	WWV-1013	1	Year					JP-VFD-C							x							
WAC-W-008	WALL AIR CONDITIONER	WWVB-015	3	Months		JP-C-WAC						X			X			X			X		

WH-W-009	WATER HEATER	WWVB-005	1	Year					JP-C-WH											X			
WH-W-010	WATER HEATER	WWVB-012	1	Year					JP-C-WH											X			
WH-W-018	WATER HEATER	WWV-1014	1	Year					JP-C-WH											X			
WH-W-019	WATER HEATER	WWVB-005	1	Year					JP-C-WH											X			
LakosFilters	SUMP SWEEPER SYSTEM LAKOS	42-001	1	Year					JP-C-Filter								X						
DA-C-018	DA WATER TANK 018	42-101	1	Year													X						
DA-C-019	DA WATER TANK 019	42-101	1	Year								X											
01-AHU-1207-1	Air Handler above 1200 Office suite	ROOF ABOVE LIBRARY										A			x			x			x		

CUP Building 42 Inventory list Government provided inventory for Contractor to utilize and manage		
Type of equipment	Spare Parts Quanity needed	Spare parts Quanity on hand
Chemical LMI pumps	2	
Liquid end assemblies	2	
LMI pump Repair kits	2	
B-1 Steam traps	2	
B-2 Steam Traps	2	
B-3 Steam Traps	2	
Thermo Dynamic traps	2	

Boiler Monthly Repair Log

National Institute of Standards and Technology
U.S. Department of Commerce

Month:
Year:
Company:

Major Repairs:

Minor Repairs:

Air Compressor Repair Summary

National Institute of Standards and Technology
U.S. Department of Commerce

Month:
Year:
Company:

Major Repair Summary:

Minor Repair Summary:

Building Water Usage Report

Month: **Year:**

Day	Start	End	Daily Total
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			
17			
18			
19			
20			
21			
22			
23			
24			
25			
26			
27			
28			
29			
30			
31			

DA Tank - Pressure - Temps				
Date	Daily DA Tank Temp	Daily DA Tank Press	Daily System Condensate Return	Daily Total Make-Up Water Flow

NIST Boulder Gas Usage				
Date	**BLR 17 Meter reading**	**BLR 18 Meter reading**	**BLR 19 Meter reading**	**BLDG. 42 Meter reading**
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				
21				
22				
23				
24				
25				
26				
27				
28				
29				
30				
31				

Boiler - Pressures - Flow								
BLR	Date	Daily M/CF Total Gas Flow	Daily GALS Total FDW Flow	Daily LBS/H Total BLR Steam Flow	Daily Lbs/H Total SYS Steam Flow	Daily PSI Total System Steam Press	Daily PSI Total BLR Steam Press	Daily LBS / H Total BLR Steam LG Flow
17								
18								
19								
17								
18								
19								
17								
18								
19								
17								
18								
19								

Boiler Temps						
BLR	Date	Daily Total Temp Stack ENT	Daily Total Temp Stack LVG	Daily Total Temp Econo IN	Daily Total Temp Econo OUT	Daily Total Temp OAT
17						
18						
19						
17						
18						
19						
17						
18						
19						
17						
18						
19						

Monthly Man Hour Report Month of:

1. All contractors, service and construction, are required to submit a monthly man-hour/ injury report. This report needs to be submitted to the EMSS Safety office no later than the 5th of each month. The report needs to contain the following information;

Contracted Work;

a) Contract Number:

b) Contract Type (Select Type): ( ) Construction ( ) Service

c) Project Name: Central Utility Plant Operation and Maintenance

d) Company Name:

e) NAICS Code for the Company performing the work:

f) Total number of man-hours worked this month (must include subcontractor hours):

g) Lost Time Injury experienced during the month:

h) Near Miss Incidents for the month:

i) Fire/Property Damage Incidents:

j) No Work Completed this Month:

2. COR's are responsible for collecting the forms from your contractor and submit to FMD Safety offic

Ft. Collins Maintenance Summary

Month:
Year:
Company:

National Institute of Standards and Technology
U.S. Department of Commerce

Monthly Maintenance Summary:

Contractor to provide Maintenance reports/checklists on the PMs completed at the Ft. Collins site, and record preventative maintenance information from the job plan inputted into Maximo Logs. Any temperature, voltage or pressure readings listed in the job plan shall be provided in the Maximo logs.

Hawaii Maintenance Summary

Month:
Year:
Company:

National Institute of Standards and Technology
U.S. Department of Commerce

Monthly Maintenance Summary:

Contractor to provide Maintenance reports/checklists on the PMs completed at the Hawaii site, and record preventative maintenance information from the job plan inputted into Maximo Logs. Any temperature, voltage or pressure readings listed in the job plan shall be provided in the Maximo logs.

After Hours Log

Date	Time	Time Alarm Cleared	Dispatch Personnel	Name of Alarm	Description	Description of Actions Taken	Stationary Engineer Initials

Steam Trap Repair / Replacement Summary

National Institute of Standards and Technology
U.S. Department of Commerce

Month:
Year:
Company:

Monthly Repair / Replacement Summary:

Discrepancies:

Chemical Testing Summary

Month:
Year:
Company:

Chemical Testing Summary:

Testing to be completed every 8 hours based on selected manufacturer's recommendations checklist and submitted as part of the monthly report.

- Chemical testing of B-17, 18 and 19 every 8 hours
- B-42 closed loop chilled water system tested monthly
- B-42 condenser water system tested every 8 hours

Contractor shall provide monthly reporting on any anomalies found in the testing for the month, and the manufacturer's recommendations on chemical adjustments or treatments to be made.

Chilled Water Repair Summary

National Institute of Standards and Technology
U.S. Department of Commerce

Month:
Year:
Company:

Major Repairs:

Minor Repairs:

CUP Building 42
Government Provided Parts

National Institute of Standards and Technology
U.S. Department of Commerce

Type of equipment	Spare Parts Quantity needed	Spare parts Quantity on hand
Chemical LMI pumps	2	
Liquid end assemblies	2	
LMI pump Repair kits	2	
B-1 Steam traps	2	
B-2 Steam Traps	2	
B-3 Steam Traps	2	
Thermo Dynamic traps	2	

Generator Preventative Maintenance

National Institute of Standards and Technology
U.S. Department of Commerce

Month:
Year:
Company:

Major Repairs:

Minor Repairs:

Job Plan : JP-C-CT-A

Description : Contractor Job Plan Cooling Tower Annual

Task Sequence	Task	Task Description	Task Long Description
0	1	OBTAIN PM	PM issued through NIST Maximo System.
0	3	SCHEDULE THE PM	Schedule the PM with the COR. NO equipment will be shut down without scheduling with the COR.
0	5	REVIEW	REVIEW ALL STEPS IN PM. REVIEW SAFTEY REQUIREMENTS FOR THIS EQUIPMENT AND SPACE. REVIEW MANUFACTURER'S INSTRUCTIONS AND EQUIPMENT HISTORY RECORD, IF AVAILABLE.
0	7	EQUIPMENT SHUTDOWN SAFETY	If shut-down is necessary; de-energize, tag, and lock out electrical circuits to motors, pumps, control circuits, etc and de-energize pressure in lines before working on equipment. Ensure that the people affected have been notified (COR to notify) and associated equipment's operation will not be affected.
0	9	CHECK WITH OPERATING OR AREA PERSONNEL	Check with operating or area personnel for any deficiencies.
0	10	CHECK STRUCTURAL MEMBERS	CHECK STRUCTURAL MEMBERS FOR BREAKS, BROKEN WELDS, LOOSE OR MISSING BOLTS. LOOK FOR EXCESSIVE RUST, PITTING, AND CORROSION BOTH INSIDE AND OUTSIDE UNIT.
0	20	SLIDING MOTOR BASE	INSPECT AND GREASE SLIDING MOTOR BASE.
0	30	CLEAN AND INSPECT	INSPECT SHEAVES, BUSHINGS, FAN SHAFTS, AND FAN HUBS FOR CORROSION. SCRAPE AND COAT WITH ZRC.
	80	PRESSURE WASH	PRESSUE WASH COOLING TOWER TANK AND EXTERIOR, MAKE SURE TOWER IS FREE OF ALL SILT AND DEBRIS FROM POWER WASHING.

Job Plan : JP-C-CT-A

Description : Contractor Job Plan Cooling Tower Annual

Task Sequence	Task	Task Description	Task Long Description
0	120	INSPECT, TEST, AND ADJUST.	INSPECT, TEST, AND ADJUST ALL ELECTRIC CONTROLS AND PROTECTIVE DEVICES. CHECK ALL ELECTRICAL CONTACTS FOR PITTING, BURNING, OR EXCESSIVE WEAR. MAKE SURE UNITS HAVE PROPER SIZE FUSES, BREAKERS, AND MOTOR CONTROL HEATERS.
0	998	COMPLETE "NEW EQUIPMENT REPORT FORM"	COMPLETE" NEW EQUIPMENT FORM" FOR DATA THAT APPLIES TO THIS PIECE OF EQUIPMENT. SEE EXAMPLES FORM AND CLASSIFICATION& ATTRIBUTE TABLE; PROVIDED BY YOUR COR
0	999	COMPLETE PM	Complete the PM through NIST Maximo System.; noting condition, suspect problems, parts used, gauge/meter readings, etc. Notify COR if any equipment requires replacement and note on PM

Job Plan : JP-C-CT-M

Description : Contractor Job Plan Cooling Tower Monthly

Task Sequence	Task	Task Description	Task Long Description
0	**1**	**OBTAIN PM**	PM issued through NIST Maximo System.
0	**3**	**SCHEDULE THE PM**	Schedule the PM with the COR. NO equipment will be shut down without scheduling with the COR.
0	**5**	**REVIEW**	REVIEW ALL STEPS IN PM.
			REVIEW SAFTEY REQUIREMENTS FOR THIS EQUIPMENT AND
			SPACE.
			REVIEW MANUFACTURER'S INSTRUCTIONS AND EQUIPMENT
			HISTORY RECORD, IF AVAILABLE.
0	**7**	**EQUIPMENT SHUTDOWN SAFETY**	If shut-down is necessary; de-energize, tag, and lock out electrical
			circuits to motors, pumps, control circuits, etc and de-energize
			pressure in lines before working on equipment. Ensure that the
			people affected have been notified (COR to notify) and associated
			equipment's operation will not be affected.
0	**9**	**CHECK WITH OPERATING OR AREA PERSONNEL**	Check with operating or area personnel for any deficiencies.
0	**20**	**INSPECT AND CLEAN**	INSPECT GENERAL CONDITION OF PAN STRAINER, ELIMINATORS, FANS, FAN SCREENS, AND DAMPERS. CLEAN.
		BLEED OFF AND FLOAT VALVES	CHECK BLEED OFF VALVE TO VERIFY PROPER OPERATION. CHECK OPERATING LEVEL IN PAN AND ADJUST FLOAT VALVE.
		WATER DISTRIBUTION SYSTEM	CHECK WATER DISTRIBUTION SYSTEM AND SPRAY PATTERN.
		BELT	CHECK BELT TENSION AND ADJUST
		VIBRATION SWITCH	INSPECT VIBRATION SWITCH ENCLOSURE FOR LOOSE WIRING AND MOISTURE. ADJUST SENSITIVITY AS NEEDED.
		CHECK AXS	INSPECT HOT WATER BASINS FOR DEBRIS AND CORROSION.
		WATER TREATMENT	Maintain water chemistry as recommended by chemical company representative.
		COMPLETE PM	Complete the PM through NIST Maximo System.; noting condition, suspected problems parts used, gauge/meter readings, etc., notify COR if any equipment requires replacement and note on PM.

Job Plan : JP-C-CT-Q

Description : Contractor Job Plan Cooling Tower Quarterly

Task Sequence	Task	Task Description	Task Long Description
0	1	**OBTAIN PM**	PM issued through NIST Maximo System.
0	3	**SCHEDULE THE PM**	Schedule the PM with the COR. NO equipment will be shut down without scheduling with the COR.
0	5	**REVIEW**	REVIEW ALL STEPS IN PM.
			REVIEW SAFTEY REQUIREMENTS FOR THIS EQUIPMENT AND
			SPACE.
			REVIEW MANUFACTURER'S INSTRUCTIONS AND EQUIPMENT
			HISTORY RECORD, IF AVAILABLE.
0	7	**EQUIPMENT SHUTDOWN SAFETY**	If shut-down is necessary; de-energize, tag, and lock out electrical
			circuits to motors, pumps, control circuits, etc and de-energize
			pressure in lines before working on equipment. Ensure that the
			people affected have been notified (COR to notify) and associated
			equipment's operation will not be affected.
0	9	**CHECK WITH OPERATING OR AREA PERSONNEL**	Check with operating or area personnel for any deficiencies.
0	20	**INSPECT AND CLEAN**	INSPECT FAN BLADES FOR CRACKS, MISSING BALANCING WEIGHTS, AND VIBRATIONS.
		BLEED OFF AND FLOAT VALVES	CHECK BLEED OFF VALVE TO VERIFY PROPER OPERATION. CHECK OPERATING LEVEL IN PAN AND ADJUST FLOAT VALVE.
		FAN SHAFT BEARINGS	LUBRICATE FAN SHAFT BEARINGS.
		EXERCISE VALVES	EXERCISE SUMP VALVES TO ENSURE FREEDOM OF MOTION.
		COMPLETE PM	Complete the PM through NIST Maximo System.; noting condition, suspected problems parts used, gauge/meter readings, etc., notify COR if any equipment requires replacement and note on PM.

Daily Report

Date:

A Report of Incidents, issues, or areas of concern over the last 24 hours.

CUP Report

Issue to Report	Area of Issue	Resolution / Comments

Site Report

Issue to Report	Area of Issue	Resolution / Comments

Metasys/ CPO-10 Report

Issue to Report	Area of Issue	Resolution / Comments

Additional Information

General Instructions for Conducting Work at the NIST Boulder Campus

OFPM Rev. C
3-30-2020

Contents

Part 1: GENERAL

1.1 INTRODUCTION:

1. This document is intended to provide the necessary information/requirements for all contractors performing work at Department of Commerce (DOC) facilities. It is understood all information in this document may not apply for the specific project described in the Statement of Work; however, it is the responsibility of the contractor to ensure the requirements are met when applicable.

2. Specifics of the activities are described in the attached Statement of Work and supporting documentation.

3. Questions about the requirements can be addressed to the Contracting Officer (CO) and Contracting Officers Representative (COR)/Project Manager designated for the specific task order.

1.2 BACKGROUND

The Office of Facilities and Property Management (OFPM) Division manages and provides services to the DOC staff in support of their research, technical, and administrative programs. DOC agencies serviced by OFPM include but may not be limited to: The National Oceanic and Atmospheric Administration (NOAA), The National Telecommunications and Information Administration (NTIA), and The National Institute of Standards and Technology (NIST).

1.3 HOURS OF OPERATION

1. DOC hours of operations are from 7:00 AM to 5:00 PM, Monday through Friday, on normal Federal workdays. Should the Contractor desire to work at times other than those specified above, the dates and hours must be scheduled at least 10 working days in advance and approved in writing by the Contracting Officer's Representative (COR).

2. Use the Main Entrance at the DOC site for materials supply and entry for:
    a. Special arrangements may be made through the COR for delivery of material/supplies, and/or overtime beyond these hours or on holidays.
    b. Check-in required at DOC Visitor's Center located at the site main entrance.

3. Employee identification shall include the name, country of citizenship, employer, and green card (if applicable) is needed for each Contractor and Subcontractor's employee working on the DOC site. These shall be submitted to the COR for review 10 days prior to the work date. After hours work will require a hard badge from the General Contractor and a COR/Federal escort 100 percent of time if accessing DOC buildings (see section 3.3). Follow the specific procedures pertaining to security and work hours prescribed by the COR/Security. Contractor vehicles must show current registration and proof of insurance.

4. Department of Homeland Security (DHS) Advisory System:
   Note: The following paragraphs describe the threat levels that may be changes at any time due to national security issues. If a change is issued by the DHS all changes in procedures/directions for entering and performing work at the site will be communicated to the contractor through the COR. Contractor's access to the DOC campus grounds may occasionally be restricted as a result of a change in the Nation's Threat Level when publicly announced by DHS.

5. National Terrorism Advisory System (NTAS)
    a. The National Terrorism Advisory System, or NTAS, replaces the color-coded Homeland Security Advisory System (HSAS). This new system will more effectively communicate information about terrorist threats by providing timely, detailed information to the public, government agencies, first responders, airports and other transportation hubs, and the private sector.
    b. It recognizes that Americans all share responsibility for the nation's security and should always be aware of the heightened risk of terrorist attack in the United States and what they should do.

6. NTAS Alerts
    a. After reviewing the available information, the Secretary of Homeland Security will decide, in coordination with other Federal entities, whether an NTAS Alert should be issued.

b. NTAS Alerts will only be issued when credible information is available.
c. These alerts will include a clear statement that there is an imminent threat or elevated threat. Using available information, the alerts will provide a concise summary of the potential threat, information about actions being taken to ensure public safety, and recommended steps that individuals, communities, businesses and governments can take to help prevent, mitigate or respond to the threat.
d. The NTAS Alerts will be based on the nature of the threat: in some cases, alerts will be sent directly to law enforcement or affected areas of the private sector, while in others, alerts will be issued more broadly to the American people through both official and media channels.
e. NTAS Alerts contain a sunset provision indicating a specific date when the alert expires - there will not be a constant NTAS Alert or blanket warning that there is an overarching threat. If threat information changes for an alert, the Secretary of Homeland Security may announce an updated NTAS Alert. All changes, including the announcement that cancels an NTAS Alert, will be distributed the same way as the original alert.
f. Contractor coordinates procedures with COR.
g. Threat Levels:
    i. Low Condition (Green): A Low Condition is declared when there is a low risk of terrorist attacks.
        1. Federal departments and agencies will implement general and agency-specific Protective Measures to assure that all facilities and regulated sectors take reasonable measures to mitigate vulnerabilities to terrorist attack.
        2. Contractor coordinates procedures with COR.
    ii. Guarded Condition (Blue): A Guarded Condition is declared when there is a general risk of terrorist attacks.
        1. In addition to Protective Measures taken in previous Threat Condition, Federal departments and agencies will implement general measures in addition to agency-specific Protective Measures to check communications with designated emergency response or command locations, update emergency response measures, and provide public with information.
        2. Contractor coordinates procedures with COR.
    iii. Elevated Condition (Yellow). An Elevated Condition is declared when there is a significant risk of terrorist attacks.
        1. In addition to Protective Measures taken in previous Threat Conditions, Federal departments and agencies may develop protective measures for the DOC campus grounds such as increased surveillance, coordinating emergency plans as appropriate with nearby jurisdictions, and implementing contingency and emergency response plans.
        2. Contractor coordinates procedures with COR.
    iv. High Condition (Orange): A High Condition is declared when there is a high risk of terrorist attacks.
        1. In addition to Protective Measures taken in previous Threat Conditions, Federal departments and agencies will implement general and agency-specific Protective Measures for the DOC campus grounds.
        2. At this Threat Level the Government may restrict access to site and facility to essential personnel only.
        3. Contractor coordinates procedures with COR.
    v. Severe Condition (Red). A Severe Condition reflects a severe risk of terrorist attacks.
        1. Under most circumstances, the Protective Measures for a Severe Condition are not intended to be sustained for substantial periods of time. In addition to Protective Measures for previous Threat Conditions, Federal departments and agencies may close Public and Government sites and facilities as a protective measure.
        2. Contractor coordinates procedures with COR.

1.4 UNFORESEEN FACILITY CLOSURES

1. When an unforeseen facility closure occurs, the Contractor will either reschedule the work or perform the work on the next regularly scheduled workday. If the facility is closed, the contractor is not to be on-site unless notified by the contracting officer they are essential. Contractor personnel may be required by the COR to remain on site to handle problems as a result of a closure, e.g., severe weather, national emergency, changes in national security levels.

2. Information about the status of the site may be obtained by calling the switch board at 303-497-3000. Usually the day's site status is available by 6 AM MST. Contractor can verify site status and sign up for email and text site notifications at the following website:

http://www.boulder.doc.gov/cgi-bin/docstatus.cgi

1.5 SANITARY FACILITIES

Toilet rooms in existing buildings may be used by the Contractor when performing work inside DOC owned buildings at the discretion of the COR. When toilet rooms in existing buildings are not available or work is being conducted outside the Contractor shall:

1. Provide temporary toilet facilities for use of all workmen.
2. Provide the COR with keys to all temporary structures.
3. Maintain toilet facilities in a clean, sanitary condition.
4. Provide adequate supplies of toilet paper, cleaning and other required items.
    a. Holding tanks for the collection of sanitary waste will not be permitted.
    b. Contractor shall install all pipes serving these trailers underground with appropriate cleanouts and provide measures to prevent them from freezing.
    c. Piping shall be run in an organized and pre-planned fashion and all piping shall be removed from the DOC campus upon demobilization of the Construction trailer(s) and the sewer main piping shall be restored to its original condition.

1.6 EQUIPMENT AND MATERIAL STORAGE

1. The contractor shall provide and store, in an area designated by the COR, all equipment, materials and required to complete the project. The materials are required to be kept undamaged, clean and dry throughout the duration of the project. The contractor shall also coordinate location of any trash receptacles, if necessary, with the COR and shall keep the construction area free of trash and debris. All equipment, materials and garbage shall be kept in a manner to avoid causing a hazard by accidental shifting, ignition or other hazardous qualities.

2. Prior to the start of work, Contractor shall meet with all Subcontractors to arrange and prepare a plot plan defining staging, storage, field office and traffic areas within lay-down/staging and trailer/parking areas designated.
    a. Submit a plan for COR's approval not less than 10 working days prior to the installation of temporary facilities.
    b. Except as specifically provided, working and storing outside these areas will not be permitted.
    c. Arrange and locate temporary structures and storage to avoid interfering with construction.

3. Within the area designated for its use, provide suitable and sufficient enclosed and covered spaces with raised flooring, to protect materials and equipment from damage by weather, construction work, or other hazards.
    a. Maintain storage and working areas in clean and orderly condition.
    b. Tie down canvas and other coverings sufficiently to be maintained in conditions of high winds.
    c. At completion of contract, contractor shall remove all storage areas and return any disturbed areas to original condition or better.

1.7 ELECTRICITY AND WATER

1. DOC may make available, at no cost to the contractor, electricity and water during the course of this project. The contractor is responsible for making connections to the existing systems. (see FAR 52.236-14, Availably and Use of Utility Services)

2. Electricity: Temporary electrical work shall meet the requirements of NFPA 70-1996 (NEC), Article 305.
    a. Temporary electrical service shall be provided from existing campus power distribution system.
        i. If the contractor chooses to use Governments electrical service, then the Contractor is responsible for connection, installation, and maintenance of all portions of the temporary electrical service, and connected equipment.
        ii. Verify connection location with COR.
    b. Make arrangements for all equipment, poles, meter, wiring, switches, outlets, etc., for all lighting and power requirements for construction purposes.
        i. Permanent existing campus power distribution system may be used.
        ii. Remove all temporary electrical equipment, when no longer needed.
        iii. Immediately preceding completion, remove and replace all damaged parts of permanent systems.
    c. Temporary electrical power derived from existing site distribution system will be paid for by the Government. Any other power generation shall be paid for by Contractor.
        i. The use of resistive heating devices is prohibited.
        ii. Turn off all lighting and/or equipment not in use.

d. Extend warranty or guarantee period on permanent systems used during construction period so they commence on date of Acceptance.
e. Provide UL approved extension cords where necessary.
f. Contractor to submit after hours lighting plan to COR for approval at least 10 working days prior to installation and use.
g. Provide temporary electrical systems which meet Government approval.
h. Contractor shall remove same from the DOC campus at such time as temporary lighting is no longer required.

3. Water:
   a. During construction/renovation/and maintenance projects the contractor, with the consent/approval of the COR, shall:
      i. Make all arrangements; install equipment, backflow preventer, piping, and outlets for an adequate supply of clean water for construction purposes.
      ii. Existing water lines may be used for temporary service.
         1. Contractor shall tie into existing domestic water line at a location designated and approved by COR, and shall provide all materials, labor and equipment necessary for the connection of all trailers requiring a domestic water connection.
   b. Contractor shall install all pipes serving these trailers underground with appropriate shut-off valves for isolation of each branch connection and shall provide measures to prevent them from freezing.
   c. Piping shall be run in an organized and pre-planned fashion and all piping shall be removed from the DOC campus upon demobilization, and the domestic water main piping shall be restored to its original condition.

1.8 TELEPHONE

Telephone service is not normally available on-site for the Contractor's use.

1.9 ROADS

1. Do not block access to existing facilities.

2. Under no circumstances will egresses be blocked. Any changes in egress shall be approved by COR.

3. Any road closures, full or partial, require notification and approval of COR a minimum of 10 working days prior to closure.

4. Contractor shall be required to remove dirt or debris from road surfaces that are a direct result of contractor operations.

1.10 PARKING

1. Personnel performing the work shall not park in any Government parking lot without prior express consent of COR.

2. Contractor acknowledges that any personnel violating this requirement do so at their own risk and will inform personnel that vehicles may be towed at their expense.

3. Parking in residential and other areas in immediate vicinity of Project site is prohibited.

4. Contractor to keep all personnel, vehicles and operations on Project site within designated limits.

5. If authorized by the COR, temporary parking areas shall be surfaced with crushed stone and maintained by the Contractor to prevent deterioration (ruts, mud, etc.) as approved by the COR.
   a. Direct personnel performing the work to use access to site designated by COR.
   b. The Contractor shall provide parking and transportation for Contractor personnel off the Project site if designated onsite parking areas are inadequate for Contractor needs.

1.11 CONTRACTOR'S USE OF PREMISES

1. Contractor must coordinate all field investigations with COR prior to accessing ceiling/interstitial spaces. Fire alarm system may need to be impaired.

2. Leave undisturbed as much of the site as practical.
    a. Land disturbed but not designated for disturbance on the drawings or by the Government shall be restored to its original condition at Contractor's cost. (See FAR 52.236-9, Protection of Existing Vegetation, Structures, Equipment, Utilities and Improvements).
    b. Leave root zones of trees undisturbed.
    c.
3. Provide warranty and/or guarantee on permanent items and systems used during construction, commencing on date of acceptance, for specified duration(s). (See FAR 52.246-21, Warranty of Construction)

4. Contractor acknowledges that there is wildlife on the Project site. Take all necessary measures to ensure that all such wildlife is protected from execution of the work. Prohibit hunting of any sort.

5. Investigation Equipment: Site investigations and field work may require the use of a stepladder and/or other investigative equipment (measuring tapes, flashlights, screwdrivers, etc.). Contractors shall provide their own stepladders and/or investigative equipment. The stepladders shall be Type 1 (industrial) or Type II (commercial).

6. Operating Equipment: Do not turn off or adjust any equipment without prior authorization from the COR. Work carefully around rotating equipment. Loosely fitting clothes (neckties, dangling jewelry, etc.) shall not be worn when near rotating equipment. Long hair shall be secured. Some equipment may start automatically. Take care when working around this equipment. (NOTE: It may become necessary to implement appropriate lock-out, tag-out procedures to control hazardous energy sources to ensure that personnel do not activate equipment during investigation, servicing, or maintenance).

7. Mechanical/Electrical Rooms: Obtain authorization from the COR prior to working in electrical and/or mechanical rooms. Lock rooms after use. Do not operate any valves, switches, or breakers without prior approval from the COR. Adhere to all OSHA Lock-Out, Tag-Out Regulations (29 CFR 1910.147).

8. It is the prime contractor's responsibility to ensure Contractor employees and Subcontractors become familiar with and obey all DOC site regulations, including fire, traffic, and security regulations. All personnel employed on the site shall keep within the limits of the work (and avenues of ingress and egress) and shall not enter any restricted areas unless required to do so and are cleared for such entry. The Contractor's vehicles and equipment shall be conspicuously marked for identification.

PART 2: SAFETY

2.1 OCCUPATIONAL HEALTH AND SAFETY

1. Contractor(s) acknowledges that by working at the National Institute of Standards and Technology (NIST), a Federal Government facility located in Boulder, Colorado, they are subject to the jurisdiction of the U.S. Department of Labor's Occupational Safety and Health Administration (OSHA). The Regional Office has authority to assign a Compliance Safety and Health Officer (CSHO) to conduct safety and health inspections regarding the Contractors operations at the NIST Boulder site. In addition to OSHA oversight, the Contractor is required to follow NIST health and safety instructions and the Office of Facilities and Property Management (OFPM) safety policies and procedures applicable to their operations.

2. **CITATIONS AND NOTICE OF PENALTY FOR NONCOMPLIANCE WITH OSHA STANDARDS**
    a. To protect NIST's interests, the Contractor acknowledges that NIST may conduct periodic safety, health and environmental inspections to assure that staff and property are not at risk of injury or damage; also, from Citations issued by OSHA for multi-employer worksites.
    b. The Contractor is required to implement and manage a job specific Safety, Health and Environmental plan. NIST will review and evaluate the Contractor's Safety, Health and Environmental plan to assure that it is site specific for the work being performed. The Safety Plan Requirements Checklist in Appendix A requires a site and job specific plan to be developed and submitted to the government for review and acceptance. No site work may be started until the Site-Specific Safety Plan is accepted. 2. In the event of an onsite emergency, first call 303-497-7777 or 7777 from a campus phone; and then call 911. The Contractor will report the following details: Location (NIST), name, building name or number, room number, and a brief description of the emergency.

2.2 CODES, ORDINANCES, REGULATIONS

Contractors are required to comply with all applicable codes, ordinances and regulations as stated in this document. Contractor to use version below, or most current version adopted by Boulder with amendments in effect on the date of the solicitation.) These include, but may not be limited to:

1. Code of Federal Regulations (CFR):
   a. OSHA General Industry Safety and Health Standards (29 CFR 1910);
   b. OSHA Construction Industry Standards (29 CFR 1926);
   c. National Emission Standards for Hazardous Air Pollutants (40 CFR 61);
   d. Resource Conservation and Recovery Act standards for hazardous waste disposal (40 CFR 260-281); and
   e. Toxic Substance Control Act rules related to asbestos and PCB's (40 CFR Part 761).
2. National Fire Protection Association (NFPA):
   a. NFPA 30 – Flammable and Combustible Liquids Code;
   b. NFPA 101 – Life Safety Code;
   c. NFPA 45 – Fire Protection for Laboratories using Chemicals;
   d. NFPA 54 – National Fuel Gas Code; and
   e. NFPA 70 – National Electrical Code.
3. Life Accessibility Standards (USFAS).
4. American National Standards Institute (ANSI) A117.1 – American National Standard Specifications for Making Buildings and Facilities Accessible and useable by the physically handicapped, latest edition.
5. Colorado Department of Public Health and Environment Regulations, Chapter 8.
6. State of Colorado Hazardous Waste Regulations 6 CCR 1007-3
7. Industrial Discharge Permit (2010-1)
8. Municipal Separate Storm Sewer System (MS4) Permit COR042002
9. Construction Permit (Air Pollutant Emissions) 09BO0159
10. Air Quality Control Commission regulation; Number 3, 5 CCR 1001-5
11. NIST ESO Site Security Specifications
12. International Building Code – IBC 2012
13. International Energy Conservation Code – IECC 2012
14. International Fuel Gas Code – IFGC 2012
15. International Mechanical Code – IMC 2012
16. International Fire Code – IFC 2012
17. International Plumbing Code – IPC 2012
18. International Electric Code Administrative Provisions – IEC 2012
19. US Department of Energy Regulations – 10 CFR 435
20. US Environmental Protection Agency Regulations – 40 CFR
21. State of Colorado Department of Public Health & Environment Regulations for Air Quality Control of Stationary Sources
22. State of Colorado Department of Labor and Employment Division of Oil & Public Safety – C.R.S. § 9-4-101
23. U.S. Army Corps of Engineers Safety and Health Requirements Manual, EM 385-1-1 (applies to fall protection only)

2.3 RESPONSIBILITY AND LIABILITY

1. The Contractor is to comply with all applicable regulations pertaining to the health and safety of their personnel and subcontractor personnel during the execution of the project. The Contractor will hold the Government harmless from any liability for any action incurred during the execution of the project by their personnel and subcontractor personnel, which results in property damage, illness, injury and/or death.

2. The Contractor is responsible for locating all underground and building utilities prior to starting work. Contractor is to stop excavation and notify COR immediately if any unexpected materials, such as utilities or concrete are encountered underground.

3. Contractor is responsible for locating all utilities 18" either side of the marked utility and shall manually locate any utility prior to mechanical excavation. If identified utilities are not located within the 36" area, the contractor shall notify the COR immediately.

4. The Contractor is responsible for confirming all dimensions and locations in the field.

2.4 PRE-CONSTRUCTION/SAFETY MEETING

The Contractor may be required to participate in a Preconstruction Conference (see FAR 52.236-26) prior to the start of work for the purpose of reviewing NIST and OFPM site safety regulations, policies and procedures; and to discuss contractual requirements. The Contractor will submit a copy of their Site-Specific Safety Plan to the Contracting Officers Representative (COR) prior to the scheduled Pre-Construction meeting for review, comment, and approval. NTP will not be issued until Site-Specific Safety Plan is approved.

2.5 SAFETY INSPECTIONS

1. The Government will conduct safety inspections of work performed by the Contractor. Unsafe conditions and/or practices by the Contractor will be brought to their attention for correction. The COR or Contracting Officers Safety Representative (COSR) has the authority to “STOP WORK” until the unsafe site or unsafe condition is deemed safe by the COR or COSR and then work can continue. If the Contractor is in non-compliance with safety rules and regulations the COR/COSR will notify the Contractor’s site superintendent/safety point of contact verbally and in writing for immediate corrective action. Immediate corrective action is required, and a written report of corrective action(s) taken is required to be returned to the COR/COSR.

2. If Contractor personnel are removed from the project as the result of unsafe actions or conduct the Government may, at any time, require a meeting with the Contractor’s management and safety personnel to discuss corrective actions.

2.6 INCIDENTS ON THE JOB

1. The Contractor is responsible for keeping the Government informed of all incidents during the execution of the project. If an on-the-job incident occurs during the execution of a project at the NIST Boulder site, the Contractor shall immediately notify the COR by phone and will submit a preliminary incident report to the COR no later than two (2) calendar days after the incident occurred. The incident report shall contain date of incident, type of incident (injury, fire, property damage, spill, etc.), a brief description of the accident, the cause of the incident, lessons learned, and corrective actions taken to prevent it from reoccurring. The contractor shall submit a final report including all the information listed above as well as any additional information as required by the CO and/or COR within seven (7) calendar days after the incident occurred.

2. The contractor is responsible for correcting any damage caused to the NIST Boulder site and field sites as a result of their construction work and returning the site to previous conditions at no cost to the Government.

3. A written report summarizing incidents for the previous month will be sent to NIST via email not later than the 5th working day of the month. The report must contain the following information:
   a. NAICS Code
   b. Total number of man-hours worked in the month
   c. Injury and illness experience for the month
   d. Near miss incidents for the month
   e. Fire or property damage incidents

4. If deemed necessary, the Government reserves the right to conduct a full investigation of all Contractor operations associated with the incident(s).

2.8 PROPER WASTE DISPOSAL

1. The Contractor will remove all materials that are not an integral part of the project, including paint, chemicals and debris in accordance with all applicable Federal, State, and/or Municipal laws, regulations, or requirements. Materials will not be

disposed of by flushing them into a drainage system. When hazardous materials from the performance of this contract are disposed of off-site, submit copies of permits from applicable Federal, State, or Municipal authorities, manifests and/or certificates-of-destruction to demonstrate that the material has been disposed as per all applicable regulations.

2. The contractor will coordinate the location of the any dumpsters provided by the contractor for the removal of waste during the project with the COR while developing the staging plan.

3. If dumpster is to be located within 100 ft of a building, the contractor is required to secure all dumpster openings with covers and locks. This shall be part of the staging plan submittal to be reviewed by NIST Security and approved by COR.

2.9 STORM WATER PROTECTION/COMPLIANCE

1. Contractors are responsible for cleanup of sediment that erodes from a job site and may be held financially responsible for removal of sedimentation from streams or storm drains.

2. Contractors must protect streams, ditches, basins, drainage swales and other water bodies from deposition of sediment from sites on which they are working.

3. Soil may not be stockpiled on paved surfaces without using best management practices such as first laying down plastic and placing straw barriers on the down gradient side of the stockpile.

4. Disturbed soil must be stabilized after 14 days have passed without work being performed in that location. Temporary stabilization is acceptable for sites that are intermittently worked.

5. Soil piles remaining after completion of work must be removed before the contractor leaves the site or within 14 days of completing work whichever is earlier.

6. All disturbed soil must be stabilized before work will be accepted as complete. Ineffective stabilization (does not prevent erosion) must be replaced.

7. Stabilization work may be delayed if the soil is frozen or covered with snow/ice.

8. Any discharge to the storm drain, other than the ones listed below, is prohibited under the Municipal Separate Storm Sewer System (MS4) permit issued to the Department of Commerce Boulder Laboratories by the U.S. Environmental Protection Agency (EPA). Note: Many of these permitted discharges do not occur at the DoC Boulder Laboratories.

    a. Discharges authorized by a separate NPDES permit;
    b. Discharges in compliance with instructions of an On-Scene-Coordinator pursuant to 40 CFR part 300 or 33 CFR 153.10(e);
    c. Water line flushing;
    d. Landscape irrigation;
    e. Diverted stream flows;
    f. Rising ground waters;
    g. Uncontaminated ground water infiltration;
    h. Uncontaminated pumped ground water;
    i. Discharges from potable water sources;
    j. Foundation drains;
    k. Air conditioning condensate;
    l. Irrigation water;
    m. Springs;
    n. Water from crawl space pumps;
    o. Footing drains;
    p. Lawn watering;
    q. Individual residential car washing (not including construction equipment);
    r. Flows from riparian habitats and wetlands;
    s. De-chlorinated swimming pool discharges;
    t. Street wash water;
    u. Power washing where no chemicals are used;

v. Roof drains;
w. Fire hydrant flushing;
x. Non-storm water discharges resulting from a spill which are the result of an unusual and severe weather event where reasonable and prudent measures have been taken to minimize the impact of such discharge;
y. Emergency discharges required to prevent imminent threat to human health or severe property damage, provided that reasonable and prudent measures have been taken to minimize the impact of such discharges; and
z. Discharges or flows from firefighting activities occurring during emergency situations.

9. Construction projects greater than one acre in area or part of a larger development plan greater than one acre in area must be permitted through the EPA electronic notice-of-intent (eNOI) system. Stormwater management on projects required to be permitted under the eNOI system must comply with the EPA Construction General Permit (CGP). See below for the permitting process.
    a. A CGP-compliant Stormwater Pollution Prevention Plan must be submitted to NIST for review. The MS4 permit requires compliance with the Best Management Practices (BMPs) specified in the Urban Drainage and Flood Control District Criteria Manual. http://www.udfcd.org/downloads/down_critmanual.htm
        i. After NIST reviews and approves the SWPPP, the contractor shall submit the NOI.
        ii. The SWPPP must conform to the requirements of the Construction General Permit. Note: Revisions will be required if the SWPPP does not comply with the CGP.
        iii. Guidance for developing SWPPPs that conform to EPA requirements may be found at: http://cfpub.epa.gov/npdes/stormwater/swppp.cfm.
    b. The contractor must register on the EPA Central Data Exchange (CDX) in order to electronically file the eNOI. To register, go to https://cdx.epa.gov/SSL/CDX/terms.asp?Referer=registration&RetrievalKey=. For more information on the NOI system, go to http://cfpub.epa.gov/npdes/stormwater/enoi.cfm.
    c. The contractor shall submit the electronic NOI (eNOI) through the CDX. Submission of a hardcopy NOI is permitted by EPA but will significantly delay the review process.
    d. The contractor shall wait 14 days for approval and/or provide any information requested by EPA. A lack of response by EPA by the end of that 14 day period is de facto approval of the contractor's eNOI.
    e. The contractor may begin land disturbing work when the NOI is approved.
    f. Upon completion of the project, the contractor shall submit a notice-of-termination (NOT) through the EPA CDX. The following requirements must be met prior to submitting the NOT.
        i. As-built drawings of the stormwater BMPs shall be provided to the COR.
        ii. Operations and maintenance (O&M) manuals for all stormwater BMPs shall be provided to the COR.
        iii. Final stabilization shall be complete per the requirements of the CGP.
        iv. All contractor-owned materials that may be considered potential pollutants shall be removed from the site.
        v. All construction BMPs shall be removed.
        vi. Items c and e may be waived if the contracting officer executes an agreement for the contractor to transfer the site to another contractor.

10. Development or redevelopment projects with a footprint greater than 5000 square feet are required to comply with Section 438 of the Energy Independence and Security Act of 2007 (EISA) and include stormwater BMPs that produce the predevelopment hydrology of the site, effectively resulting in a runoff coefficient comparable to grassland (approximately 0.4). This includes both the quantity and quality of runoff. This applies to the design and construction of new facilities or plans for redevelopment of facilities at the end of the life cycle. This requirement does not apply to repairs made to operable facilities such as roof repairs and resurfacing of roadways or parking areas. This requirement may also be excluded from a project if a separate project or phase of a large project is planned to meet the requirements of EISA Section 438. The COR and contracting officer will make this determination.

11. The dewatering of construction sites is covered under the MS4 permit if the groundwater to be removed meets the water quality standards in 5 CCR 1002-31 and 38 and has a total suspended solids content of less than 30 mg/L. There is no known or suspected contamination of groundwater at the DoC Boulder Labs. Discovery of suspected contamination (unusual odor, color, etc.) must be reported to the COR immediately. Dewatering and discharge to the storm sewer system may not continue until sampling and analyses have been performed and compliance with 5 CCR 1002-31 and 38 have been verified. Plans for construction projects involving excavation at or near the water table must include a plan to comply with these requirements. Portable tanks may be necessary to contain water until analytical results have been received. The use of portable tanks to contain water prior to receipt of analytical results may be required. At minimum, settling of suspended

solids will be necessary. Note: Dewatering, treatment and discharge of contaminated groundwater requires a separate permit. Obtaining a dewatering permit from EPA may take up to six months.

12. On construction projects required to be permitted under the EPA eNOI system, designs must be reviewed by a landscape architect or landscape planner to ensure that revegetation will meet the requirements of the CGP and control the growth of noxious weeds. All other construction projects must comply with NIST/DoC Boulder Labs requirements for revegetation.

13. Failure to comply with applicable regulations or permit terms may result in work stoppage and/or other contractual actions.
    a. On construction sites regulated under the CGP and eNOI system, contractors must begin repairs on deficient (improperly installed, missing or ineffective) stormwater BMPs on the day on which the deficient condition is first observed and complete repairs by the end of the following day unless replacement of the BMP is required. The CGP allows 7 days for replacement of BMPs. Any failure to adhere to this requirement must be noted in the SWPPP log and reported to the COR.
    b. For projects not regulated under the CGP and eNOI system, if deficient stormwater protection exists at the fault of the contractor, the contractor shall make repairs according to the schedule in Item 1, aside from the requirement to document in the SWPPP log.
    c. If corrective actions are the responsibility of the government and require separate contract actions, acquisition of contractor services and/or procurement of materials will be initiated by NIST AMD within 72 hours of receipt of a defined requirement and evidence of availability of funds.

2.10 FIRE PREVENTION/PROTECTION

1. The contractor will comply with all fire prevention recommendations of the International Fire Code - 2003 edition, and the National Fire Protection Association (NFPA) Regulations which are incorporated herein by reference. The Contractor will also comply with the applicable recommendations of NFPA Chapter 10 entitled “Portable Fire Extinguisher” and for each area of construction where combustible materials and/or flammable liquids are present; the Contractor must provide their own fire extinguishers. The fire extinguishers should be located in easily accessible places where the work is being performed. The Contractor will maintain fire ratings in walls by ensuring wall penetrations made by the Contractor are filled with a UL approved fire stopping material and applied consistently with manufacturer recommendations.

2. The Contractor shall obtain a "**Hot Work Permit**" from the COR for use of open flame devices, such as blow torches, portable furnaces, tar kettles or gas and electric welding and cutting equipment. Hot Work permits should be requested at a minimum of 48 hours prior to the start of work. The Contractor will be held liable for any loss of Government property damage, injury and loss of life due to fire. Failure to comply with fire prevention measures as described by the terms of this statement of work may result in the Contractor paying penalties for damages caused by negligence. Duration of hot work permits shall not exceed one week – in some circumstances upon written request to the COR a Hot Work Permit may be issued on a weekly basis (i.e. Monday morning to close of business on Friday). Prior to the commencement of work, fire watches and the signed Hot Work Permit shall be posted in the work area. All Hot Work Permits must be returned to the COR once hot work is complete or the permit has expired.

3. The Contractor is bound by Executive Order 13058, Protecting Federal Employees and the Public from Exposure to Tobacco Smoke in the Federal Workplace. Smoking is prohibited within 25 feet of building entrances/exits/roll-up doors and building HVAC systems fresh air intakes. Contractor smoking areas must be designated for long term construction and the use of NIST provided smoking areas is required while on NIST property.

2.11 EMERGENCY RESPONSE PROCEDURES FOR THE UNSCHEDULED MINOR DISTURBANCES OF ASBESTOS-CONTAINING MATERIAL (ACM)

1. ***Prior to commencement of the contract, the COR (or Safety, Health and Environment Office personnel) will review locations of known asbestos containing materials (ACM) with Contractor. However, there may be unidentified ACM in the work area.*** All work, including contact with and handling of hazardous materials, the disturbance or dismantling of structures containing hazardous materials and/or the disposal of hazardous materials will comply with the applicable federal, state and municipal safety and health requirements (40 CFR 761,40 CFR 61, CDPH&E Chapter 8), including training, certifications, medical surveillance, sampling and inspections. The DOC Boulder – Engineering, Maintenance, Safety &

Support Division Emergency Response Procedures for the Unscheduled Minor Disturbances of Asbestos-Containing Material (ACM) are outlined below.

2. When an identified or suspect VAT/mastic, TSI, surfacing, or miscellaneous ACM is identified or undergoes an unscheduled or inadvertent minor disturbance resulting in damage to or exposure of the material, the individual causing or discovering the disturbance will immediately implement the following actions:
   a. **Do not** attempt to clean up or repair the damage.
   b. Notify occupants in the immediate area regarding the occurrence and vacate the area as quickly as possible to prevent any further disturbance of the material.
   c. Notify occupants in any immediately adjacent areas regarding the occurrence.
   d. Secure the immediate area where the damage or spill occurred by closing and locking doors, if possible, or by initiating any other actions that would prohibit entry to the area.
   e. Contact the COR immediately after securing the area to provide notification regarding the damage or spill. If the COR cannot be immediately contacted, call the OFPM Service Desk (303-497-3191). Include the following information, if possible:
      i. Building number/room number or description of the area where the damage or spill occurred;
      ii. Description of the material that was damaged or spilled and a brief summary of how it occurred;
      iii. Approximate time of the occurrence;
      iv. Name and telephone number of the individual causing or discovering the damage or spill;
      v. Description of the methods implemented to secure the area from access of personnel and/or further disturbance.
   f. OFPM will dispatch trained and certified emergency response personnel to the area of the damage or spill to conduct the proper emergency clean-up or repair actions as soon as possible.
   g. Entry to the immediate area and any adjacent areas designated by the OFPM personnel shall be strictly prohibited until the area is deemed uncontaminated and safe for re-occupancy.
   h. The COR shall arrange to perform tests to determine if the material is hazardous.
   i. If the material is found hazardous and additional protective measures are needed, a contract change may be required, and the Contractor will submit a change order request to the COR for review.

PART 3: PERSONAL/PHYSICAL SECURITY & PROPRIETARY INFORMATION

3.1 GENERAL

The Contractor/subcontractor will be responsible for all security aspects of work performed under this contract. Security is an individual responsibility.

3.2 CONTRACTOR'S SECURITY POINT OF INTEREST

The Contractor will provide to the COR the name, address and telephone number of a "Point of Contact" for receipt or notification of security related matters at or before the Preconstruction Conference.

3.3 PERSONAL SECURITY & EMPLOYEE IDENTIFICATION

1. The Contractor will fill out and require each of the Contractor's/subcontractor's employees performing contract work to fill out, all forms that may be necessary for security as specified by the COR, the CO, and Department of Commerce Security Department. These forms will be submitted to the Government via a secure website provided by the COR. **THIS INFORMATION IS PERSONAL IDENTIFIABLE INFORMATION AND SHALL NOT BE EMAILED**. Two ID's are required, one of which must be a current picture ID and Green Card if required. The Contractor shall submit these forms as soon as possible after award but no later than thirty days following Notice to Proceed (NTP).

2. DoC issued identification must be worn above the waist in PLAIN SIGHT at all times. Allowing other employees to use the security ID for any reason other than access to the NIST site is cause for immediate removal from the site. A successful security background investigation must be completed prior to the issuance of formal ID's and keys.

3. The contractor shall provide to the COR the names of individuals needing access so they can be scheduled for the security training and review of their paperwork. All paperwork must be filled out prior to the scheduled time for submittal of the

forms. **Any Contractor Employee attempting to obtain a badge without the prior scheduling with the COR will be turned away.** Each construction area requires at least one Contractor employee to obtain a Personal Identity Verification (PIV) badge. For each construction area, a PIV badged Contractor employee shall be on-location daily during the duration of the project. The Contractor shall make an effort to obtain these identification badges within the first thirty days following NTP.

4. **ALL individuals' names needing access for work outside of normal business hours work MUST be provided to the COR, in writing, 10 working days prior to the scheduled work. Approval of request, by NIST Security, is required prior to after-hours access.**

5. **The COR MUST have, in writing, twenty-four hours' notice of deliveries by vendors. The company making the delivery, the driver's name, and the location the delivery is to be made MUST be provided. Access into the buildings WILL NOT be allowed by delivery personnel. Any unscheduled deliveries may not be allowed on site.**

BADGE TYPE	ACCESS	REQUIREMENTS (minimum)	DURATION	NOTES
29 Day Paper	Site only, escort required inside of all buildings	Visitor Registration system entry	29 days	Advance notice required, not intended for daily access
179 Day Paper	Site only, escort required inside of all buildings	DN52 form required, company name, Prime or sub-contractor employee	179 days	Turned in on Wednesdays, takes one week to process
Site badge (less than 6 months)	Building access, no escort required	NAIS form, fingerprints	Up to 6 months	
Site badge (more than 6 months)	Building access, no escort required	NAIS form, fingerprints, background check	Over 6 months, until PIV badge is issued	This badge is issued for individuals awaiting a PIV badge
PIV badge	Building access, no escort required	NAIS form, fingerprints, background check, e-Qip	Up to 5 years with annual renewal by COR	**THIS BADGE PROCESS TAKES APPROXIMATELY SIX MONTHS**

*This is a brief badge summary, subject to change. Confirm current requirements with COR.

PART 4: PROJECT PROCEDURES

4.1 GENERAL DESCRIPTION

1. The contractor will supply all management, supervision, labor, materials and equipment required for the successful completion of the project for the entire length of this contract. The contractor is required to have an authorizing employee on-site at all times while work is being performed.

2. This project must be completed within the contracted period of performance (per FAR 52.211-10). With initial proposal and prior to commencement of work the contractor must provide detailed project schedule in Microsoft Project or other scheduling software to the COR for acceptance. Final schedule shall be agreed upon after NTP at post-award meeting.

3. Notify the COR of the work progress and any required inspections three working days in advance of all inspections. Upon substantial completion, request a punch list inspection with the COR and CO five working days in advance. COR will submit to CO and Contractor a written list of items requiring completion or correction. Contractor will respond to list with an anticipated schedule of completion of outstanding items.

4. The Contractor shall also be responsible for all materials delivered and work performed until completion and acceptance of the entire work, except for any completed unit of work which may have been accepted under the contract.

5. **The coordination of activities between the COR, the contractor and the building occupants is critical in an effort to minimize disruption of the research and damage to ongoing experiments. The notification of building occupants is the COR's responsibility.**

6. Entrance into laboratories is normally not required. If entrance into the area is required the contractor shall contact the COR or his/her designee a minimum of 5 working days prior, and arrangements with the owner/occupant to ensure work to be performed will not disrupt or damage technical experiments/projects in the specified area will be coordinated.

7. When special entrance requirements are deemed necessary the COR and or his/her designee will make arrangements for contractor personnel to have the necessary briefings and training prior to entrance into the specified area.

8. To the extent possible, the contractor must assign specific staff to this site and this contract.

4.2 PROJECT DOCUMENTATION PROCEDURES

All submittals shall be submitted to the COR in a timely manner to allow ten working days for review. The return of disapproved submittals and their resubmission is not a basis for time extension. All Requests For Information (RFI) shall be submitted in writing and will be returned by the COR in a timely manner. Notify the COR via RFI with any conflicts associated with this Scope of Work, drawings and any appendixes in a timely manner.

4.3 PERMIT REQUIRED OPERATIONS DURING CONSTRUCTION OR MAINTENANCE ACTIVITIES

1. To ensure safety as a priority, to ensure individuals performing certain operations have the appropriate training and equipment, and to prevent unnecessary utility outages DOC has the following permits which are required when performing tasks/functions to support this contract/project.

2. The policies and procedures can be obtained from the Government COR/Operations Supervisor and are required prior to the work being performed.

3. The activities requiring permits are:
    a. Roof Access Permit (48-hour advance notice to COR)
    b. Penetration Permit - walls, floors, ceilings, and roofs (48-hour advance notice to COR)
    c. Hot Work Permit (48-hour advance notice to COR)
    d. Fire Protection System Impairment Permit (5 working days advance notice to COR)
    e. Excavation Permit (48-hour advance notice to COR)

4. All organizations performing work on behalf of the DOC are also required to have a current Lock-out/Tag-out policy and all employees must be currently indoctrinated on company procedures. If necessary, OFPM has a policy/procedure that can be obtained from the COR/ Operations Supervisor.

5. Maintain a record set of all inspections, permits or other authorizations necessary for the work to proceed.

4.4 MAINTENANCE OF WORK SITE

Because this facility has 24/7 operation, work sites shall be kept in an organized, safe and clean manner (see FAR 52.236-10). Debris and garbage shall be cleaned at the end of each work day. Any materials or equipment stored in approved areas shall be left in an orderly manner. Regular maintenance of working areas shall be the responsibility of the contractor (ie. outside areas shall be mowed if necessary, hallways shall be swept and kept clear of trip hazards). The Contractor shall perform final cleaning and detailing of work areas at completion of the project (see FAR 52.236-12).

4.5 WORK REQUIRING ADVANCE NOTICE AND COORDINATION

NIST is a functioning laboratory facility operating on a 24/7 schedule. As such, 10 working days' notice to the COR is required for any activity that could potentially impact NIST employee safety, work or facility operation outside of the work area. These activities include, but are not limited to, utility outages; parking lot or road closures; work impacting access to corridors or spaces outside of the work area; loud work such as core drilling, jack-hammering, etc.; work creating vibration; and work that may produce odors. Any outages impacting significant research activities will require a minimum of 10 and up to 25 working days' notice to COR for coordination unless otherwise noted in contract.

Appendix

Roof Access Permit

Penetration Permit

Hot Work Permit

Fire Protection System Impairment

Excavation Permit

Roof Access Permit

Please complete and provide to BSHED prior to accessing rooftops. Refer to the NIST Boulder Roof Access Process policy for additional information. Note: A Project Specific Work and Fall Protection Plan may be required based on the complexity of the project.

Section I. General Information	Request Date:	Requesting Division:	
	Requestor's Name:	Requestor's Phone:	
	Location of Work (Bld & Wing):		
	Name(s) of Personnel Authorized to Perform Work:		
	Project Duration:	Start Date:	Finished Date:
Section II. Operation(s) Conducted	☐ Changing Filters	☐ Electrical Troubleshooting	☐ Reroofing/ Roof Inspection
	☐ Demolition	☐ Equipment Check/Calibration	☐ Rigging
	☐ Drilling	☐ Equipment Install/Repair	☐ Sand/Bead Blasting
	☐ Electrical Installation/Modifications	☐ Painting	☐ Welding/Cutting (Hot Work)
	☐ Other (Describe):		
Section III. Work Tools & Materials	☐ Carts/Dollies	☐ Hand/Power Tools	☐ Mechanical Equipment
	☐ Cranes	☐ Hazardous Chemicals	☐ Pipe Threaders
	☐ Electrical Tools	☐ Instruments	☐ Scaffolding
	☐ Generators	☐ Ladders	☐ Welders/Torches
	☐ Other (Describe):		
Section IV. Safety Precautions	**Control of Hazardous Energy (Lockout/Tagout):** ☐ Electrical Shutdown ☐ Exhaust Shutdown ☐ RF Energy Shutdown ☐ Other:	**Fall Protection:** ☐ Personal Fall Arrest ☐ Personal Fall Restraint ☐ Guard Railings ☐ Warning Lines ☐ Other:	**Fire Prevention:** ☐ Barriers ☐ Detection Sensor Protection ☐ Hot Work Permit ☐ Welding Curtains ☐ Other:
	Safety Training: ☐ Fall Protection ☐ Lockout/Tagout	☐ PPE	☐ Other:
	☐ Other Special Precautions (Describe):		
Section V. Approvals	The following acknowledgements are required prior to safe access:		

Representative	Printed Name and Signature	Date
Division Chief, Group Leader, or COR of Requesting Party		
BSHED Designee		
OFPM Designee (acknowledgement)		
Affected Laboratory Contact (if applicable)		
Affected Laboratory Contact (if applicable)		

Roof Access Permit - Project Specific Work and Fall Protection Plan

This plan shall describe, in detail, the specific practices, equipment, and methods used to protect workers from fall hazards while working on NIST's roofs. The plan shall be updated as conditions change.

General Information	Description of the project or tasks to be performed:		
Section II. Potential Fall Hazards	Identify all fall hazards (check all that apply):		
	☐ Floor/Wall Openings	☐ Ladders	☐ Open-sided Floors/Roofs
	☐ Open-sided Ramps/Runways	☐ Scaffolds	☐ Skylights
	☐ Other (Describe):		
Section III. Fall Controls	Methods of fall protection to be used (check all that apply):		
	☐ Catch Platform	☐ Covers (Floor & Wall Openings)	☐ Engineered Anchors
	☐ Guardrails	☐ Horizontal Life Lines	☐ Personal Fall Arrest System
	☐ Personal Fall Restraint System	☐ Safety Net	☐ Warning Lines
	☐ Other (Describe):		
Section IV. Plan Specifics	**Describe procedures for assembly, maintenance, inspection, disassembly of fall protection system to be used:**		
	Describe procedures for handling, storage, and securing tools, equipment, and materials:		
	Describe methods of overhead protection for workers who may be in, or pass through work area:		
	Describe methods to be implemented for prompt, safe removal of injured worker(s):		

Section V. Training	Employees who received fall protection training on the above site specific fall protection work plan.	
	Printed Name	**Date**
	Name & Title of Trainer:	
Section VI. Additional Information	Provide any additional information or drawings as necessary.	

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NIST S 7101.04

U.S. DEPARTMENT OF COMMERCE
NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY

NIST / OFPM SURFACE PENETRATION PERMIT
WORK PERFORMED USING STANDARD SURFACE PENETRATION METHODS

WORK DESCRIPTION

Scope of Work

Workers	Date	Building Manager / Facilitator or Supervisor

PROJECT EVALUATION

- ☐ A building review and walk down of accessible sides of the surface that will be disturbed has been performed to identify:
- ☐ Have "As-Built" drawings been reviewed? Explain:
- ☐ Are there visible indications that utilities and equipment might be present? Explain:
- ☐ Are there utilities and equipment that might be disturbed during the work activity? Explain / Attach Mitigation Controls:
- ☐ Will dust or debris be a housekeeping issue? Explain / Attach Mitigation Controls:
- ☐ Do noise limits need to be established? Explain / Attach Mitigation Controls:
- ☐ Is a Fire Protection System (FPS) impairment required? Explain / Attach Impairment:
- ☐ Is there asbestos containing material present (ACM), lead based paint (LBP) or silica present? Explain / Attach Survey:

The project will be performed using one of the following standard surface penetration methods. Additional utility locate measures are not necessary. Check the method(s) to be used:

- ☐ Hanging pictures or board where the length of the nail or screw will not fully penetrate the wall material
- ☐ Penetrations into drywall, sheetrock, or similar construction material with the following tools:
- ☐ Drill fitted with a bit stop to prevent penetration beyond the depth of the material
- ☐ Utility knife or other cutting tools where the blade is adjusted to prevent penetration beyond the depth of the material

PROJECT CONTROLS AND INSTRUCTIONS

Comply with the following controls and instructions when making surface penetrations.

X STOP WORK immediately if:

- You encounter unexpected utilities
- Your tools meet unexpected resistance / smoke
- You damage utilities during the work activity
- You encounter Suspect Asbestos Containing Material

- ☐ Use a ground fault circuit interrupter when using corded electrical power tools. Test before use.
- ☐ Do not penetrate into a single layer of drywall, sheetrock, or similar construction material unless both sides can be visually inspected to verify the absence of utilities that may be disturbed.
- ☐ Once initial penetration is complete, use a fiber optic scope, or equivalent technique approved by the OFPM Safety, Health and Compliance Group to verify the absence of utilities prior to proceeding with work.

Surface Penetration Permit Approved by:

Printed Name / Title:	Signature Date:

NISTSafety

Permit No. XXXX

NIST

OFPM

Engineering, Maintenance & Support Services Division

Hot Work Permit

This Hot Work Permit is required for any operation involving open flame, heat producing, or spark creating work. This work includes: welding, cutting, brazing, grinding, soldering, thawing pipe, torch applied roofing, or chemical welding.

SECTION I. Preparation.

Date:________ Bldg./Rm. Number:__________ Hot Work by: NIST Employee ☐ Contractor ☐

Description of work to be performed:____

____

____

Hot Work performed on: ____

Section II. Work Process (check all that apply):

☐ arc welding ☐ open flame cutting ☐ power saw metal cutting ☐ drilling

☐ gas welding ☐ soldering ☐ open flame heating ☐ grinding

Section III. Fire Safety Requirements (check all that apply):

- ☐ Flammable liquids at least 35 feet from edge of hot work zone
- ☐ Combustible liquids at least 35 feet from edge of hot work zone
- ☐ Flammable/combustible solids with an NFPA 704 flammability hazard relocated 35 ft.
- ☐ Electrical tools rated for appropriate Class I or II atmosphere
- ☐ Hot slag provisions required
- ☐ Control of open flames required
- ☐ Floor swept clean of combustibles (35 ft. min.)
- ☐ Work coordinated with others in the area
- ☐ Ductwork blanks required
- ☐ Overhead spark/slag collection required
- ☐ Hand held fire extinguisher required (minimum rating of 2A:40B:C)
- ☐ Fire watch available and on station during hot work and for at least 30 minutes after completion of the last hot work
- ☐ Impairment Permit Required
- ☐ Fumes to be controlled
- ☐ Arc-welding screens installed
- ☐ Fall protection required
- ☐ Ingress/egress planned
- ☐ Barricades required
- ☐ UV/IR detectors impaired
- ☐ Area to be wetted
- ☐ LOTO required
- ☐ CSE Permit required
- ☐ Sprinklers serviceable
- ☐ Sprinklers impaired

Section IV. Approvals/Concurrence. (Sign, print name, date, and phone)

I certify by signing this permit all requirements have been taken as required. When the job is completed, person performing hot work marks the card portion "Complete," and is responsible for returning it to EMSS Safety.

COTR\NIST Supv:______________________ OSHE/EMSS Safety Rep: ______________________

Print Name______________________ ______________________

Telephone Number:______________________ ______________________

Person Performing Hot Work:______________________ Fire Watch:______________________

Print Name ______________________ ______________________

Telephone Number:______________________ ______________________

Permit Expiration: From: Date/Time ______________________ To: Date/Time: ______________________

Emergencies Call 911 and 303-497-7423

NIST BOULDER FIRE PROTECTION IMPAIRMENT PERMIT

Instructions: This form is used to request impairments of the fire alarms, sprinklers, fire hydrants, and water supply/agent supply. The Contracting Officer Technical Representative (COTR) shall submit this completed form prior to doing work in any NIST Building. Complete a separate FIRE PROTECTION IMPAIRMENT PERMIT request for each location if multiple locations are required.

PART I. Initial Request Information (Originator).

Originator: ____ Originators Work Phone #: ____ Originators Cell Phone #: ____
Work to be performed in (Please be specific): Building: ____ Location/Room #: ____
Contractor: ____ Contractors Work Phone #: ____

Description of work: ____

Work Activities Include (Check all that apply):

Dust/Fume Generation ☐ Adding/Removing Fire Alarm Devices ☐ Electrical Outage ☐ Water Outage- Sprinkler or Hydrant ☐

Welding/Soldering/Grinding ☐ Sprinkler Mods ☐ Clean Agent Mods ☐ Demolition/remodeling ☐ Maintenance to FP System ☐

Designated Construction Zone: ☐ /Duration ____

Duration of Impairment:	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday
Date:							
Start Time:							
End Time:							

TYPE OF IMPAIRMENT REQUEST: Planned ☐ Emergency: ☐ Fire Alarm: ☐ Water System: ☐ Clean Agent System: ☐
Send Initial FIRE PROTECTION IMPAIRMENT PERMIT Request to: fireimp-mail@nist.gov.

PART II. Impairment Requirements (Impairment Coordinator).

Requirements (impairment options): ____

Impairment to be verified by: Contractor: ☐ COTR: ☐ IC: ☐

Risk Determination: RAC 1 ☐ RAC 2 ☐ RAC 3 ☐ RAC 4 ☐ RAC 5 ☐

Impairment Coordinator returns this FIRE PROTECTION IMPAIRMENT PERMIT to Originator

PART III. Impairment Verifications.

A. Impairment Verification by Originator:

Impairment Signs Posted: ☐ Impairment Applied: ☐ Verified by (Insert Name): Contractor: ____ /Date: ____
COTR: ____ /Date: ____
IC: ____ /Date: ____

Return this impairment verification FIRE PROTECTION IMPAIRMENT PERMIT to: fireimp-mail@nist.gov

B. System Restored by Originator:

Impairment Removed: ☐ Impairment Signs Removed: ☐
Return this system restoration verification FIRE PROTECTION IMPAIRMENT PERMIT to: fireimp-mail@nist.gov

Impairment Coordinator: ____ /Date: ____
Rev Date 6/2014

Appendix D – Excavation Permit

Excavation Permit Form

Date /Time of Excavation: ___________________________________

Supervisor or COTR over seeing Excavation: ______________________

Contact Information: _______________________________________

Contractor/ Gov. Person performing Excavation: ____________________

Contact Information:____

Project Description/ Work Order #/ Reason for Excavation:

____

____

____

____

____

Site leader for the excavation Single POC is:_______________________

Cell Phone #_________________________

Locates have been performed and inspected by Excavation leader and all safety issues such as shoring, hand digging requirements and tamping are in place. The site leader named above is responsible for all aspects of this excavation and has reviewed with all individuals the safety requirements and the scope of work required.

YES [ ] NO [ ] Date/Initial; _____________________________

I have reviewed this permit request with the COTR/Supervisor and appointed Site Competent Person and approve this excavation:

EMSS Operations Manager/designee: _____________________date_____

2 Copies of Permit are required, 1 - onsite at Excavation 2- on file at EMSS Safety Group.

NIST SAFETY PLAN REQUIREMENTS CHECKLIST

CONTRACT NAME	**NIST Boulder Central Utility Plant, Operations, Maintenance and Repair**
CONTRACT NUMBER	**1333ND21CNB190017**
COR	**Chris McKinney, Chris Colburn, Danielle Swarts**
DATE	
NIST CHECKLIST PREPARER	

All checked items must be included in a written safety plan and be accepted by the government prior to any work starting. All requested documentation must be submitted. Job Hazard Analysis (JHA) for jobs with few task are expected to accompany the safety plan, jobs with multi-task can be submitted sequentially as the job progresses. **RETURN CHECKLIST COPY WITH REQUIRED INFORMATION LOCATION IDENTIFIED IN THE RIGHT MARGIN BELOW**

Written Description Required

YES	NO	N/A	WRITTEN REQUIREMENTS	*Identify location of in submitted Safety Plan.
x			Contractor shall comply with the OSHA General Duty Clause 5(a)1 and 29 CFR 1926 & 29 CFR 1910 as applicable. Additionally the contractor is obligated to comply with USEPA, CDPH&E, NIST Boulder Laboratories Security Office, Office of Safety, Health, & Environment, and the Office of Facilities and Property Management.	
			The following are the minimal requirements that must be covered within the written safety plan:	
x			Company's Health, Safety, & Environmental Policy.	
x			Subcontractor Safety Management	
x			Emergency Response Plan	
x			Hazard Communications Policy	
x			Accident Investigation Policy to include Root Cause Analysis	
			The following must be addressed if it is within the scope of work of the contract, this will include the providing of special licenses/training certificates that employees have completed to be designated as a competent person for the project. **Job Hazard Analysis (JHA) must be submitted prior to any work beginning or a permit is issued.**	
x			**PPE & Life Safety (29 CFR 1926 Subpart E)** site minimum PPE consist of ANSI Approved safety glasses, hardhats, work boots, high vis shirts/vest. Then as required hearing protection, gloves, fall protection gear, and emergency eyewash for corrosive chemicals and concrete work.	

x				**Fire Protection (29 CFR 1926 Subpart F)** to include use/storage of flammable and combustible materials, use of LP Gas, use of temporary heating devices.	
x				**Material Handling (29 CFR 1926 Subpart H)** Site plan for material storage. Copies of training certificates for all operators of material hoist, personnel hoist, aerial lifts. Cranes and derricks require a site and lift plan prior to lift.	
x				**Tools-Power & Hand(29 CFR 1926 Subpart I):** Contractor shall provide information concerning Assured Grounding Policy. Due to the site's purpose the contractor will ensure that all electrical power tools are properly connected to GFCI's.	
x				**Welding/Cutting/Hot Work (29 CFR 1926 Subpart J):** Contractor shall ensure that all welding/cutting/hot work is done in accordance with applicable equipment safety measures. This will include use of screens and/or blanks. Contractor will coordinate to ensure COR has submitted appropriate Impairment permits and request Hot Work Permit from Safety Group. All hot work must have Hot Work Permit posted in area.	
x				**Scaffolding & Ladders (29 CFR 1926 Subpart L):** Contractor shall ensure that the proper ladder is being used correctly. Scaffolding other than rolling type, will be erected by a Competent Person, with scaffold grade planking, guardrails, and contractor will address fall protection measures to be used. Contractor will maintain copies of all daily scaffold inspections	
x				**Fall Protection (29 CFR 1926 Subpart M):** Contractor shall provide a fall protection plan and a fall rescue plan. All floor and wall openings will be guarded with approved guardrails/barricades. All employee's effected by the fall protection plan must have attended training IAW 1926.503(b) and provide copies of training certificates. Contractor must ensure that a licensed PE has designed all fall protection systems. For roof work, contractor must obtain a Roof Permit from the OFPM Safety Office.	
		x		**Excavation & Trenching (29 CFR 1926 Subpart P):** Contractor must perform Utility Locates prior to beginning. Contractor will provide site drawing of proposed excavation/trench. Excavation/trench greater than 4ft deep must be protected from cave-in, contractor to provide method of protection. If greater than 20ft in depth protection	

				system/method must be designed by a licensed PE. Contractor must perform daily inspection of excavation/trench, to included hazardous atmosphere if they exist. Contractor must provide certificate of training for Competent Person and also a rescue plan in the event of a cave-in. Contractor will work with COR to secure an excavation permit from EMSS Safety Group. All excavation/trench's must be protected by fencing or barricades.	
x				**Electrical (29 CFR 1910 Subpart S, 29 CFR 1926 Subpart K, NFPA 70E)** Contractor shall provide training documentation in accordance with the standards outlined in NFPA 70E, Article 110.6E. Contractor shall ensure workers have appropriate PPE to perform assigned work.	
x				**Toxic & Hazardous Substances (29 CFR 1926 Subpart Z) (29 CFR 1910 Subpart Z)** Contractor shall comply with standards set forth in all sections of 1910.1001, 1926.1101, all applicable Appendix's and CDPH&E Reg 8 .	
x				**Permit Required Confined Space (29 CFR 1910.146):** Contractor will provide training certificates for all entrants, attendants, rescue personnel. Contractor will provide permit entry system/methods, rescue plan, ventilation methods, additional PPE to be used, LOTO methods used, & air monitoring methods used. Contractor will provide completed copy of permit after work is completed. A copy of the permit must be posted at entrance will work is ongoing.	
x				**Control of Hazardous Energy (Lockout/Tagout)(29 CFR 1910.147 & 1926.417):**Contractor must adhere to NIST supplemental LOTO Procedures for existing equipment. COR & Contractor will acquire procedures from the Safety Group prior to work beginning. Contractor will provide training certificates for all personnel performing LOTO work. If new equipment is being installed, contractor will utilize their own LOTO procedure until equipment is tied into NIST services.	
x				**Lift Plan -** submit lift plan that provides at a minimum: visual representations of the area(s) being hoisted from/to; boom length or boom angle; radius; gross capacity; net load weight; travel path (i.e. proximity to immovable objects, powerlines, workers in the area)	
				General Safety Measures:	
x				**Environmental Protection:** If required Contractor will obtain Storm water Permit and	

				submit a SWPP for acceptance. All waste generated will be disposed of properly. Contractor is responsible for all waste, general & hazardous. Contractor will perform general housekeeping IAW 29 CFR 1910 & 1926.	
				Contractor shall take all necessary precautions to prevent injury to the public, building occupants, and property. This will include the use of hazard signage, fences, barricades, and any other approved methods. Contractor shall not create any obstructions when storing materials or performing assigned work. Access & egress must be maintained at all times for occupants and employees. Contractor shall provide a site drawing for locations used to store materials & equipment. This will include the use of secondary containment for hazardous materials. MSDS's must be present and all containers in use must be properly labeled with associated hazards. **When requested the contractor shall provide copies of air and noise monitoring and specialty toxic material usage (Stainless steel welding, etc)**	
x				NIST utilizes a **3-Stikes & Out** Policy in regards to safety violations. If safety violation is observed OFPM Safety office and Contractor Safety manager will correct (**Strike 1**). If same employee has another safety violation OFPM Safety office & Safety Manager will again correct (**Strike 2**). If that employee has another safety violation (**Strike 3**) the OFPM Safety office will have the employee removed from site and have employees badge privileges will be revoked by DOC Police.	

Site Specific Permits Needed

- Impairment Permit
- Hot Work Permit
- Roof Permit
- Excavation Permit
- Penetration Permit
- Storm water Protection Permit
- Energized Electric Work Permit
- Tower Climbing Cert's

Periodic updates to the submitted safety plan or JHA's may be required if scope or techniques differ from original submission

CONTRACTOR SAFETY REPRESENTATIVE NAME & PHONE NUMBER

APPENDIX 12.10.E: Exchange of Lockout/Tagout Program

The lockout/tagout standard requires that NIST exchange energy control procedures with outside employers who service and/or maintain NIST equipment/machines

that require lockout/tagout. This form is used to notify both parties that they must comply with the restrictions and prohibitions of those procedures. This form shall be completed by the Contracting Officer's Representative in conjunction with the outside employer's representative. The exchange of information must occur before service/maintenance activities begin. If NIST workers will also be working on this equipment or in surrounding areas, this document must be attached to the Energy Control Procedures.

I. Identification of Outside Employer:

Name: ______________________________________ Phone: ___________________

Address: ____

Project Name/Equipment: ____

II. □ Check here to indicate that energy control procedures for the equipment/machine have been exchanged.

Comments: ____

____

III. After comparing the two Lockout/Tagout programs, note any additional restrictions/prohibitions below:

____

____

IV. Affected Personnel (listed below) shall understand and comply with these differences (use additional paper as needed and attach).

(Printed Name) (Signature)

__________________________________ __________________________________

__________________________________ __________________________________

__________________________________ __________________________________

__________________________________ __________________________________

V. Acknowledged acceptance of the provisions of this form:

Outside Employer Representative: ____________________________ ______________

(Signature) (Date)

NIST Contact: _______________________________ ______________

(Signature) (Date)

Primer on Safety Plan Requirements/Addendums for COVID-19

For all service contracts taking place on NIST facilities, it is required that safety plans include an addendum specific to COVID-19 safety measures to address safe workplace requirements for all workers and NIST staff.

What follows is a listing or requirements to be included in a safety plan addendum by all contractors:

1. Overarching Guidance:

    a. Contractors are to immediately report any cases of suspected or confirmed COVID-19 cases immediately to NIST COR.

    b. Contractors/contractor employees must comply with NIST COVID-19 specific safety guidance.

    c. How will the contractor ensure compliance with NIST requirements, CDC and OSHA guidelines?

    d. What physical workplace changes will be implemented to maximize worker and visitor safety?

    e. What requirements will be placed on changes in worker behavior on the project site?

    f. What training and communications will be utilized to ensure knowledge and understanding of updated requirements when coming to work?

    g. Contractor response to workers who have been on site who test positive for COVID-19

2. Physical Workplace Requirements:

    a. Signage

    b. Cleaning & Disinfection Stations

    c. Changes in cleaning and disinfection protocols of equipment used in performance of the contract and contractor-controlled workspaces.

3. Worker Behavior:

    a. Requirements for staying home when sick.

    b. Worker health checks

    c. Social Distancing

    d. Face Coverings

    e. Gathering restrictions & how they will be dealt with for common activities of groups of more than 10 people

    f. Work Breaks and Lunch Revisions

    g. Reducing physical contact with others

    h. Changes in AHA’s & JHA’s for COVID-19 specific measures

    i. Shared resources (equipment, supplies, etc.) revisions

4. Training & Communications:

a. Mandatory back to work training for employees on new requirements/expectations and documentation to track training completion.

b. Info to employees on changes in the workplace to mitigate the spread of germs/virus

c. Plan for how to communicate information to employees on COVID-19 specific changes in the workplace beyond training

5. NIST specific requirements can be accessed at the website below. The contractor shall ensure compliance with the most current NIST Visitor and Contractor Protocols for Mitigating COVID-19 Exposure on NIST Campuses, to include use of the most current COVID-19 Self-screening checklist.

https://www.nist.gov/about-nist/visit/nist-visitor-and-contractor-protocols-mitigating-covid-19-exposure-nist-campuses

PWS Attachment 11 - Emergency Generators			
Generator Number	Description/Nomenclature	Location	
EMG-C-019	EMERGENCY GENERATOR, CUMMINS, Model DGHE-5750399 Serial Number B060887377	42-001	BOULDER
EMG-C-021	EMERGENCY GENERATOR, GENERAC INDUSTRIAL POWER 3 PHASE, 277/480 VAC, 451 A Model 12521860500, Serial Number 2108765	42-X-S	BOULDER
EMG-C-022	EMERGENCY GENERATOR, GENERAC INDUSTRIAL POWER 3 PHASE, 277/480 VAC, 451 A Model 12521860500 Serial Number 2108764	42-X-S	BOULDER
EMG-C-023	EMERGENCY GENERATOR, GENERAC INDUSTRIAL POWER 3 PHASE, 277/480 VAC, NATURAL GAS Model 12521860500 Serial Number 2108766	42-X-S	BOULDER

WWVh Field Sit, Kekaha, HI Lighting Summary								
Room	# of fixtures per space	Fixture/Lamp Type						
		T-8/4'	T-8/2'	T-12/4'	Biaxial CF26DT	Sodium SOX 35W	Incandescent bulb (60W)	LED
Reception	2	6						
Office1	2	6						
Office2	2	6						
Janitor Closet	1				1			
Bathroom	5	2	3		1			
Lunchroom	2	4						
Hallway	5	10						
Lab	15	45						
Screen Room 1	5			10				
Screen Room 2	6			12				
Transmitter Room	20	60						
Generator Room	12			24				
Shop	8	16						
Garage	4						4	
Exterior wallpacks	8					8		
Exit/Emergency	3							X
Exit	1							X
Security/Motion	3							X
Emergency	7							X

Performance Work Statement (PWS)
Operations and Maintenance Services
Central Utility Plant (CUP) - NIST Boulder, CO Campus
&
NIST Facilities in Ft. Collins, CO & Kauai, HI

Attachment 13
Contract Transition Requirements
September 2020

a) Services under this contract are vital to the Government and must be continued without interruption. In accordance with contract clause 52.237-3, Continuity of Services, both the incoming and outgoing contractors shall provide services to facilitate an orderly and efficient transition to the contractor selected for the award of this contract and the subsequent follow-on Central Utility Plant (CUP) Operations and Maintenance contract.

b) The transition period between contractors shall be the first 30 days of the new contract and the final 30 days of the existing contract. During the 30 day transition the incumbent contractor shall cooperate and provide support with the transition of the successor contractor as outlined below.

c) Incumbent Contractor Responsibilities

i) During the 30 day transition the incumbent shall fully staff the CUP per the contract requirements, shall perform all contract requirements, and is fully responsible for contract performance.

1) All performance requirements related to CLIN X001 shall remain in place to include chemical treatment, preventative maintenance, minor repairs etc. The Contractor shall ensure that the plant is operable to meet daily load requirements for Steam, Chilled Water, and laboratory-grade air.

2) The incumbent shall continue to meet all contractual obligations for the preventative maintenance for the 30 days for CLINS X002, X003, and any remaining authorized work orders or emergency work orders issued under CLIN X004.

3) The incumbent shall remain the first responder to alarms.

ii) The successor contractor is authorized to staff the CUP facility and Boulder Site with one stationary engineer and one mechanic during each 8-hour shift for purposes of observation and familiarization with current operations and processes, to include witnessing responses to alarms. This transition is referred to as "shadowing". The incumbent contractor shall accommodate the successor contractor staff in the CUP and on the Boulder Site and cooperate with the successor contractor during the 30-day transition or shadowing period.

iii) The incumbent contractor shall coordinate a date during the 30-day transition in which the transition of chemical storage shall be transitioned between contractors.

iv) The government will provide general Maximo and Metasys training to the successor contractor.

v) The incumbent shall be responsible for providing training on the following:

1) Boilers:
How to swap the boilers from manual to automatic
Go through a manual transfer to the generators at the CUP
Startup/shutdown of a boiler
Transferring from one boiler to another

2) Chilled Water System:

Chiller rotation – How to rotate the chiller to the one you want when you are doing maintenance
Review of the flushing of the strainers for the cooling towers
Rotation of cooling towers
How to reverse the blades of the cooling towers for a defrost cycle

3) Air Compressors/Dryers:
Swapping the air compressors from condenser water to domestic water and back.
Changeover of air compressors and air dryers for manual rotation

4) How to operate the plant in manual mode (without Metasys):
Chillers/Cooling Towers
Boilers
Air Compressors

5) Water chemistry – demonstrate for successor contractor how water chemistry testing/monitoring is currently completed.

iv) During the 30 day transition the incumbent shall coordinate and work with one stationary engineer and one mechanic during each 8-hour shift. Allowing the successor contractor to observe the plant in operation 24/7 shall be key in the transition. Allowing observation in the swing and mid shifts will allow the successor contractor the ability to witness out of hours responses.

v) The incumbent remains first responder to alarms shall take the on-duty mechanic from the successor contractor with them on the alarm response.

vi) In addition, the incumbent contractor shall insure the successor contractor participates in all mechanical room walks and any chemical checks for the existing equipment on the Boulder site.

d) Successor Contractor Responsibilities

i) The successor contractor shall staff, at a minimum, one stationary engineer and one mechanic per 8-hour shift. The successor contractor shall obtain a NIST email, all required network/system access (Metasys/Maximo), and be able to access, view Metasys and acknowledge or respond to alarms with the incumbent contractor.

ii) During the 30 day transition the successor contractor shall observe the incumbent contractor in the operation of the CUP 24/7. In addition, they shall respond with the incumbent contractor on-shift mechanic the for alarm responses.

iii) The successor contractor shall obtain Maximo and Metasys training during the 30-day transition from the appropriate NIST personnel. Trainings will be offered between the hours of 7:00-3:30; the contractor shall make available all personnel for these trainings.

iv) During the 30 day transition a date shall be established to have the mechanic(s) from the successor contractor at the Boulder site for an onsite review of the equipment that is maintained under this contract. An orientation on what preventative maintenance and repairs have been performed shall be completed with a NIST representative. The successor contractor mechanic shall accompany the incumbent contractor mechanic during their monthly PMs during the 30-day transition for observation.

v) During the 30 day transition a date shall be established to have the mechanic(s) from the successor contractor walk each of the main buildings on the NIST site with the appropriate building manager. This will include Buildings: 1, 2, 3, 24, 81, and 92. The intent will be to identify the different types of HVAC systems operating in the buildings so as to have familiarity when responding to alarms out of hours.

vi) During the 30 day transition a date shall be established to have a mechanic from the successor contractor at the Ft. Collins site for an onsite review of the equipment that is maintained. An orientation on what preventative maintenance and repairs have been performed shall be completed with a NIST representative.

vii) During the transition between the incumbent contractor and the successor contractor the successor contractor shall not utilize any of the incumbent contractor's property such as microwave, refrigerator, or personal computers utilized to perform work on their contract.

viii) During the 30 day transition the successor contractor shall make appropriate arrangements to install their chemical condenser water and boiler water treatment equipment. The successor contractor shall coordinate installation during the 30 days in which the chemical condenser water treatment equipment and its operation shall be transitioned between contractors. At the end of the 30 days the successor contractor shall begin providing chemicals immediately and have their chemical supply on hand. During this 30 days, the successor contractor shall provide the chemical parameters to align with their intended chemical treatment plan to the COR for review.

ix) In the event the incumbent contractor is awarded the follow-on contract there will be no transition period and performance shall commence as stated in the follow-on contract.